KU LEUVEN

FACULTY OF SCIENCE

Maximising Edge Density Subject to Connectivity Constraints

Erdős Problem 915, from Proof to Search

Louis Vandenbruwaene

Promotor: Dr. Stijn Cambie
Reader: Prof. Jan Goedgebeur
KU Leuven

Thesis presented in
fulfillment of the requirements
for the degree of Master of Science
in Mathematics

Academic year 2026–2027

Foreword

This thesis studies how many edges or arcs a network can contain when no pair of vertices is allowed to have too many independent routes between them. The starting point is Erdős Problem 915, an extremal question about edge-disjoint paths in simple graphs. The work begins with the necessary background to formulate the problem and all its variants. It then discusses their answers or highlights the difficulty behind unsolved cases. Afterwards, computational research on small graphs confirms these values or provides evidence for conjectures. This is accomplished by measuring connectivity with a maximum-flow routine and by searching for extremal graphs with metaheuristics. Completed computations either prove the extremal value in a finite search space or produce a lower bound, while unfinished searches provide only evidence for conjectures. At the end, the thesis synthesises the results and presents the open problems that remain. This work is theoretical in nature, but the computational methods and the resulting conjectures are of interest to anyone who uses networks and wants to understand the relationship between connectivity and size.

I am grateful to my promotor, Dr. Stijn Cambie, for his day-to-day guidance, corrections, and mathematical feedback throughout the project. I could not ask for a more patient and enthusiastic supervisor. I thank my reader, Prof. Jan Goedgebeur, whose graph theory implementation knowledge pushed the boundaries of the computational work significantly. I also thank the people who discussed early versions of the arguments, checked examples, or helped clarify the presentation. Above all I thank my girlfriend Sara, whose patience and encouragement carried me through the long stretches of this work.

Contribution Statement

All the work in this thesis is my own, except where I explicitly say otherwise. I used artificial intelligence to implement the code, suggest and verify proofs, format LaTeX and check spelling. Where a result comes from the literature I say so and cite it. The theorems I use but do not claim as mine are those of Mader, Bártfai, Bollobás, Leonard, Sørensen and Thomassen, Menger, Whitney, Gomory and Hu, Mantel, Tutte, and Baranyai. Related hypergraph-connectivity work includes Bérczi, Chandrasekaran, Király and Kulkarni [1]. Proofs are located in Appendix A and all computational work is available in my personal repository [2].

A result whose proof I have not checked line by line carries the badge [AI] in its heading, both in Chapter 1 where it is stated and in Appendix A where it is proved. Those proofs were written by an AI model, and for several of them I am not the right person to check them. A badged result should be read as a conjecture that comes with a proof attached which is likely correct, rather than as a theorem I can vouch for myself. The proofs that carry no badge are the ones I have worked through myself. Two of them I wrote out in full: the Gomory-Hu double count for the simple undirected edge-disjoint case (Theorem 1.2) and for the undirected multigraph edge-disjoint case (Theorem 1.3), together with the three counting lemmas they run on (Lemmas A.5 to A.7). The other unbadged results of Appendix A are ones whose proofs I have since read and checked. Results taken from the literature carry no badge either, since their proofs are published and are cited where the result is stated. Definitions, constructions, conjectures and open questions state nothing that a proof could settle, so they carry no badge.

The summary of my original results is:

Variant	Contribution
Hypergraph, vertex separation	Exact value at $m = 2$. The $m = 3$ universal bound and its stated attainment range, via a new incidence-rank lemma (Theorems 1.9 and 1.10).
Directed multigraph	Exact value for all n, m, plus quadratic-branch classification for $n \geq \max(8, 2m)$ (Theorems 1.13 and A.48).
Directed simple graph, arc separation	Unconditional $n^2/4+\Theta_m(n)$ bound and an $O(\sqrt{m}\,n^{3/2})$ structural distance from a one-way bipartite graph (Proposition A.29 and theorem 1.11).
Transferred quadratic bounds	$k_m^{\mathrm{dir}}(n) = n^2/4+\Theta_m(n)$ and, for fixed r, m, leading term $(m-1)n^2/(4(r-1))$ for directed hypergraphs under both separations and all orientations (Theorems 1.12, 1.14 and A.64).

Short Summary

Erdős Problem 915, posed by Bollobás and Erdős [3], asks for the maximum number of edges in a graph on n vertices such that no two vertices are connected by m edge-disjoint paths. In modern notation this asks for

$$\ell_m(n) = \max\{|E(G)| : |V(G)| = n,\ \lambda_G^{\max} \leq m - 1\}.$$

Mader proved that, for simple graphs and all $n \geq 2$ and $m \geq 2$,

$$\ell_m(n) \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$$

and provided a construction achieving this bound when $(m-1)|(n-1)$. This thesis revisits this theorem and studies the corresponding extremal problem under three independent changes: the graph model (simple graphs, multigraphs, hypergraphs and multihypergraphs), the path-separation notion (edge-disjoint or internally vertex-disjoint) and the presence or absence of directions. The simple edge-disjoint case is proved independently here using Gomory-Hu trees [4], and a general extremal construction resolves it for all n and m. Other rigorously settled variants include the multigraph value $L_m(n) = (m-1)(n-1)$, the equality of the simple edge and vertex problems for $m \leq 4$, due to Bártfai [5], Bollobás [6] and Leonard [7], and their divergence at $m = 5$, shown by Sørensen and Thomassen [8]. The directed case is resolved for $m = 2$: the value $\ell_2^{\text{dir}}(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ is proved by a long induction and for $m \geq 3$ the hand proof gives way to a conjecture, after an explicit ten-vertex graph rules out the naive guess.

The centrepiece is then a unified program, which represents non-hypergraphs with a multiplicity matrix and converts the vertex-disjoint problem to an edge-disjoint one by a standard vertex-splitting construction. The hypergraphs are handled as a list of hyperedges, which may use head and tail sets to represent directed hyperedges. Then a maximum-flow algorithm measures connectivity between all pairs of vertices and reports the maximum local connectivity. At each point of the program, optimisations are applied to reduce the search space and traverse it more efficiently. When these optimisations are exhausted and the computation becomes intractable, the program uses metaheuristics to search for extremal graphs, which are then checked for correctness.

The hypergraph vertex problem is settled completely at $m = 2$. At $m = 3$ the formula $\lfloor 2(n-1)/(r-1) \rfloor$ is a universal upper bound for multihypergraphs and is exact whenever $r \geq 3$ and $2 \leq \binom{n-2}{r-2}$, the upper bound following from a new rank bound on incidence graphs proved by way of Tutte's triconnected decomposition. On the directed side the multigraph problem closes completely, $L_m^{\text{dir}}(n) = (m-1)\max(2(n-1), \lfloor n^2/4 \rfloor)$ for every n and every m, by deleting a sparse reachability-preserving subgraph at a time rather than a vertex at a time. For every fixed $m \geq 3$, the extremal value for simple digraphs, arc and vertex alike, is $n^2/4 + \Theta_m(n)$ with

no assumption on the shape of the extremiser, and at $m = 2$ it differs from $n^2/4$ by only an $O(1)$ correction. For fixed r and m, the directed hypergraph, which had only bounds a factor of four apart, has its leading term settled at $(m-1)n^2/\big(4(r-1)\big)$, so its bipartite construction is asymptotically optimal.

Abbreviations and Symbols

MILP	mixed-integer linear program
SPQR	the triconnected decomposition tree (S series, P parallel, Q single edge, R rigid)
[AI]	in a theorem heading: the proof of that result was written by an AI model and is not verified line by line by the author (Contribution Statement)

$G = (V, E)$	graph with vertex set V and edge set E
$D = (V, A)$	directed graph with vertex set V and arc set A
n	number of vertices
m	forbidden-connectivity parameter, with all local connectivities at most $m-1$
r	hyperedge size (every hyperedge spans r vertices)
$\lambda_G(u, v)$	maximum number of edge-disjoint u-v paths
$\lambda_G^{\max}$	maximum local edge-connectivity over all vertex pairs
$\kappa_G(u, v)$	maximum number of internally vertex-disjoint u-v paths
$\kappa_G^{\max}$	maximum local vertex-connectivity over all vertex pairs
$\ell_m(n)$	simple undirected edge-disjoint extremal function
$L_m(n)$	undirected multigraph edge-disjoint extremal function
$k_m(n)$	simple undirected vertex-disjoint extremal function
$\ell_m^{\mathrm{dir}}(n)$	simple directed arc-disjoint extremal function
$L_m^{\mathrm{dir}}(n)$	directed multigraph arc-disjoint extremal function
$k_m^{\mathrm{dir}}(n)$	simple directed vertex-disjoint extremal function
$\ell_m^{(r)}(n)$, $k_m^{(r)}(n)$	r-uniform hypergraph edge- and vertex-disjoint extremal functions
$M(n)$	$\max\bigl(2(n-1), \lfloor n^2/4 \rfloor\bigr)$, the directed multigraph value per unit of $m-1$
$M^*(n)$	the m-free directed multigraph optimum, $L_m^{\mathrm{dir}}(n) = (m-1)\, M^*(n)$, equal to $M(n)$ by Corollary A.41
$B_{p,q}$	one-directional complete bipartite digraph, all arcs from a p-part to a q-part
$\mu(u, v)$	multiplicity of an edge or arc between u and v
$h_m(b)$	most edges a feasible 2-connected block on b vertices can carry (Theorem A.13)
c_m	per-vertex growth constant of the simple undirected vertex problem, $c_m = \lim_{n\to\infty} k_m(n)/(n-1) = \sup_{b\geq 2} h_m(b)/(b-1)$ (Theorem A.13)

Contents

Chapter 1

The Problem and Its Variants

A network is useful not only because it connects, but because it connects in more than one way. A single road between two towns is a liability and a second independent road is insurance. The mathematical content of that intuition is the number of routes between two points that share no segment and the question this thesis circles is how many connections a network can hold before some pair of points is forced to have too many such routes. This chapter gives the foundation for the question and its variants and goes over the results that can be proven using logical arguments.

1.1 Graphs and routes

To state the question precisely we need only a few ideas. A *(multi)graph* G is a finite set of points, called *vertices*, together with a multiset of *edges*, each joining two distinct vertices. Asking the two ends to be distinct rules out loops, which some texts do admit and which play no role in anything below. We write $V(G)$ for the set of vertices, $E(G)$ for the multiset of edges, and $\mu(u,v)$ for the *multiplicity* of a pair, the number of copies of the edge joining u and v. The *size* $|E(G)| = \sum_{u \neq v} \mu(u,v)$ counts the edges with their multiplicity, and it is the very quantity this thesis tries to make as large as possible under certain constraints. If $\mu(u,v) \leq 1$ for every pair, so that no edge is repeated, G is a *simple graph*. A repeated edge joins two vertices exactly as a single edge does, so what separates the two notions is the multiplicity rather than the number of ends. One pictures such a simple graph by representing the vertices as dots and the edges as lines between them.

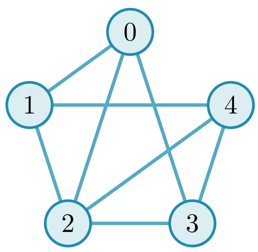


Figure 1.1. The simple graph $K_5 - \{04, 13\}$ has five vertices and eight edges, where K_n is the complete graph on n vertices.

We write $n := |V(G)|$ for the number of vertices and define the *degree* of a vertex as the number of edges meeting it. Unless specified otherwise, graphs are simple. A *path* from a vertex u to a vertex v is a sequence of distinct vertices that starts at u, ends at v, and joins each consecutive pair by an edge. Two paths between the same endpoints are *edge-disjoint* if no edge belongs to both. The largest number of pairwise edge-disjoint paths between u and v is their *local edge-connectivity* $\lambda_G(u, v)$. The pair with the most such paths determines the value for the whole graph:

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v).$$

A theorem of Menger [9], recalled in Appendix A, gives this quantity a second face: $\lambda_G(u, v)$ also equals the least number of edges one must remove to cut every route from u to v (Figure 1.2). The maximum number of edge-disjoint paths between a pair is exactly the number of edges in the minimum cut between them when every edge has unit capacity.

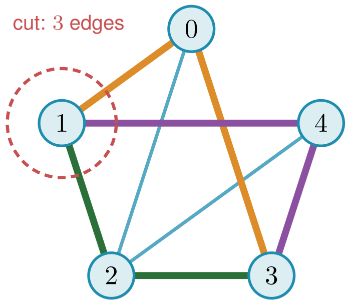


Figure 1.2. Menger's theorem on Figure 1.1: three edge-disjoint 1–3 routes (coloured) meet the minimal three-edge cut around vertex 1, so $\lambda(1, 3) = 3$.

1.2 Erdős Problem 915

The constraint at the heart of this thesis is a cap on connectivity. Requiring $\lambda_G^{\max} \leq m - 1$ says that no pair of vertices is joined by as many as m edge-disjoint paths, or equivalently that every pair can be separated by removing fewer than m edges. The question is how many edges a graph can hold while staying under that cap.

Question 1.1 (Erdős Problem 915 [10]). How many edges can a graph on n vertices have if no two of its vertices are joined by m edge-disjoint paths?

Writing the answer as an extremal function,

$$\ell_m(n) = \max\{\, |E(G)| : |V(G)| = n,\ \lambda_G^{\max} \leq m - 1 \,\},$$

the problem asks for $\ell_m(n)$, the most edges possible under the cap. Mader proved that no graph under the cap has more than $\lfloor m(n-1)/2 \rfloor$ edges (the upper bound) in the simple-graph case and provided the shared-clique construction, which achieves the (lower) bound but is only defined when $m-1$ divides $n-1$. The brackets $\lfloor \cdot \rfloor$ round their content down to the nearest whole number and their ceiling counterpart $\lceil \cdot \rceil$ rounds up. Both recur throughout, because an edge count is always an integer while the formulas that bound it need not be.

Theorem 1.2 (Mader [11]). *For all $n \geq 2$ and $m \geq 2$,*

$$\ell_m(n) \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

Notice how $n \geq m$ is not needed for the inequality itself, and is generally needed to attain equality, since a graph on $n < m$ vertices cannot exceed the edge count $\binom{n}{2}$ of the complete graph on it, which falls short of $\lfloor m(n-1)/2 \rfloor$ once m exceeds n. The floor can still let a small case through by coincidence: K_2 attains $\lfloor 3/2 \rfloor = 1$ at $n = 2$, $m = 3$. The proof given in Appendix A was discovered independently of Mader and uses the Gomory-Hu tree (Theorem A.2). The shared clique graph is then generalised to an extremal construction that exists for all values of $n \geq m$ (Theorem A.10). The very same Gomory-Hu tree returns for the multigraph and hypergraph variants (Theorem 1.3, Proposition 1.8). Hence its properties are recorded here. Every graph has a weighted spanning tree such that $\lambda_G(u, v)$ equals the lightest edge on the tree path from u to v. All $\binom{n}{2}$ local connectivities are therefore encoded by just $n - 1$ weights. The largest of them, $\lambda_G^{\max}$, is simply the heaviest edge of the tree. The proof (Appendix A) uses a double counting argument on that tree.

1.3 Variants of the problem

So far the problem has been stated for simple graphs, but the same question remains unanswered in a more general setting. Each assumption made so far will be toggled to create a lattice of variants and the thesis will explore each node of that lattice.

The first is *arity*: an edge joins exactly two vertices and a *hyperedge* joins $r \geq 2$ vertices. Hence such *hypergraphs* use *Berge paths*, which aligns with the path definition above. The size of a hypergraph counts its hyperedges with their multiplicity, exactly as the size of a graph counts its edges. The second is *multiplicity*: a *simple* graph or hypergraph carries at most one copy of each edge, so $\mu \leq 1$ throughout, while a *multigraph* allows parallel copies. The third is *direction*: a directed edge, called an arc, is an ordered pair of vertices and a directed hyperedge has one tail and $r - 1$ heads. These three toggles create a lattice of eight models, shown in Figure 1.3.

The digraphs do not have a Gomory-Hu tree, but they do have a directed analogue of Menger's theorem. A digraph has three degree counts at each vertex. The *out-degree* $d^+(v)$ counts the arcs leaving v and the vertices they reach are its *out-neighbours*. The *in-degree* $d^-(v)$ counts the arcs entering v and the vertices they come from are its *in-neighbours*. The *total degree* $d(v) = d^+(v) + d^-(v)$ adds the two. Figure 1.4 shows a digraph carrying three arc-disjoint routes between one chosen pair of vertices. The directed form of Menger's theorem says that route count equals the size of the smallest arc-cut separating the pair, and the figure also marks the out-degree and in-degree that bound both quantities from above, which is exactly how Menger reads in a one-way network. A directed r-hyperedge has one tail and $r - 1$ heads. In the figures, one colour denotes one hyperedge, drawn either as a star or as a line oriented away from its tail. It is a single edge on several vertices, not a path of ordinary edges. A directed *Berge route* enters that edge at its tail and leaves at one of its heads.

The final assumption that will be considered is *separation*: routes may be required either to avoid sharing edges, as above, or to be internally vertex-disjoint, sharing no intermediate vertex. This gives the local vertex-connectivity $\kappa_G(u, v)$ and its maximum $\kappa_G^{\max}$. Whitney's inequality $\kappa_G(u, v) \leq \lambda_G(u, v)$ [12] records that avoiding shared vertices is at least as demanding as avoid-

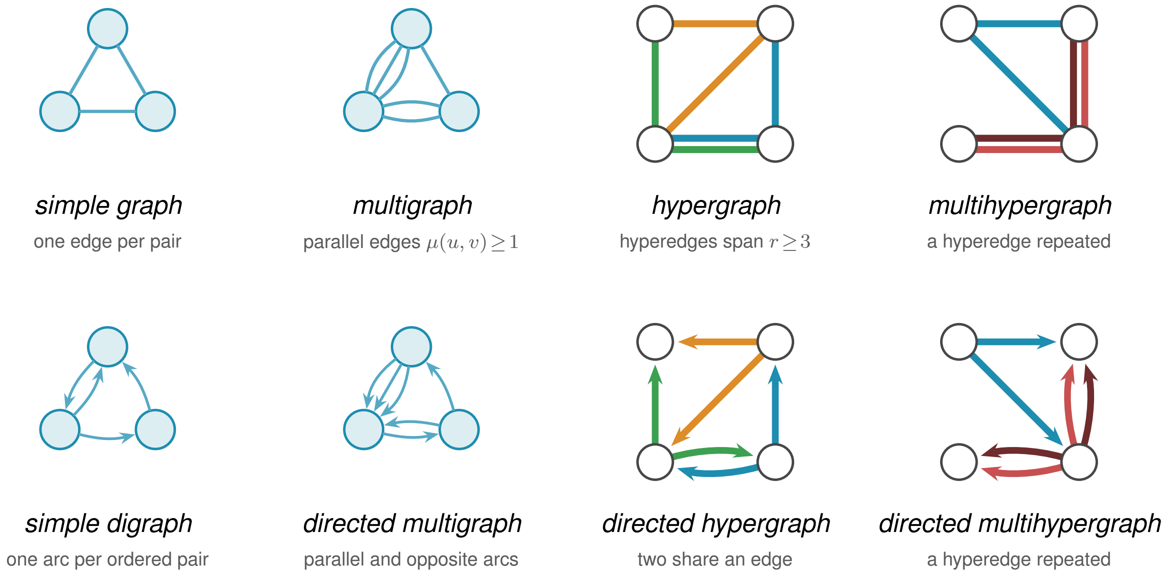


Figure 1.3. The eight models side by side, with the four undirected on top and the four directed below. The simple graph is a triangle, the multigraph has parallel edges, the hypergraph has three hyperedges, and the multihypergraph has one repeated hyperedge. The directed versions have arcs instead of edges, with the same pattern of repetition and hyperedges. In the two multihypergraph panels the two shades of red are the two copies of one hyperedge, not two different hyperedges.

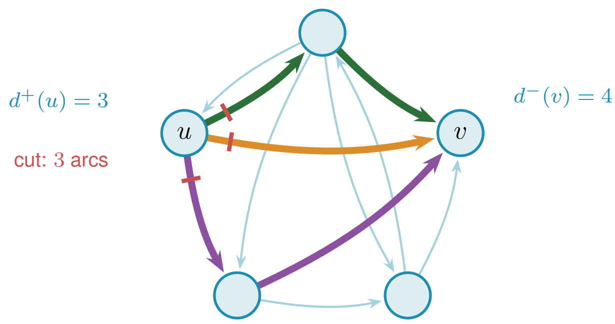


Figure 1.4. Three arc-disjoint $u \to v$ routes (coloured) and the corresponding three-arc directed cut (red) in a *simple* digraph. Thus $\lambda(u, v) = 3$, meeting the degree bound $\min(d^+(u), d^-(v)) = 3$.

ing shared edges and Figure 1.5 shows the gap on a graph where two routes are edge-disjoint yet forced through one common vertex.

In the vertex-disjoint variants, multiplicity of an edge is irrelevant compared to its existence, so parallel edges never raise κ. Therefore they never break the connectivity constraints either. Counted with multiplicity, the maximum number of edges could simply be infinite and the extremal question would be empty. The multigraph vertex variants are thus posed on the underlying simple graph, which is why those two rows of Table 3.1 reduce to their simple counterparts and why every figure in this thesis reports the reduced problem for them.

With all this in mind, the thesis considers multihypergraphs and directed multihypergraphs for both edge- and vertex-disjoint routes. Ordinary graphs ($r = 2$) and simple hypergraphs (no repeated hyperedges) are special cases. The final status map appears in Figure 3.3. The notation for all their extremal functions is summarised in Table 1.1.

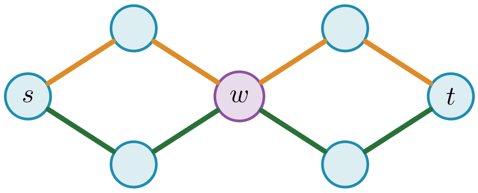


Figure 1.5. The separation axis: two edge-disjoint s–t routes both pass through w, so $\lambda(s,t) = 2$ but $\kappa(s,t) = 1$.

Table 1.1. Notation for the extremal functions.

Model	edge-disjoint routes	internally vertex-disjoint routes
Simple graph	$\ell_m(n)$	$k_m(n)$
Multigraph	$L_m(n)$	$K_m(n)$
r-uniform simple hypergraph	$\ell_m^{(r)}(n)$	$k_m^{(r)}(n)$
r-uniform multihypergraph	$L_m^{(r)}(n)$	$K_m^{(r)}(n)$
Simple digraph	$\ell_m^{\mathrm{dir}}(n)$	$k_m^{\mathrm{dir}}(n)$
Directed multigraph	$L_m^{\mathrm{dir}}(n)$	$K_m^{\mathrm{dir}}(n)$
Directed r-uniform simple hypergraph	$\ell_m^{\mathrm{dir}\,(r)}(n)$	$k_m^{\mathrm{dir}\,(r)}(n)$
Directed r-uniform multihypergraph	$L_m^{\mathrm{dir}\,(r)}(n)$	$K_m^{\mathrm{dir}\,(r)}(n)$

1.4 Proven bounds

Theorem 1.3. *For undirected multigraphs on n vertices, $L_m(n) = (m-1)(n-1)$.*

Appendix A gives the full proof which also uses the Gomory–Hu tree. The lower bound is attained by a spanning tree with every edge thickened to $m-1$ parallel copies.

In simple graphs the edge and vertex problems give the same answer for $m \le 4$, and three papers together say so. The vertex values came first: k_3 from Bártfai [5] and k_4 from Bollobás [6]. Leonard [7] then proved that the two separations agree throughout $2 \le m \le 4$, in the same paper that settled the edge value at $m = 5$, a year before Theorem 1.2 covered every m at once.

Theorem 1.4 (Bártfai [5], Bollobás [6], Leonard [7])**.** *For all $n \ge m$ and $m \le 4$,*

$$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor .$$

By Whitney's inequality every edge-feasible graph is vertex-feasible, so $k_m(n) \ge \ell_m(n)$ (Figure 1.6). One is tempted to expect the equality forever. This, however, is not the case, and Sørensen and Thomassen proved that the vertex problem diverges from the edge problem at $m = 5$.

Theorem 1.5 (Sørensen-Thomassen [8])**.** *For simple graphs,*

$$k_5(n) = \left\lfloor \frac{5(n-1)}{2} \right\rfloor = \ell_5(n) \quad (6 \le n \le 13), \qquad k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 4 \quad (n \ge 14).$$

The two separations therefore part company at $n = 14$, and $k_5(n) > \ell_5(n)$ for every $n \ge 14$.

Sørensen and Thomassen state the second formula for every $n \geq 6$ apart from $n = 7$ and $n = 12$, and supply those two values separately. On the small range the two formulas agree except at exactly those points, where the second reads one too few at $n = 7$ and one too many at $n = 12$, so splitting the statement by range removes both exceptions. Remark A.12 records the counting convention that turns their numbers into these.

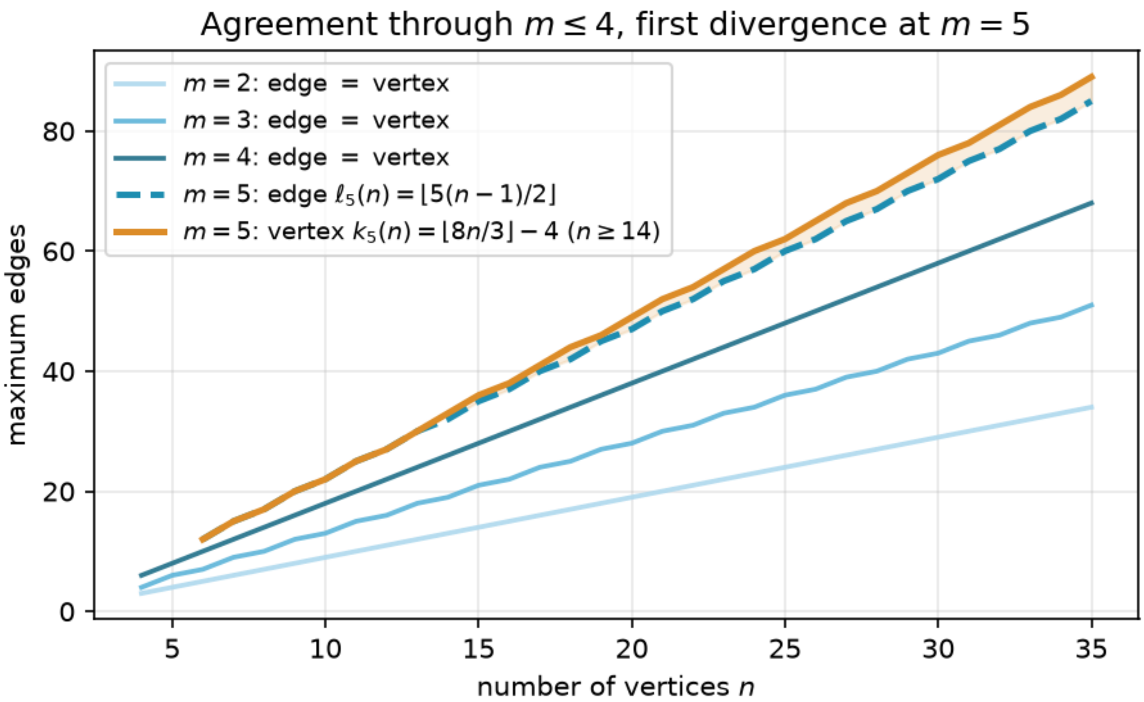


Figure 1.6. The edge and vertex extremal functions for $m = 2, 3, 4$ (blue shades, single curve per m since they agree) and $m = 5$ (dashed blue for edge, solid orange for vertex). The vertex function diverges from the edge function at $n = 14$ and grows faster thereafter.

Theorem 1.6 (Directed arc value at $m = 2$, [AI]). *For simple digraphs,*

$$\ell_2^{\text{dir}}(n) = \max\left(2(n-1),\ \left\lfloor \frac{n^2}{4} \right\rfloor\right).$$

Two constructions give the lower bound, one on each branch, and Figure 1.7 draws both. The first is a hub joined to every other vertex in both directions, which reaches $2(n-1)$ arcs. That branch has no unique extremiser, since any tree on n vertices with every edge taken in both directions reaches $2(n-1)$ arcs too, the tree offering each pair only its one path in each direction, and the bidirected star is the case of it drawn here. The second sends every arc from one half of the vertices to the other and none back, which reaches $\lfloor n^2/4 \rfloor$. Section A.3 defines both and checks that neither carries a second route between any pair. The upper bound is an induction on n, proved in Section A.4 together with the three lemmas it runs on: delete a vertex of minimum total degree, which carries at most the average number of arcs, and apply the bound to what remains. The base cases $n \leq 7$ are left to exhaustive search rather than hand argument, and Chapter 2 is what runs them. The same induction never mentions the separation, so it survives the change to κ verbatim once the base cases are re-run.

Theorem 1.7 (Directed vertex value at $m = 2$, [AI]). *For simple digraphs, $k_2^{\mathrm{dir}}(n) = \ell_2^{\mathrm{dir}}(n)$.*

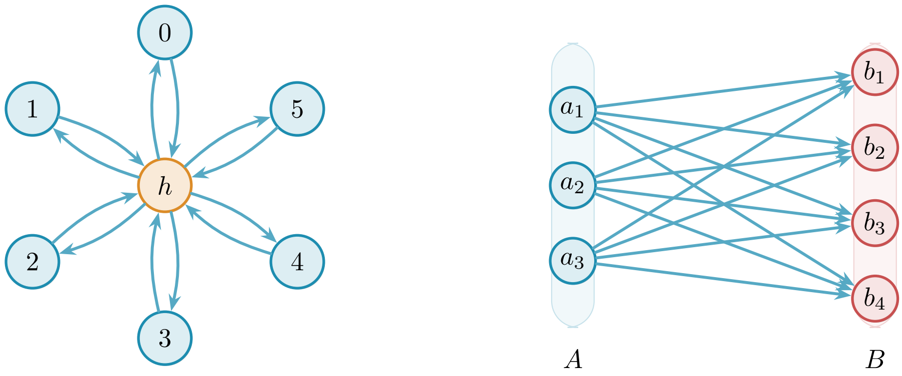


Figure 1.7. The two branches tie at $n = 7$, where each holds 12 arcs at $\lambda^{\max} = 1$. Left: the bidirected star, a hub joined to six spokes both ways, $2(n-1) = 12$. Right: the one-directional complete bipartite digraph $B_{3,4}$, every arc from A to B, $\lfloor n^2/4 \rfloor = 12$.

The hub leads for $n < 7$ and the bipartite wall leads for $n > 7$. Notice how the directed value is quadratic in n for large n, while the undirected value is linear. This is because digraphs can be lopsided in one direction without creating routes back.

Proposition 1.8 (Hypergraph edge bound, [AI]). *An r-uniform multihypergraph on n vertices with Berge edge-connectivity at most $m-1$ has size at most $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$. There are simple hypergraphs attaining the bound when $m - 1 \le \binom{n-2}{r-2}$, and multihypergraphs attaining it when $(r-1) \mid n-1$.*

The proof is the multigraph argument again, run through the hypergraph Gomory-Hu tree, and Appendix A gives it. It has to be re-run rather than deduced from Theorem 1.3, because the obvious reduction points the wrong way. Replacing every hyperedge by a spanning star on its r vertices gives a multigraph of size $(r-1)|E(\mathcal{H})|$, but two routes may then take two different star edges of one hyperedge, which a family of Berge routes forbids. The star multigraph can therefore carry more connectivity than the hypergraph it came from, and a feasible hypergraph need not produce a feasible multigraph to count. The star hypertree of Figure 1.8 attains the $m = 2$ case: a hub joined to disjoint blocks of $r-1$ vertices, Berge connectivity one everywhere, $\lfloor (n-1)/(r-1) \rfloor$ hyperedges, exactly as a tree attained the multigraph bound. For $r \ge 3$ the bound is attained by *simple* hypergraphs, with no repeated hyperedge, whenever $m - 1 \le \binom{n-2}{r-2}$ (Theorem A.52). That is a real difference between the two edge sizes. For ordinary graphs simplicity is what separates Mader's $\lfloor m(n-1)/2 \rfloor$ from the multigraph value $(m-1)(n-1)$, and one edge size higher the two are equal.

Theorem 1.9 (Hypergraph vertex value at $m = 2$, [AI]). *For $r \ge 2$ and $n \ge 1$, the maximum number of hyperedges of an r-uniform multihypergraph on n vertices with $\kappa^{\max} \le 1$ is*

$$K_2^{(r)}(n) \;=\; \left\lfloor \frac{n-1}{r-1} \right\rfloor ,$$

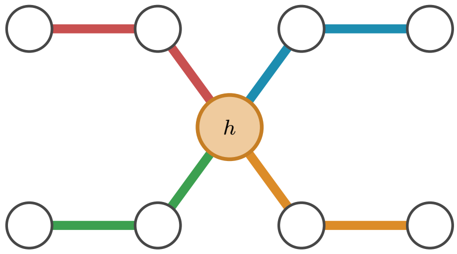


Figure 1.8. The $r = 3$, $n = 9$ star hypertree: four hyperedges meet only at the hub h, attaining $\lfloor (n-1)/(r-1) \rfloor = 4$ with Berge connectivity one.

attained by the star hypertree of Figure 1.8, which is simple. In particular $K_2^{(r)}(n) = k_2^{(r)}(n) = \ell_2^{(r)}(n)$: the vertex and edge problems agree at $m = 2$ for every uniformity, and repeated hyperedges never help.

Theorem 1.10 (Hypergraph vertex value at $m = 3$, [AI])**.** *For $r \geq 2$, every r-uniform multihypergraph on n vertices with $\kappa^{\max} \leq 2$ has at most $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ hyperedges. For $2 < r < n$, which is exactly the range where $2 \leq \binom{n-2}{r-2}$, the bound is attained by a simple hypergraph, so*

$$k_3^{(r)}(n) \;=\; \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor \;=\; \ell_3^{(r)}(n).$$

There are multihypergraphs attaining it when $(r-1) \mid (n-1)$ instead, by repeating one hyperedge rather than needing $\binom{n-2}{r-2}$ distinct ones, so $K_3^{(r)}(n)$ takes the same value on that range (Proposition A.57).

Both proofs read Berge routes as ordinary paths in the incidence graph, and Appendix A gives them. The $m = 3$ case is the harder one and needs a rank bound proved through Tutte's triconnected decomposition. At $m \geq 4$ that route stops for want of a 4-connectivity analogue, and the value is open. Proposition 1.8 counts hyperedges, but it does not say how to *measure* the Berge connectivity of a hypergraph we are handed, and that measurement is the new part. The metro picture shows why. An ordinary edge is a one-stop link between a single pair, so policing edge-disjointness means watching each pair on its own. A hyperedge is a whole line: it stops at all r of its vertices and so touches $\binom{r}{2}$ pairs at once, yet like a line that can run only one train at a time it may serve only one route. When several routes want to board the same line at different stations, the checker must wave exactly one through and turn the rest away and it must do this for every line simultaneously. That bookkeeping is the one new piece of machinery the next chapter builds and it is why the hypergraph rejoins the family only once the checker is in place.

Theorem 1.11 (The quadratic bound with a linear error, [AI])**.** *For all $m \geq 2$ and $n \geq 2$, every simple digraph D on n vertices with $\lambda_D^{\max} \leq m-1$ satisfies*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)(n-1),$$

so for each fixed m,

$$\ell_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \qquad (m \geq 3),$$

and, for $m \geq 3$, both the leading constant $\frac{1}{4}$ and the order *of the second-order term of Conjecture 3.1 hold unconditionally. At $m = 2$, Theorem 1.6 instead gives $\ell_2^{\mathrm{dir}}(n) = n^2/4 + O(1)$ as $n \to \infty$.*

The proof is in Appendix A. Its engine is a count of two-step routes: in a feasible digraph a vertex cannot have too many out-neighbours that reach one another, since the short detours through them would breach the cap, and that alone forces the arc count down to the wall plus a linear correction.

The routes this count produces are internally vertex-disjoint, not merely arc-disjoint. That is what lets the whole argument be re-run under the stricter separation, and Whitney's inequality alone would not justify the transfer, since it points the other way.

Theorem 1.12 (Directed vertex problem, leading constant and order, [AI])**.** *For all $m \geq 2$ and $n \geq 2$, every simple digraph D on n vertices with $\kappa_D^{\max} \leq m - 1$ satisfies*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)(n-1),$$

and consequently, for each fixed m,

$$k_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \qquad (m \geq 3).$$

Theorem 1.13 (Directed multigraph value, every n and m, [AI])**.** *For all $n \geq 2$ and $m \geq 2$,*

$$L_m^{\mathrm{dir}}(n) \;=\; (m-1)\,M(n), \qquad M(n) := \max\!\Big(2(n-1),\; \left\lfloor \frac{n^2}{4} \right\rfloor\Big).$$

The extremisers include the doubled trees on the linear branch and the balanced one-way bipartite multigraph on the quadratic one. On the linear branch these are constructions exhibited rather than a classification, and Appendix A says explicitly that not every linear-branch extremiser has been classified. The proof deletes a *subgraph* at a time rather than a vertex. Inside any feasible multigraph there is a sparse skeleton that alone preserves which vertices reach which, and removing one copy of each of its arcs costs every pair at least one route. Peeling off $m-1$ such skeletons therefore accounts for every arc, and the parity obstruction that stopped the vertex-deletion induction never arises. Equality also classifies the quadratic extremisers once $n \geq \max(8, 2m)$ (Theorem A.48), and only the smaller range remains open.

Theorem 1.14 (The directed hypergraph constant, [AI])**.** *Let $r \geq 2$ and $m \geq 2$. Every forward directed r-uniform multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa_{\mathcal{H}}^{\max} \leq m - 1$ satisfies*

$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1} \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)^2(n-1).$$

The same bound therefore holds under $\lambda_{\mathcal{H}}^{\max} \leq m-1$, which is the stronger hypothesis by $\kappa \leq \lambda$. Hence for fixed r and m both the edge- and the vertex-disjoint maxima are $\big(1 + o(1)\big)(m -$

$1)n^2/\big(4(r-1)\big)$*, and the bipartite construction of Proposition A.60 is asymptotically optimal for both.*

The proof, in Appendix A, forms the one-step shadow digraph whose arc $u \to v$ records that some hyperedge has tail u and head v. A matching argument transfers the hypergraph route cap to the shadow's two-step route budget, after which Lemma A.33 controls the shadow and a tail–head incidence count converts its arcs back to hyperedges. Since the bipartite construction already attains that leading term, it is asymptotically optimal.

For the directed simple digraph problem of Conjecture 3.1, the matching upper bound for fixed $m \geq 3$ is open, though not entirely: Theorem 1.11 proves unconditionally that the value is $n^2/4 + \Theta_m(n)$, so what the conjecture still claims beyond what is known is the coefficient of the linear term rather than the shape of the answer.

Chapter 2

Certifying and Discovering Bounds by Machine

The previous chapter gave insight into the problem and some variants, but left most of the harder questions open. This chapter will serve as a sanity check for the given theorems by testing them on small graphs through brute-force enumeration. It will also be the foundation of conjectures for the open cases, by discovering extremal graphs or at least sufficiently dense graphs. Pattern recognition on the discovered values, or even a family of extremal graphs, then lets conjectures be made, although those statements are naive because they are based on small graphs only. Larger graph spaces will be scoured by a heuristic, namely tabu search. A guided search is needed there because only a small fraction of the search space is extremal: assigning edges at random gives a binomially distributed edge count, so almost every graph lands far from the cap. The archive is the repository at `https://github.com/louisvandenbruwaene/thesis` and every file path quoted in this thesis is relative to it, with the commands and the audit trail collected in Section A.11. Below is a diagram of the program's architecture, showing how the modelled graphs are fed to the checker and how the results are used to either certify or discover bounds.

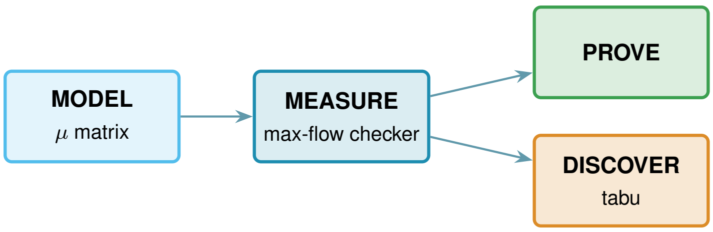


Figure 2.1. A single model feeds the exact max-flow checker. Certification turns that measurement into finite upper bounds, while heuristic search uses it to discover lower bounds. The badge colours on the code cards match the boxes here.

2.1 Modelling

All non-hypergraphs are represented using one multiplicity matrix μ: it is symmetric when undirected and binary when simple. Hypergraphs are stored as (multi)sets of hyperedges, which are sets of vertices with a tail distinguished in the directed model. Figure 2.5 shows this encoding for a directed multigraph and hypergraph. Matrix or tensor representations for hypergraphs are thus not recommended due to their size and sparsity. All possible graphs of a given size will later be checked to see if they are extremal for the given size and connectivity restraint. Matrices are iterated as a depth-first search with cells taking values between 0 and $m-1$, and the diagonal is always zero. Hypergraphs are implemented similarly from the ground up as there are no matrix tools for them. Notice how depth-first search is memory-efficient.

2.2 Measuring

Given n and m, walk through every graph on n vertices, measure the largest local connectivity $\lambda^{\max}$ or $\kappa^{\max}$ of each with a max-flow computation and keep the densest graph that stays at or below the cap $m-1$. If the enumeration cannot finish within the time limit, the best graph so far is returned. It is therefore important to distinguish between the densest graph possible and the densest graph found. Here they are called exact (upper) bounds and lower bounds respectively. In the vertex-disjoint and hypergraph cases the graph will be transformed before the flow can be measured. This ensures that every variant shares the same integer max-flow engine no matter the implementation. Thus the whole question reduces to a flow computation: A *maximum flow* asks how much can be pushed from one point to another through a network whose links each carry a limited capacity. Menger's theorem (Theorem A.1) already turned $\lambda(u, v)$ into the number of edges/arcs in a minimum cut. For a simple graph, $\lambda(u, v)$ is therefore also the capacity of a cut over a network with unit capacities. For a multigraph, the parallel copies can be combined into one link whose capacity is their multiplicity $\mu(u, v)$. In the directed case these capacities follow the direction of the arcs, while in the undirected case they are available in either direction. The max-flow/min-cut theorem of Ford and Fulkerson [13] says that this smallest cut has exactly the value of a maximum flow. So the route count we are after is just a maximum-flow value: measure the flow and you have measured the connectivity.

The routine that runs this flow and reports the route count is the connectivity *checker*. The program ships an optional compiled C helper that runs this same flow more efficiently.

2.2.1 Transformations

This max-flow computation is built for directed multigraphs (including simple digraphs) with edge-disjoint paths. That is why the other variants will have to be transformed into such a flow network. Undirected multigraphs have a natural identification with a directed multigraph, obtained by replacing every edge with a pair of arcs going in opposite directions. To transform a multihypergraph, replace each distinct hyperedge with a helper gate, an entry point and an exit point joined by a single arc whose capacity is the multiplicity of the hyperedge. Every member of the hyperedge then flows into the entry and out of the exit at unlimited capacity, so the one arc between them is the only bottleneck (Figure 2.3b). A directed hyperedge is wired the same way, except that only its tail reaches the entry and only its heads leave the exit, which is what forces a route to cross it from tail to head. The directed hypergraph's flow network is just a restricted version of the undirected one. Another representation for multihypergraphs is a separate gate with capacity one for each hyperedge.

The hypergraph Menger theorem (Theorem A.50) proves that the resulting maximum flow is exactly the Berge connectivity. Figure 2.2 shows four gate-disjoint routes in a whole hypergraph. For the vertex-disjoint hypergraph checker, the ordinary vertices are split as before and the helper point is added to the split network with infinite internal capacity.

Lastly, all vertex-disjoint versions can then be transformed into an identical edge-disjoint question by splitting each vertex v into an in-copy v^{in} and an out-copy v^{out} joined by a single unit-capacity arc (Figure 2.3a). Every arc $u \to v$ of the digraph becomes $u^{\text{out}} \to v^{\text{in}}$ in the split network. Routing through v now consumes the one unit of capacity on its internal arc, so two paths sharing v as an intermediate vertex compete for that unit and cannot coexist in any maximum flow. A max-flow from u^{out} to v^{in} in the split network therefore equals the number of internally vertex-disjoint u-v paths in the original graph. Notice how the undirected or hypergraph transformation is done beforehand and the gates aren't split by this step.

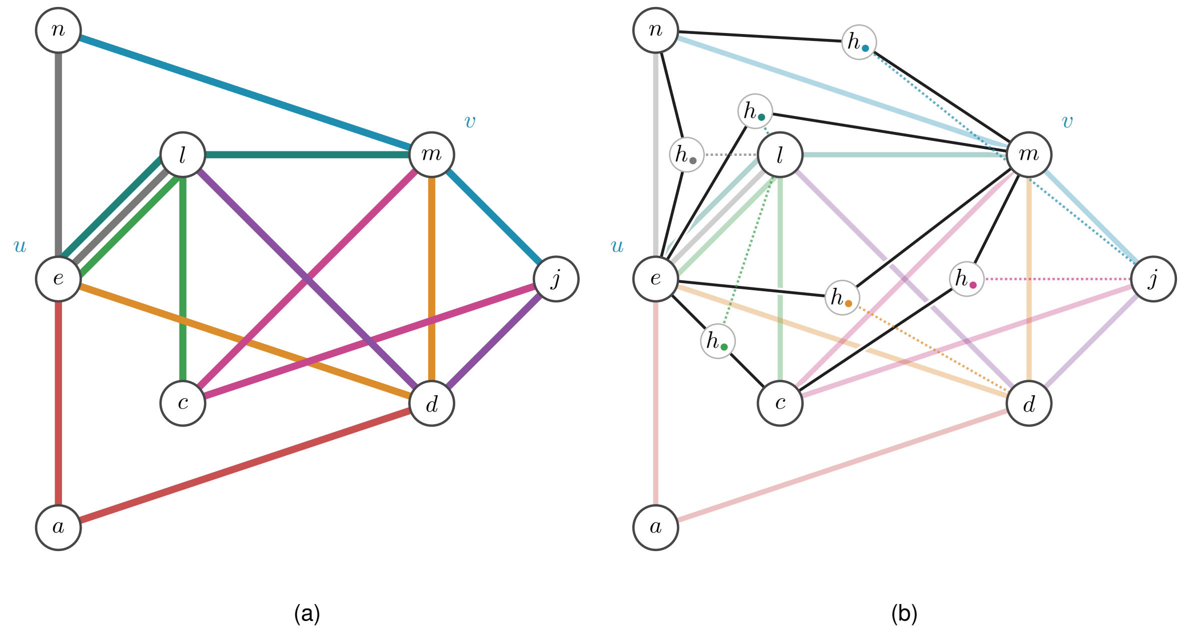
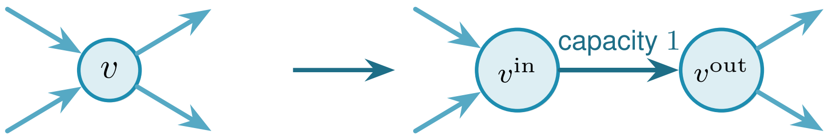


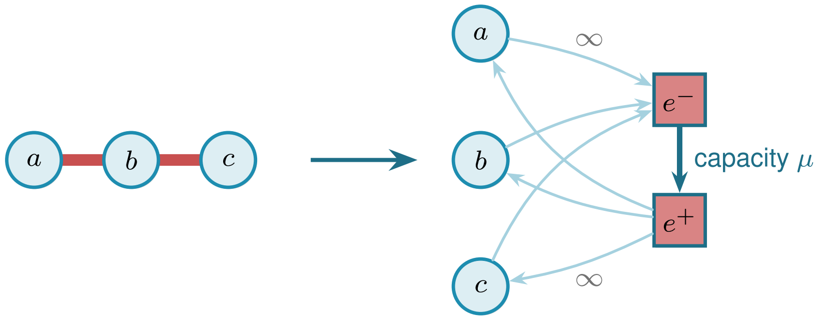
Figure 2.2. An undirected 3-uniform hypergraph and its gate network for u, v. The four black routes use distinct capacity-one gates (they are drawn as one point to reduce visual clutter), so the checker reads $\lambda(u, v) = 4$. A fifth would reuse a gate. The helper points connect to the hyperedge's third member by dotted lines.

2.3 Checking every graph of a fixed size

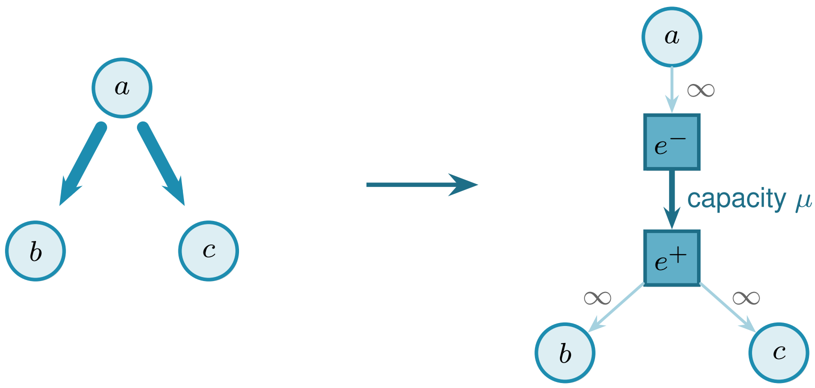
The connectivity checker works on all variants but measures only one graph at a time. All graphs of a fixed number of vertices must be enumerated to discover the bound for a single connectivity constraint. A simple graph on n vertices is a choice of $\mu(u, v) \in \{0, 1\}$ on each off-diagonal entry, a multigraph a choice in $\{0, \ldots, m - 1\}$, and a digraph varies every ordered pair rather than only the upper triangle. A hypergraph is not a matrix at all, so it is enumerated over its own list: a subset of all $\binom{n}{r}$ possible vertex sets of size r. In the directed model, the tail is specified with a pointer and the multihypergraphs have multiplicities for each hyperedge. Figure 2.5 shows how a directed multigraph and hypergraph are represented. Using these representations for all graph variants means iteration traverses all possible representation states easily. Walking those possibilities, measuring each candidate with the checker and keeping the densest feasible graph settles a bound exactly. If the enumeration hits its deadline before finishing, the returned graph is



(a) Split each vertex v into v^{in} and v^{out} joined by a unit-capacity arc to transform vertex-disjointness into edge-disjointness.



(b) Transform a hyperedge into a gate with capacity equal to the multiplicity of the hyperedge.



(c) The directed hyperedge is transformed into a gate with one tail and two heads. This gate has capacity equal to the multiplicity of the hyperedge and every route must enter at the tail and leave at a head.

Figure 2.3. The building blocks of the connectivity checker: vertex splitting, hyperedge gates and directed hyperedge gates.

labelled as a lower bound. On the small cases this exhaustive mode independently reproduces every value proved by hand in Chapter 1.

The vertex count dominates every other factor. The candidate count is (values per cell)$^{(\text{number of cells})}$, and raising n grows the number of cells itself, quadratically for graphs and as $\binom{n}{r} = \Theta(n^r)$ for fixed-rank r-uniform hypergraphs, cubically in the $r = 3$ experiments plotted here. Raising the connectivity budget only enlarges the base (two states per cell for a simple graph, more for a multigraph), and adding direction only multiplies the number of cells: it decides both directions of a pair independently, turning that pair's two states (edge or none) into four rather than doubling them. Which of the latter two costs more depends on m and the model rather than following one fixed order. Directed hypergraphs are especially numerous because each hyperedge also requires a choice of tail and heads. These are precisely the variants whose answers Chapter 1 could not prove. The separation, by contrast, does not change the enumeration: edge

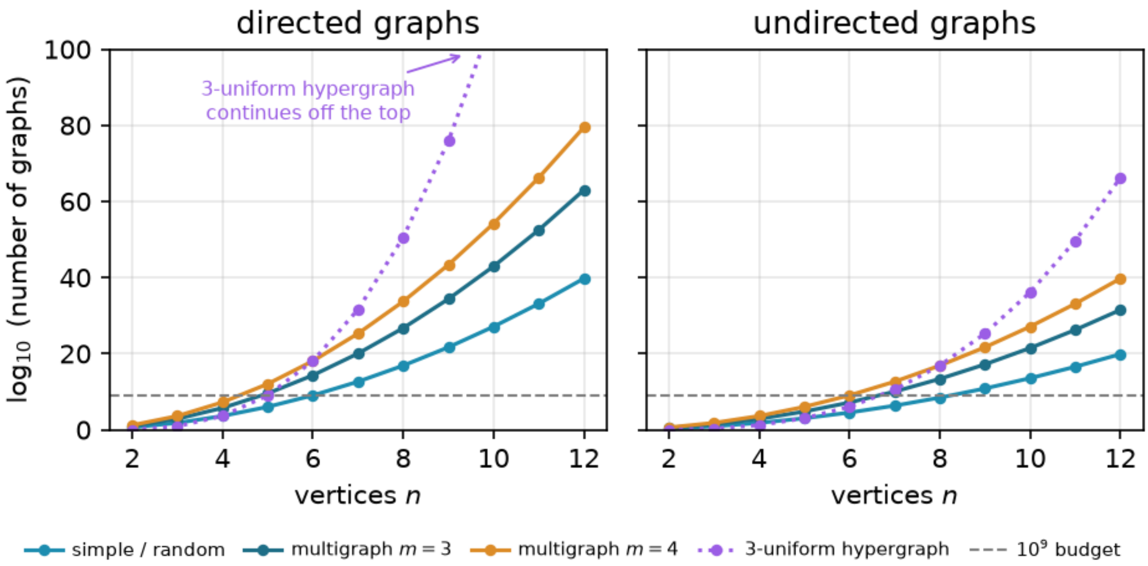


Figure 2.4. Candidate counts for blind enumeration on a logarithmic scale. Every model crosses the generous 10^9 budget in single-digit order, and directed objects grow fastest because they have more independent entries to decide.

and vertex disjointness search the same graphs and differ only in output.

The number of graphs the checker must visit grows as $2^{\binom{n}{2}}$ for simple undirected graphs and $2^{n(n-1)}$ for digraphs, and capping multiplicities at $m-1$, so that each run ranges over the m values $\{0,\ldots,m-1\}$, raises the base from two to m, giving $m^{\binom{n}{2}}$ undirected multigraphs and $m^{n(n-1)}$ directed ones. The enumeration is exact wherever it finishes and it does so only for the smallest sizes. Figure 2.4 measures exactly how fast the search space grows.

2.3.1 Pruned search

The enumeration step will skip whole families of hopeless graphs, via a pruned search. The base cases $n \leq 7$ of the directed $m = 2$ theorem (Theorem 1.6) are the finite checks the induction of Appendix A depends on, and they sit just past where blind enumeration of digraphs is tractable. They can be reached by enumerating more cleverly. Every variant offers a fixed list of edges its graphs are allowed to use, whether those are pairs, ordered pairs, or oriented sets of r vertices, and a graph is a choice of how many copies each of them gets. Rather than writing out whole choices one after another, fix an order on that list and decide the edges one at a time, leaving the undecided ones absent. Each partly built graph then stands for the whole family of graphs agreeing with it so far, and a pruning rule discards that entire family if it is deemed to bear no fruit.

Adding an edge never lowers a connectivity, since a family of disjoint routes survives intact when more edges are laid around it (Proposition A.4). Every completion of a partly built graph only ever adds, so the partly built graph is a floor under all of them. The moment it breaks the cap, so does every completion, and the branch is abandoned.

A second pruning method takes the edges already placed and adds the most the undecided

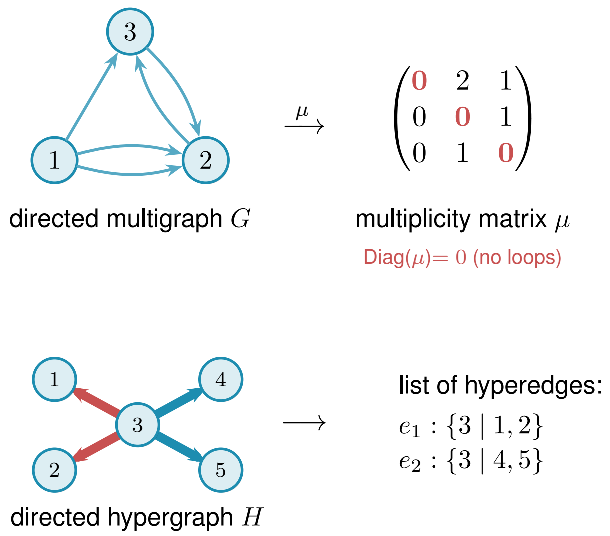


Figure 2.5. The two encodings, shown for the directed variants. The digraph is represented by the multiplicity matrix μ, where μ_{ij} counts arcs from i to j. The directed hypergraph is a list of hyperedges instead, where the tail of each is isolated. Multihypergraphs repeat at least one entry in the list of hyperedges. Since a matrix (or tensor) representation would need n^r entries and those would be almost all zero, it is avoided for efficiency.

ones could ever contribute, which is how many are left times the largest multiplicity allowed. That sum is an upper bound on every completion, so if even it cannot beat the best feasible graph found so far, the branch is dropped unexplored (Figure 2.6). Only infeasible graphs and graphs that cannot beat the incumbent are ever discarded, so a run that finishes returns exactly the maximum the basic walk would have returned.

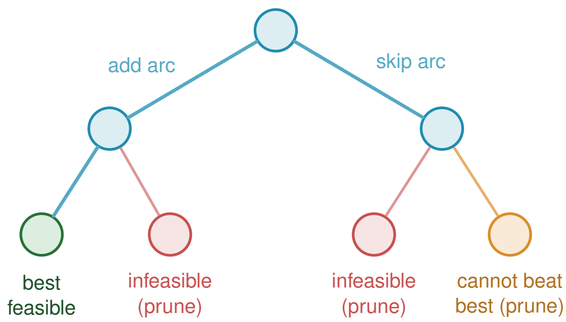


Figure 2.6. Each level of the partially built graph fixes one adjacency. Infeasible partial graphs (red) and branches unable to beat the incumbent (orange) are discarded.

2.3.2 Generating search

Iterating over all possible graphs and measuring each one's connectivity repeats many isomorphic cases. Another method, suggested by Prof. Jan Goedgebeur, instead generates all non-isomorphic graphs. It uses the `geng` program from the `nauty` toolkit of McKay and Piperno [14] to perform the difficult task of generating one representative of each isomorphism class. The program is called as a subprocess, and its `graph6` output is decoded. For a given n, `geng` produces all non-isomorphic *undirected* graphs. The program then treats each such graph as a support graph and assigns directed arc multiplicities to its edges. Consequently, the search automatically inherits the support graph's isomorphism reduction. For each support edge $\{u, v\}$, considered in a fixed order, it assigns the ordered pair of multiplicities $\mu(u, v)$ and $\mu(v, u)$. Each lies between 0 and $m - 1$, and the two cannot both be zero because every support edge must represent at least one arc. After each assignment, the checker evaluates the partially decorated multigraph. If any pair already exceeds the allowed connectivity, that branch is stopped immediately: adding more arcs cannot decrease connectivity. This is the same monotonicity-based pruning used in Figure 2.6. Decorations that reach the target arc count are retained and converted to canonical form, which ensures that each isomorphism class is reported only once. Because different support graphs produce disjoint search spaces, they can be processed independently on separate processor cores. The related orientation tools `directg` and `watercluster2` are not used. They assign only one orientation to each edge, so they cannot represent the bidirected edges or the repeated arc multiplicities that occur in the extremal graphs.

Table 2.1 times three ways of proving $\ell_2^{\mathrm{dir}}(n)$ for a simple digraph on a single core: a blind sweep over every matrix assignment with no pruning at all, the depth-first pruned search of Figure 2.6, and the geng-generated pipeline above. The blind sweep is left out past $n = 4$, where $2^{n(n-1)}$ assignments already cost 6.3 s. Each further vertex multiplies the assignment count by 2^{2n}, so $n = 5$ would run for about an hour and $n = 6$ for weeks, the same growth Figure 2.4 plots. Below $n = 5$ both remaining methods are dominated by fixed overhead, Python function-call overhead or subprocess start-up overhead, rather than by search cost, and neither number there is a meaningful comparison. From $n = 5$ the gap widens with every vertex: generation is 5.5 times faster at $n = 5$, 20 times faster at $n = 6$ and 68 times faster at $n = 7$, where the pruned search takes 1156.6 s against the generated pipeline's 17.0 s. All three methods return the same proved value at every n where they finish.

n	$\ell_2^{\mathrm{dir}}(n)$	blind (s)	pruned (s)	generated (s)
3	4	0.05	0.003	0.009
4	6	6.3	0.008	0.008
5	8	–	0.22	0.039
6	10	–	14.5	0.73
7	12	–	1156.6	17.0

Table 2.1. Wall-clock time to prove $\ell_2^{\mathrm{dir}}(n)$ by three methods, single core. The blind sweep has no pruning. The pruned search is Figure 2.6's depth-first branch and bound. The generated pipeline is the geng-seeded search above. A dash marks a run not attempted at that size.

2.4 Proving every m at once

The search so far looks at one pair (n, m) at a time, but for the directed multigraph a faster method can be used: it rescales the multiplicities by a factor of $m - 1$ and then checks the relevant thresholds for a fixed n in one pass. The general theorem of Theorem 1.13 has now replaced this route in `solve`, but the older formulation is still a useful independent finite check.

Divide each multiplicity of a feasible directed multigraph D by $m - 1$. This produces weights $w(u, v) \in [0, 1]$ for which every pairwise maximum flow is at most one and whose total weight equals the arc count of D divided by $m-1$, as Figure 2.7 illustrates. Thus, with

$$M^*(n) = \max\Big\{\textstyle\sum_{u\neq v} w(u,v) : w(u,v)\in[0,1],\ \mathrm{maxflow}(s,t;w)\le 1 \text{ for all } s\neq t\Big\}$$

for the largest total weight of such a matrix, scaling gives

$$L_m^{\mathrm{dir}}(n) \ \le\ (m-1)\,M^*(n) \qquad \text{for every } m\ge 2.$$

For the reverse inequality, an integral matrix is needed whose weight equals $M^*(n)$, since a $\{0,1\}$ matrix survives multiplication by $m-1$ as integer arc multiplicities untouched. Construction A.36's spanning tree taken to be a star already supplies one on the linear branch: the *double star*, one hub bidirected to every other vertex, $2(n-1)$ arcs at multiplicity one. For $n\le 6$ the linear branch $2(n-1)$ is the larger term of $M(n)$, so the double star already realises $M^*(n)$ there, and the historical optima for those n take exactly this form. Past the crossover at $n=7$ (Theorem 1.6) the quadratic branch $\lfloor n^2/4\rfloor$ catches up and then overtakes it, and the maximiser becomes Construction A.36's other half, the bipartite digraph, instead: no single shape witnesses $M^*(n)$ at every n. What generalises is not one construction but the fact that *some* $\{0,1\}$ matrix always attains $M^*(n)$, which Corollary A.41 proves for every n by a different route: an explicit $\{0,1\}$ construction gives attainment, and clearing denominators from a rational optimum rules out any fractional matrix doing better, rather than exhibiting a shape for the optimum itself. Consequently, for every n and every $m\ge 2$, the maximum number of arcs in a directed multigraph with pairwise maximum flow at most $m-1$ is exactly $(m-1)M^*(n)$: scaling that integral maximiser by $m-1$ gives the lower bound, and the preceding scaling argument gives the upper bound. The finite optimisation is retained only as an independent check of the cases $n\le 6$.

$$L_m^{\mathrm{dir}}(n) = (m-1)\,M^*(n) \qquad \text{for every } m\ge 2.$$

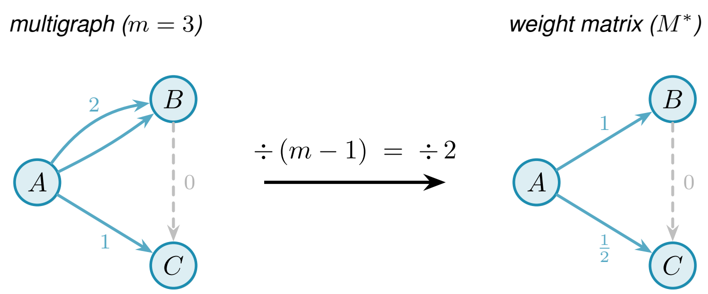


Figure 2.7. Scaling at $m=3$: dividing multiplicities by $m-1=2$ turns the integer multigraph into a weight matrix with entries in $[0,1]$ and pairwise maximum flow at most one. Hence one m-free optimisation bounds every threshold.

The model itself is a mixed-integer program with one chosen cut per ordered pair, and it is written out, with the theorem it settled and the record of what was run, in Section A.6.4.

2.5 The heuristic search

So far, iterating all graphs of a fixed size that meet the connectivity constraint and counting their edges was a blind walk. Apart from pruning, there is nothing that steers the walk towards better candidates because they all need to be checked anyway. From a certain point onwards, that will become impossible due to the massive number of possible graphs. The goal then becomes finding the densest possible graphs in such an efficient and guided way that they are likely to be (almost) extremal. The guide that pulls the walk towards a (local) maximum is called a heuristic. To avoid getting stuck in a local maximum, a metaheuristic allows worsening moves to discover new optima.

To implement the metaheuristic, give each graph one number: its energy

$$\Phi(G) = -|E(G)| + \Lambda \cdot \max\bigl(0,\ c^{\max}(G) - (m-1)\bigr),$$

which rewards edges (arcs, hyperedges) and penalises any excess connectivity, where $c^{\max}(G)$ denotes $\lambda^{\max}(G)$ or $\kappa^{\max}(G)$, matching whichever separation the run measures. The search starts from the empty graph and moves by toggling one adjacency: an edge, an arc, or a hyperedge, depending on the model. The method used is *tabu search* [15], which escapes a local optimum by forbidding the pairs it has just touched. Among the moves not forbidden, the walk always takes the one that lowers Φ most, even when every candidate raises it. This is unlike a greedy walk that only ever accepts an improving move. After each move the pair it changed is put on a tabu list for a fixed number of steps, so the walk cannot immediately undo what it just did and cycle back into the optimum it came from. A move that would beat the best graph seen so far overrides the tabu flag regardless of its status, an *aspiration criterion* that keeps the tabu list from ever blocking genuine progress, and the walk perturbs itself when it stalls.

Simulated annealing [16], which accepts a worsening move with the probability given by the Metropolis rule [17] and lowers that probability as the run cools, is an alternative on the same energy and the same neighbourhood, but at equal wall-clock budget it is the weaker engine here. In the equal-budget comparisons reported here, tabu reaches denser candidates: at $n = 7$, $m = 3$ it reaches the 24-arc directed multigraph extremiser against annealing's 20, and the 18-arc simple digraph extremiser against annealing's 16. Every discovery value this thesis reports therefore comes from tabu search. Annealing is kept only as an independent cross-check in the program's self-test, where agreement between two unrelated walks is worth more than the speed of either.

It is fair to ask why the walk is allowed above the cap at all. Reaching every feasible graph does not require it. Removing an edge can only lower $\lambda^{\max}$ and $\kappa^{\max}$, never raise them, so every subgraph of a feasible graph is again feasible, and the empty graph is feasible trivially. From any feasible target we can therefore delete edges one at a time down to the empty graph without ever crossing the cap, and reading that path backward builds the target up the same way. The feasible region is connected through the empty graph by single-edge moves that stay inside it, so a walk that never went infeasible could in principle visit every feasible graph there is. We let it go infeasible anyway, because reachability and efficiency are different things. A feasible graph can be locally maximal, meaning every edge we might add creates a forbidden pair, and yet still sit far below the densest feasible graph. To improve such a graph the walk has to add an edge and then remove a different one, a swap whose first half is infeasible. A walk confined to feasible graphs could only find that better graph by retreating most of the way back to the empty graph and climbing a different slope, a detour far longer than the budget allows. Allowing a brief excursion above the cap lets the walk trade one load-bearing edge for a better one in two moves rather than dozens. Going above the bound is never the goal. It is the shortcut the walk takes on its way back down to it. A single walk depends on where it started, so either engine is run several times, each time from a different seed, until the wall-clock budget runs out, and the densest witness of all the runs is kept. Independent restarts share nothing, so they can also be spread across processor cores.

2.5.1 Measuring only on the possible bounds

Removing an edge, arc, or hyperedge can never raise $\lambda^{\max}$, while adding one raises it by at most one. The same holds for $\kappa^{\max}$. This relies on the monotonicity and unit-step bounds of Proposition A.4. Therefore the max-flow algorithm only needs to be run every once in a while, depending on how close to the constraint the graph sits. A removal applied to a feasible graph leaves it feasible, and an addition made while the graph sits strictly below the cap can lift $\lambda^{\max}$ by at most one, so in both cases the energy is known without measuring anything. Only an addition made at the cap itself needs the checker. The search therefore carries a running upper bound on $\lambda^{\max}$ and resynchronises it exactly only when that bound is what a step turns on. Section A.10 gives the bookkeeping, and the regression test that holds the fast walk to the naive one step for step.

2.6 Rediscovering the known extremisers

Before the search is pointed at the open cases, it is run on the settled ones. Run from the empty graph on the cases settled in Chapter 1 or by the exact methods above, and told only to pack in edges without breaching the connectivity cap, the tabu search arrives unprompted at the named constructions and matches the established value in every tested case. Table 2.2 reports the densest graph it found against the value the theory predicts.

The graphs the search finds are the named constructions themselves, not merely graphs of the right size. Figure 2.8 draws three of them. The undirected simple run at $m = 2$ settles on a spanning tree, the only shape with $n - 1$ edges and no cycle. The undirected multigraph run settles on a star with every spoke doubled to multiplicity $m - 1$, an instance of the multi-tree extremiser of Theorem 1.3. The directed multigraph run settles on the same star, bidirected and doubled to the same multiplicity, the thickened tree of Construction A.36. The simple directed run finds the bidirected star at $n = 4$ and, at $n = 7$, the tying value of twelve arcs at which the hub and the one-directional wall meet (Figure 1.7), the crossover Theorem 1.6 predicts. The search carries none of these names. It lands on them because, under the connectivity cap, there is

Construction	Variant	n	m	Search	Theory
spanning tree (simple undirected)	$\ell_m(n)$	6	2	5	5
multiplicity-$(m-1)$ star (multi undirected)	$L_m(n)$	6	3	10	10
bidirected star (simple directed)	$\ell_m^{\mathrm{dir}}(n)$	4	2	6	6
bidirected star (simple directed)	$\ell_m^{\mathrm{dir}}(n)$	6	2	10	10
hub–bipartite tie (simple directed)	$\ell_m^{\mathrm{dir}}(n)$	7	2	12	12
double star (multi directed)	$L_m^{\mathrm{dir}}(n)$	4	3	12	12
double star (multi directed)	$L_m^{\mathrm{dir}}(n)$	5	3	16	16

Table 2.2. Validation by rediscovery. For each variant the tabu search, run from scratch with a handful of restarts, recovers exactly the extremal value proved or certified earlier. The "Search" column is the lower bound it found, and the "Theory" column is the proved or certified value.

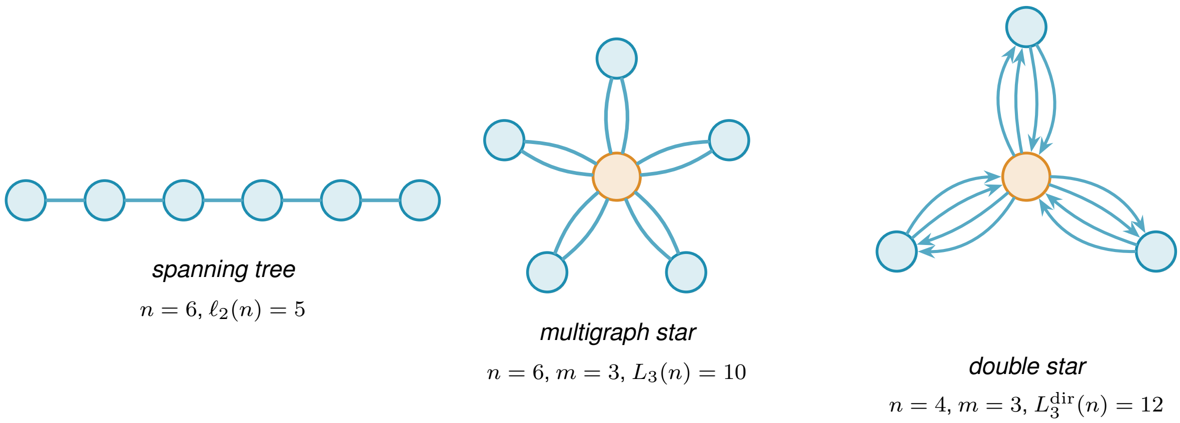


Figure 2.8. Three of the shapes Table 2.2 names, each drawn at the size the table reports. Figure 1.7 draws the fourth, the bidirected star tying the one-directional wall at $n = 7$.

nowhere denser to go.

Past $n = 7$ the search reaches its own limit before the checker does. Repeated tabu runs at $n = 8$, $m = 2$ stall at the hub's arc count, because dismantling a finished hub and rebuilding the one-directional wall (Figure 1.7) is far more than the single-arc moves of the walk can tunnel through, and the true value there comes from the construction and the checker rather than from the search. The same double star of Figure 2.8, at $n = 7$ and $m = 4$, has 36 arcs at $\lambda^{\max} = 3$, matching $2(n-1)(m-1)$, a further witness past where the search itself reaches. Neither is a proof of optimality. Each is an exactly measured feasible graph, and therefore a lower bound, the status every witness returns.

Chapter 3

Synthesis, Results, and Open Problems

This thesis began by proving certain variants using logical arguments, confirmed those results for small graphs and then searched bigger graphs for dense examples. This chapter collects the output of those programs, interprets it and provides conclusions for all variants.

3.1 The directed frontier

The directed arc problem is a race between a hub with $m(n-1)$ arcs and an augmented bipartite construction with $\lfloor (n+m-2)^2/4 \rfloor$ arcs (Constructions A.15 and A.17). They cross near $n = 9$ at $m = 3$.

Conjecture 3.1. *For simple digraphs and all $n \geq m \geq 2$,*

$$\ell_m^{\mathrm{dir}}(n) = \max\left(m(n-1),\ \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor\right).$$

At $m = 2$ this is Theorem 1.6, and the first branch is the directed hub of Construction A.15. Figure 3.1 plots the two branches at $m = 3$ and the size at which the second overtakes the first.

Exhaustive search proves the values $6, 9, 12, 15$ for $n = 3, 4, 5, 6$ when $m = 3$, for both separation conditions (Remark A.23), and does not settle $n = 7$. Past the sizes exhaustion reaches, a general upper bound would follow from the structural claim of Conjecture A.28: that every extremal example other than the hub splits into a part A and a part B with every arc running from A to B and none returning. One returning arc is enough to break the count, since it lets a route leave B through A and exceed the internal budget of $m-2$. Ruling that arc out for every extremal example is the step that is missing.

The conjecture is therefore correct in its leading term and in the size of its error. This is proved in Theorem 1.11, without any assumptions about the shape of the extremal graph. The only unknown is the coefficient of the linear error term: it lies somewhere between the conjectured $(m-2)/2$ and the proved $4(m-1)$. The discussion below considers what the bound can and cannot tell us about the *shape* of an extremal graph. A weaker form of the same argument answers that separate question.

The directed vertex problem also has the same leading term and error order, by Theorem 1.12. Appendix A treats it separately from the arc problem. It remains open whether the two values are

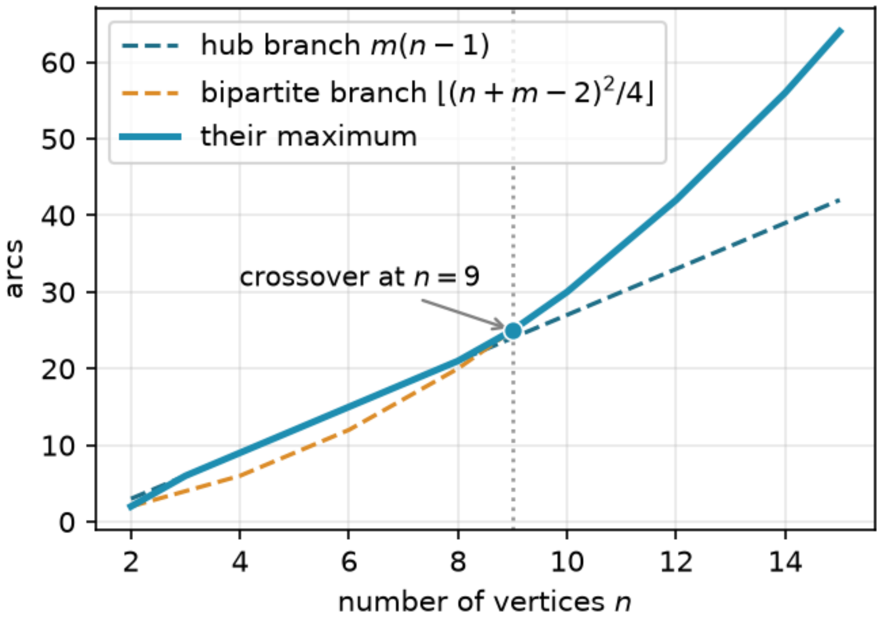


Figure 3.1. The two lower-bound branches of Conjecture 3.1 at $m = 3$, and the maximum they define.

exactly equal for $m \geq 3$.

If parallel arcs are allowed, the problem is completely solved: Theorem 1.13 gives the exact value for every $n \geq 2$ and $m \geq 2$.

3.2 Why direction makes the value quadratic

In an undirected graph every edge contributes to the degree of both its endpoints, and Mader's theorem turns this symmetry into a hard cap of $\lfloor m(n-1)/2 \rfloor$ edges, linear in n. A digraph does not have this symmetry. In Construction A.16 all arcs go from one part of the vertices to the other and none come back, so a route can cross from the first part to the second only once. That construction holds $\lfloor n^2/4 \rfloor$ arcs at $\lambda^{\max} = 1$, and no undirected graph can carry so many edges at connectivity one once n is large. The linear cap therefore does not survive the change of model: already at $m = 2$ and $n = 8$ the one-directional construction holds 16 arcs against the hub's 14, so the directed value follows the larger of the two branches rather than the linear one alone.

Construction A.17 carries the same idea to every m by letting each vertex of the larger part receive $m-2$ extra in-arcs from within that part, drawn in Figure 3.2. These arcs provide $m-2$ short detours for each pair and raise $\lambda^{\max}$ to $m-1$, while the count increases to $\lfloor (n+m-2)^2/4 \rfloor$. Its $m = 3$, $n = 10$ instance holds thirty arcs where the hub branch gives twenty-seven.

The quadratic term is what the directed upper bound has to account for. The argument must explain why no configuration has more arcs than the bipartite construction. This requires finding the partition, ruling out arcs in the reverse direction, and controlling the internal structure of the larger part B. That structural task is what Conjecture A.28 has not yet resolved for $m \geq 3$. Proposition A.29 and Theorem 1.11 show that this bipartite structure is the right one, using two

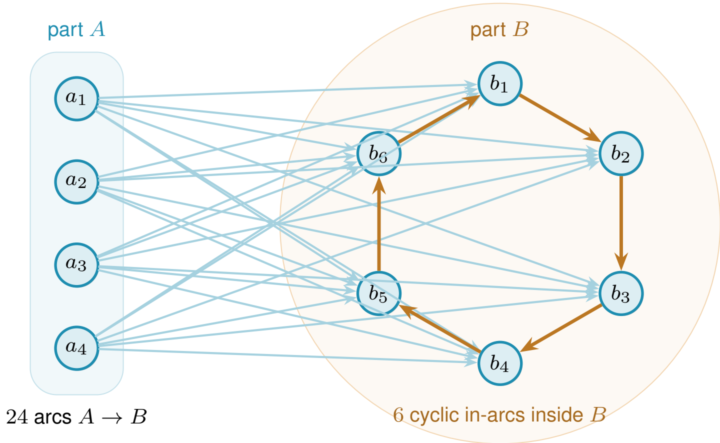


Figure 3.2. The augmented bipartite digraph of Construction A.17 at $m = 3$, $n = 10$. Every cross-part arc runs from A to B, and inside B each vertex takes one arc from its cyclic predecessor, which is $m - 2$ arcs per vertex. A pair a_i, b_j then has the direct arc and one detour, so $\lambda^{\max} = 2$, and the count is $24 + 6 = 30$ against the hub branch's 27.

different measures. In *size*, the extremal value lies within $O_m(n)$ of the wall's count, so what is left open is a constant inside the second-order term rather than the phenomenon. In *shape*, every feasible digraph becomes a subgraph of a one-directional bipartite graph after deleting $O(\sqrt{m}\, n^{3/2})$ arcs, a weaker error term because it comes from the count that does not induct on a low-degree vertex. The undirected problem has no analogous obstacle, because Mader's degree-counting argument closes it in a few lines. The quadratic phenomenon is therefore the reason the directed and undirected problems differ in difficulty.

3.3 Summary of the sixteen variants

Figure 3.3 sorts all sixteen structural variants by how much is known, and Table 3.1 states the strongest result for each. Each entry is either proved here, supported by a result from the literature, conjectured with a proved lower bound, or proved asymptotically. Every value rests on a proof or on a completed exhaustion, never on a machine certificate alone, and the markings follow the standard set in Chapter 2.

Here $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$. Simple-graph formulas assume $n \geq m$. Below that range the complete object gives the trivial maximum. The $m = 5$ vertex formula is the one exception, needing $n \geq 6$ rather than $n \geq m$: at $n = 5$ it reads 9 where K_5 is already feasible at $\kappa^{\max} = 4$ and gives $k_5(5) = 10$. The edge-hypergraph bound is attained when $(r-1) \mid (n-1)$ for multihypergraphs, and for simple hypergraphs when $r \geq 3$ and $m - 1 \leq \binom{n-2}{r-2}$. The $m = 3$ vertex-hypergraph bound is attained by a simple hypergraph when $2 < r < n$, and by a multihypergraph whenever $(r-1) \mid (n-1)$ (Proposition A.57). Full statements and exceptions are in Theorems 1.2, 1.4, 1.5, 1.9 to 1.14 and A.52 and proposition 1.8.

3.3.1 Values at $m = 3$ and $m = 6$

Table 3.1 and Figure 3.3 give each variant a category. The four grids and the table that follow give the same sixteen variants as numbers, one plotted point or table cell per (n, m), for a reader

Model	Edge or arc-disjoint	Vertex-disjoint
Undirected		
simple graph	$\lfloor m(n-1)/2 \rfloor$, proved	the edge value for $m \leq 4$, and at $m = 5$ the edge value up to $n = 13$ then $\lfloor 8n/3 \rfloor - 4$ from $n = 14$, while it is open for $m \geq 6$
multigraph	$(m-1)(n-1)$, proved	the simple value
r-hypergraph	$\leq \lfloor (m-1)(n-1)/(r-1) \rfloor$, with equality in the range below	$\lfloor (n-1)/(r-1) \rfloor$ at $m = 2$ and $\leq \lfloor 2(n-1)/(r-1) \rfloor$ at $m = 3$, while it is open for $m \geq 4$
r-multi-hyper	the same bound, with equality when $(r-1) \mid (n-1)$	the same bounds, with repeated hyperedges settling cases missed by the simple model at $m = 3$
Directed		
simple digraph	$M(n)$ at $m = 2$ (proved), and $n^2/4 + \Theta_m(n)$ for $m \geq 3$ with the exact value conjectured	$M(n)$ at $m = 2$ (proved), and $n^2/4 + \Theta_m(n)$ for $m \geq 3$ with equality to the arc value open
multigraph	$(m-1)M(n)$, proved	the simple digraph value
r-hypergraph	$(1 + o(1))(m-1)n^2/[4(r-1)]$ for fixed r, m, leading term proved	the same leading term for fixed r, m
r-multi-hyper	the same leading term, the bound proved with repeats allowed	the same leading term for fixed r, m

Table 3.1. The sixteen variants and the strongest result known for each. The two multigraph vertex rows read “the simple value” because parallel copies do not raise κ, under the counting convention of Section 1.3. Ranges of validity, including the one at $m = 5$, are stated below the table.

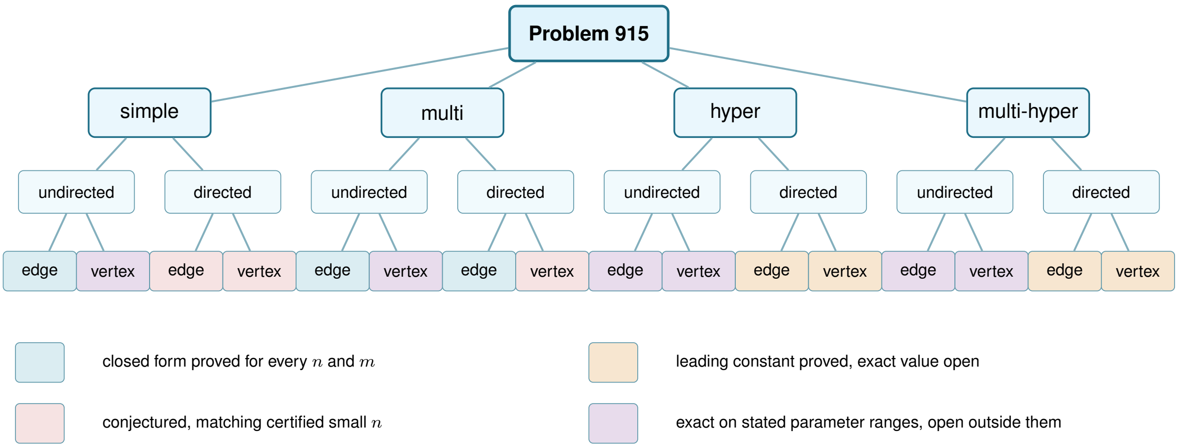


Figure 3.3. Status of the sixteen variants. Colour marks the status of a leaf: blue a closed form proved for every n and m, purple an exact value proved on stated parameter ranges and open outside them, red a conjectured exact formula with a proved lower bound, and amber a proved leading term with the exact value unknown. The model and direction rows are labels and carry no status, since each of them now covers two leaves that differ. Table 3.1 is authoritative for exceptions and conventions.

who wants a value instead.

Proposition 1.8 bounds every simple hypergraph from above. It does not claim the bound is attained outside the ranges stated there. The $m = 6$, $n = 6$, $r = 3$ cell in Figure 3.7 is where non-attainment is present: exhaustion over all $125\,970$ twelve-hyperedge simple 3-uniform hypergraphs on six vertices finds every one infeasible, so the bound of 12 is never reached and the true maximum is 11. The smallest instance of the same gap is $n = r = m = 3$, where a simple hypergraph carries one hyperedge against the bound's prediction of two, while a multihypergraph attains it.

Table 3.2 gives the same sixteen variants at both m as numbers, colour-matched to the four grids above.

3.3.2 Orientations of a directed hyperedge

The directed hypergraph rows of Table 3.1 rest on one bound, Theorem 1.14, which pins the leading term. The table uses the *forward* convention throughout: a hyperedge has one tail and $r-1$ heads. Reversing every hyperedge makes the one-head *backward* convention equivalent to it (Lemma A.61), and Theorem A.64 extends the same leading term to every other way of splitting a hyperedge into tail and head sets, under both separations. The orientation therefore adds no asymptotic row to Table 3.1. The conventions may still differ for small numbers of vertices, and whether they agree exactly once $n > r$ is open.

3.4 Contributions and limitations of the program

The program replaced separate methods for each case with one shared approach. Before it, each variant required its own computation and hand proof. Now the same interface represents every variant in two ways: with a multiplicity matrix for graph models and with vertex sets for hypergraph

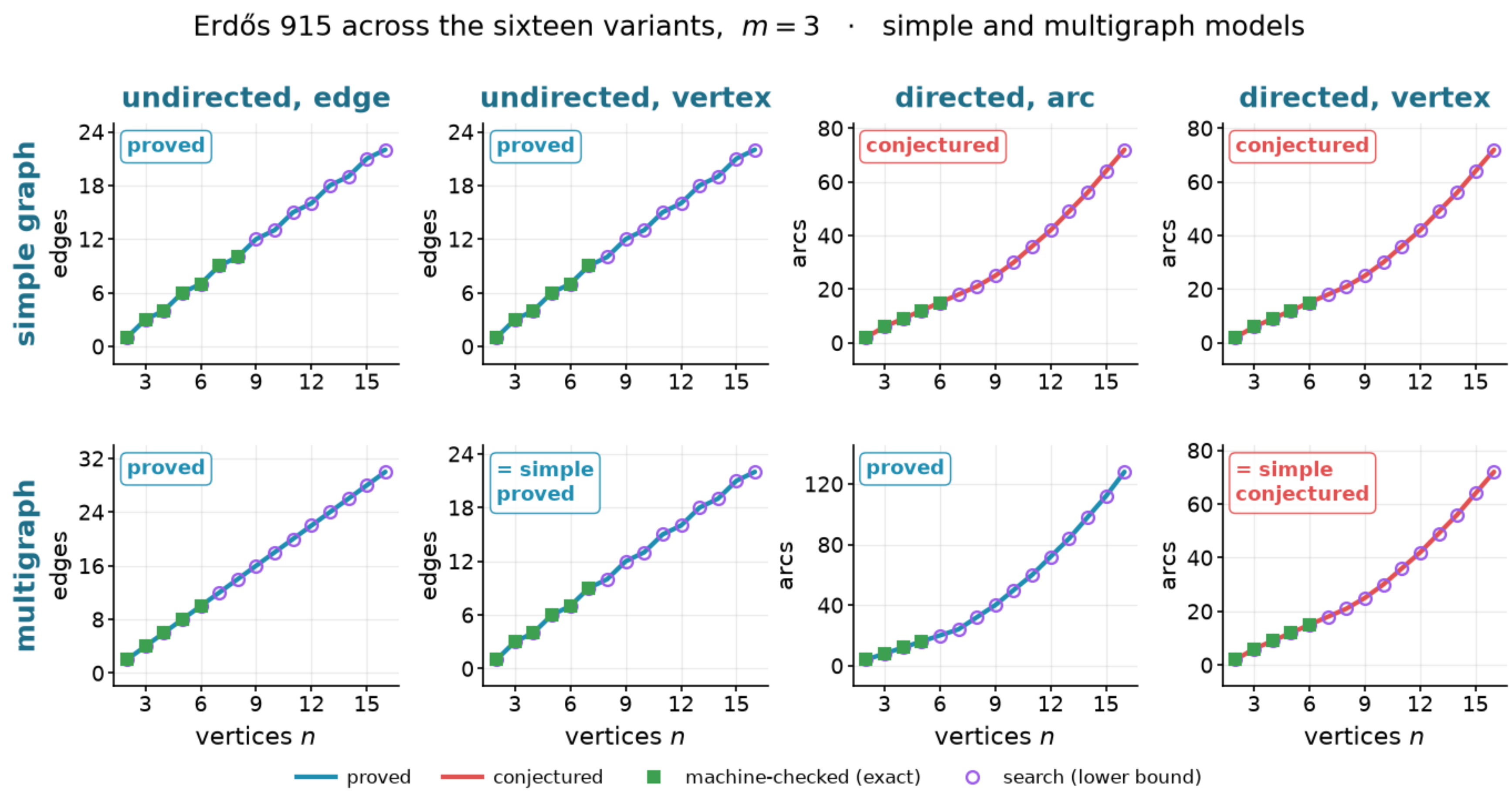


Figure 3.4. The eight graph-model variants at $m = 3$: simple and multigraph rows, undirected and directed columns. A blue curve is a theorem, a red one a conjecture and a yellow one an interpolation, filled squares are exact finite values, and open circles are witnesses. Every panel is drawn from that one vocabulary. Where a conjecture is a maximum of two counts, as Conjecture 3.1 is, the curve plotted is the maximum itself. Figure 3.5 continues with the hypergraph rows at the same m, and Figures 3.6 and 3.7 redraw both halves at $m = 6$.

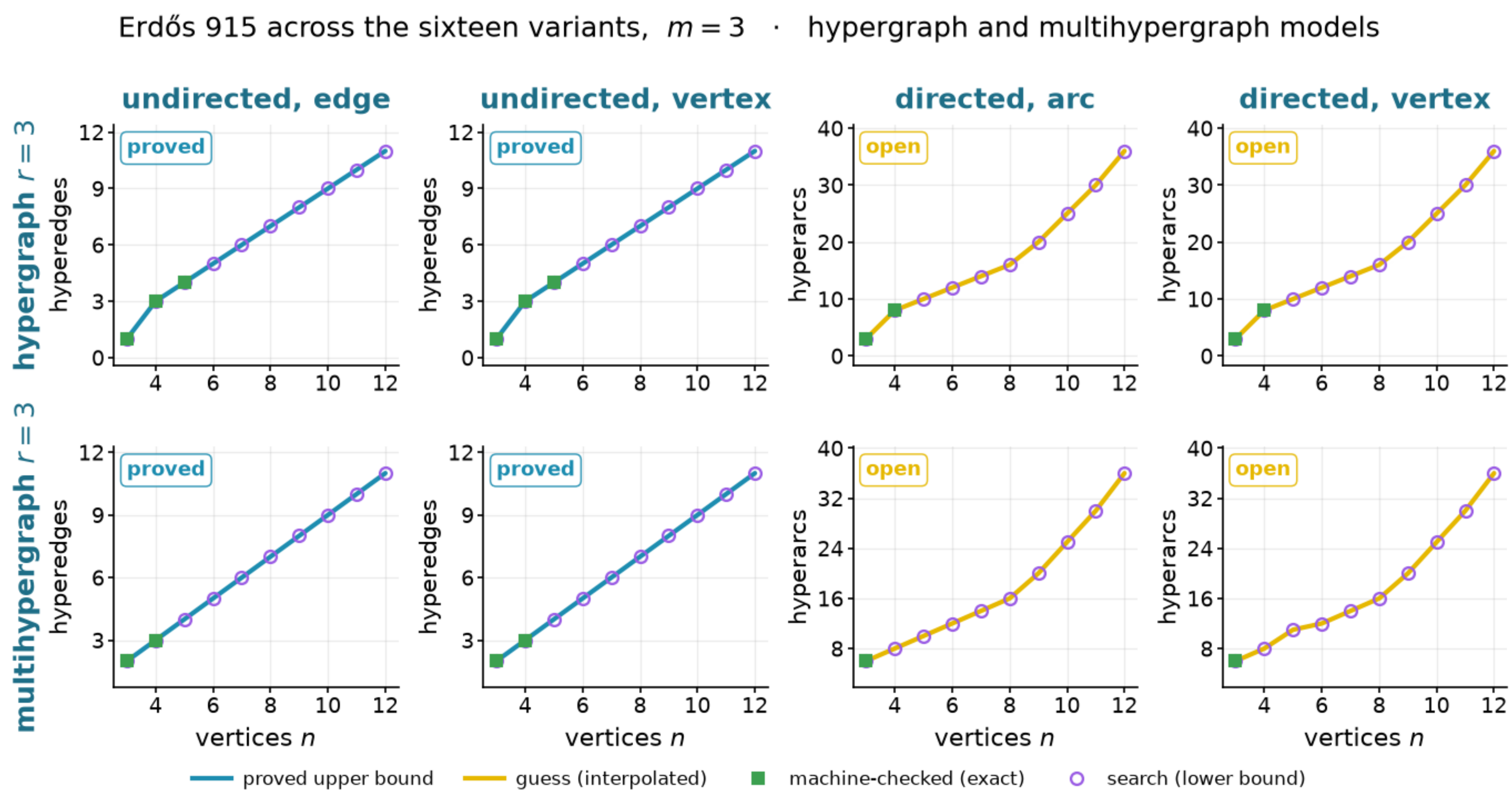


Figure 3.5. The eight hypergraph-model variants at $m = 3$, continuing Figure 3.4: hypergraph and multihypergraph rows, undirected and directed columns, with the same colour and marker key. The blue curve here is Proposition 1.8, a proved upper bound rather than a proved value, which is why the key names it as one. It is attained only inside the ranges stated there, so a square or a circle below it marks a gap that is open rather than an error.

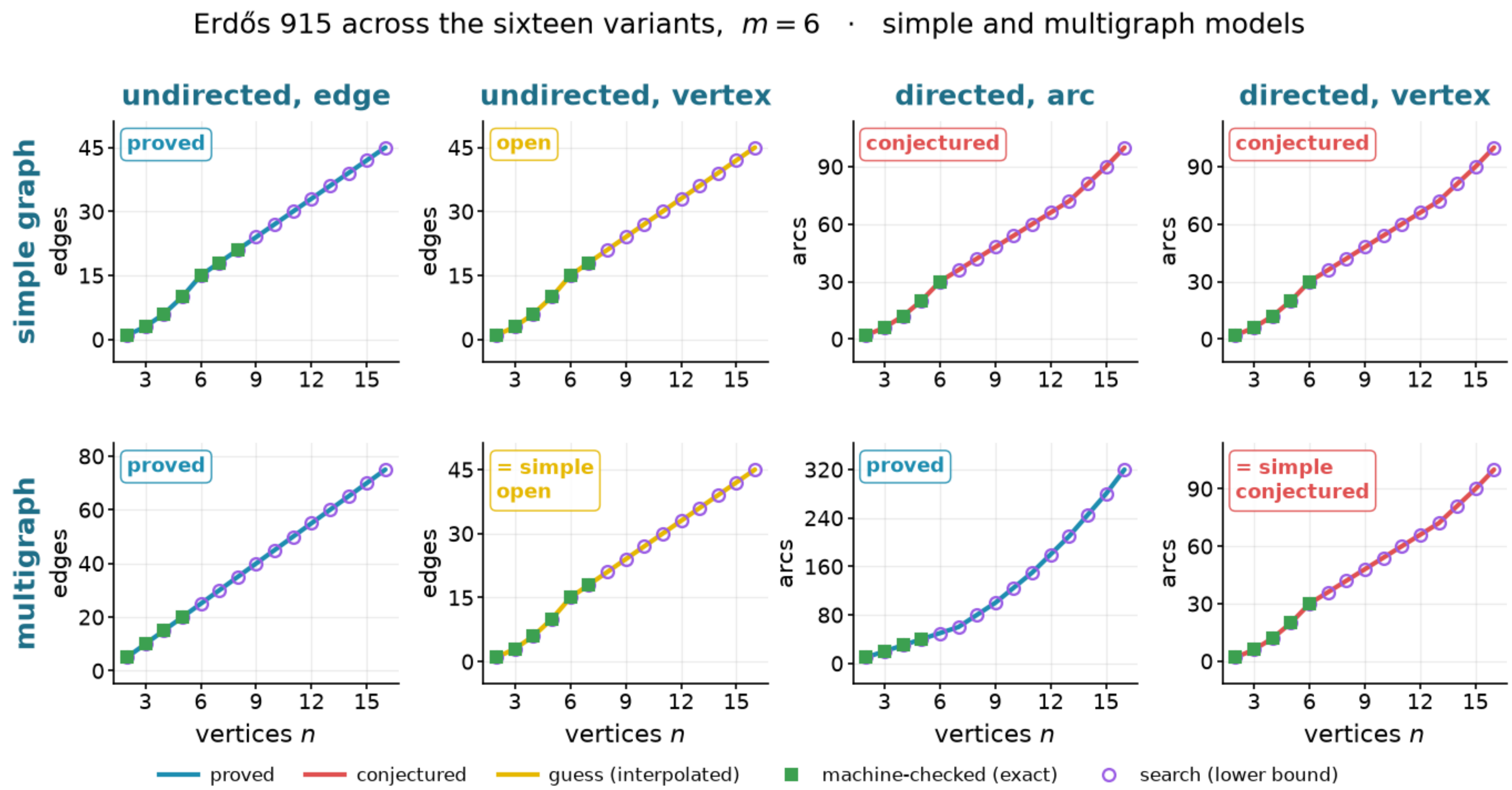


Figure 3.6. The same eight graph-model variants at $m = 6$, to be read against Figure 3.4. Raising m widens every feasible band, and the exact squares thin out as the enumeration range falls behind the constructions.

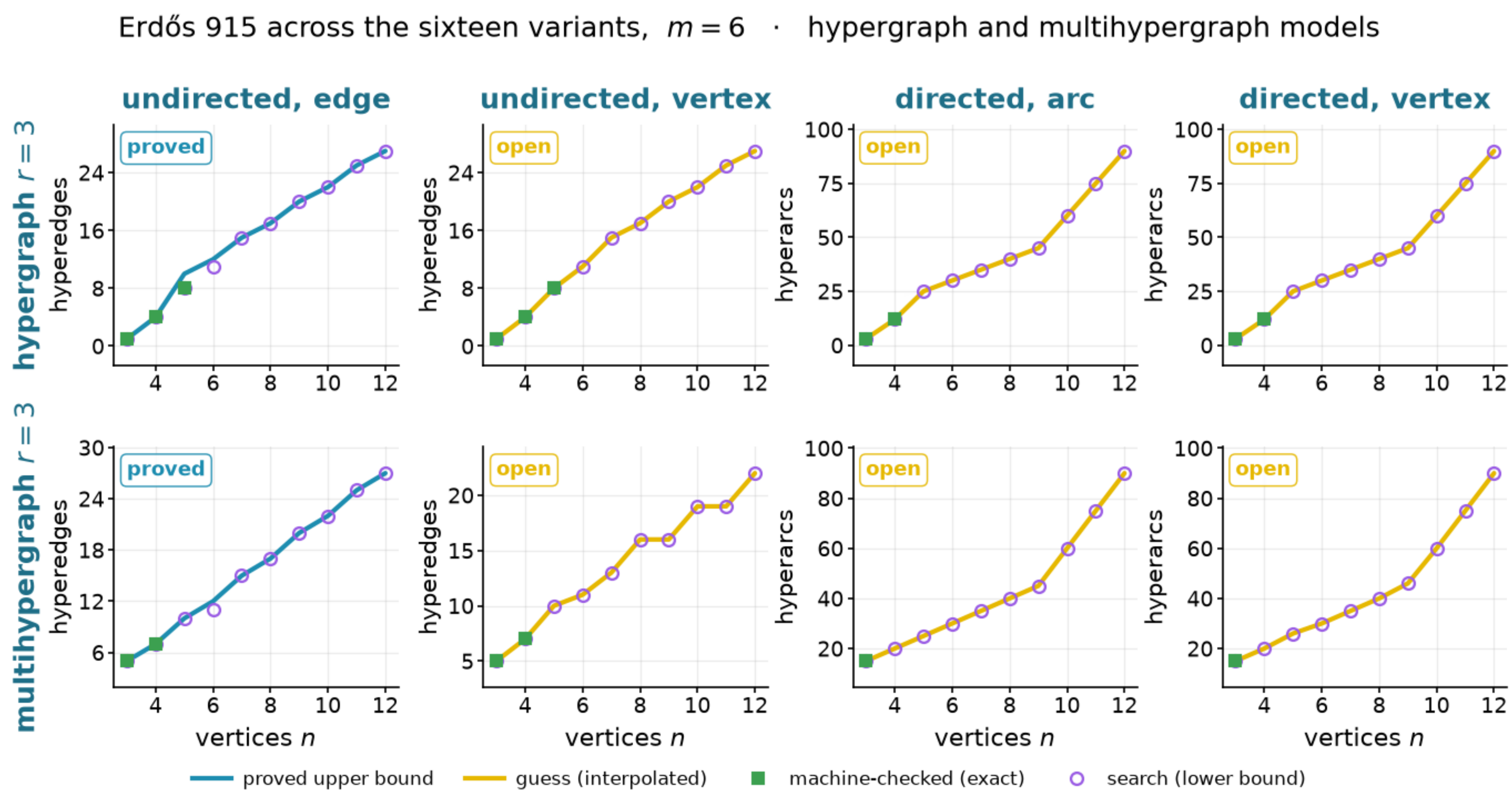


Figure 3.7. The same eight hypergraph-model variants at $m = 6$, continuing Figure 3.6 and to be read against Figure 3.5. The blue curve is again the upper bound of Proposition 1.8, and at this m it runs visibly above what has been reached: the exact square at $n = 5$ is 8 against the curve's 10, and at $n = 6$ the curve reads 12 where the true maximum is 11.

Model	Variant	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$
$m=3$								
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	4	6	7	9	10
	undirected, vertex ($k_m(n)$), proved	1	3	4	6	7	9	10
	directed, arc ($\ell_m^{\mathrm{dir}}(n)$), conjectured	2	6	9	12	15	≥18	≥21
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), conjectured	2	6	9	12	15	≥18	≥21
multigraph	undirected, edge ($L_m(n)$), proved	2	4	6	8	10	12	14
	undirected, vertex ($k_m(n)$), = simple, proved	1	3	4	6	7	9	10
	directed, arc ($L_m^{\mathrm{dir}}(n)$), proved	4	8	12	16	20	24	32
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), = simple, conjectured	2	6	9	12	15	≥18	≥21
hypergraph $r=3$	undirected, edge, proved		1	3	4	5	6	7
	undirected, vertex, proved		1	3	4	5	6	7
	directed, arc, open		3	8	≥10	≥12	≥14	≥16
	directed, vertex, open		3	8	≥10	≥12	≥14	≥16
multihypergraph $r=3$	undirected, edge, proved		2	3	4	5	6	7
	undirected, vertex, proved		2	3	4	5	6	7
	directed, arc, open		6	≥8	≥10	≥12	≥14	≥16
	directed, vertex, open		6	≥8	≥11	≥12	≥14	≥16
$m=6$								
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	6	10	15	18	21
	undirected, vertex ($k_m(n)$), open	1	3	6	10	15	18	≥21
	directed, arc ($\ell_m^{\mathrm{dir}}(n)$), conjectured	2	6	12	20	30	≥36	≥42
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), conjectured	2	6	12	20	30	≥36	≥42
multigraph	undirected, edge ($L_m(n)$), proved	5	10	15	20	25	30	35
	undirected, vertex ($k_m(n)$), = simple, open	1	3	6	10	15	18	≥21
	directed, arc ($L_m^{\mathrm{dir}}(n)$), proved	10	20	30	40	50	60	80
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), = simple, conjectured	2	6	12	20	30	≥36	≥42
hypergraph $r=3$	undirected, edge, proved		1	4	8	≤12	15	17
	undirected, vertex, open		1	4	8	≥11	≥15	≥17
	directed, arc, open		3	12	≥25	≥30	≥35	≥40
	directed, vertex, open		3	12	≥25	≥30	≥35	≥40
multihypergraph $r=3$	undirected, edge, proved		5	7	10	≤12	15	17
	undirected, vertex, open		5	7	≥10	≥11	≥13	≥16
	directed, arc, open		15	≥20	≥25	≥30	≥35	≥40
	directed, vertex, open		15	≥20	≥26	≥30	≥35	≥40

Table 3.2. Figures 3.4 to 3.7 as numbers, one table for both m and every model. Blue is proved, red conjectured, gold open, matching each figure's curves, and a bold green entry is machine-checked exact regardless of the row's colour. A plain (non-bold) proved entry is exact: either its closed form is a value theorem, or, in an undirected hypergraph row, a construction reaches the row's proved upper bound, by the attainment range of Proposition 1.8 where that range covers the point and by a search witness the checker verified where it does not. In the latter rows a $\leq$-prefixed entry is a proved upper bound for which no matching construction is known at that point. A $\geq$-prefixed entry, in any conjectured or open row, is a verified construction rather than a proved optimum. At $m=6$, $n=6$ that is the hypergraph edge row's ≤ 12, where the true maximum, 11, is established by the full enumeration in the paragraph above rather than by the timed exhaustion behind this table's exact entries. A blank cell means the model does not exist at that n: no $r=3$ hyperedge exists on two vertices. Raising m from 3 to 6 also moves the undirected simple vertex row, and its multigraph reduction, from proved to open (Theorem 1.4, valid only for $m \leq 4$).

models. An exact checker tests connectivity, a certifier proves upper bounds for small cases, and a tabu search finds examples that give lower bounds. The rediscovery in Section 2.6 shows that this approach works: when applied to solved cases, it recovers the known answers.

These three tools provide different levels of evidence. The checker measures a given graph exactly, and a complete search proves the best value for a finite case. The certifier is also a finite check, but it is not a reusable proof unless it comes with a replayable bound. Heuristic search finds examples and lower bounds only. Thus, a value is settled only when the lower and upper bounds agree. Directed vertex values were checked separately from arc values because Whitney's inequality gives a witness, but not an upper bound. Every computational claim in this thesis can be rerun from the repository. The principal checks run from `program/erdos915_unified.py`. The two block sweeps in Appendix A use separate scripts in `research_notes/scripts/`, which are named where their results appear and intentionally provide independent implementations. Section A.11 gives the environment, commands, and audit evidence. No theorem here depends only on an unchecked solver result.

The program also has limits. The search finds examples but does not prove upper bounds. The certifier proves upper bounds only for a fixed size, and only as a finite solver check rather than as a reusable certificate (Section 2.3). The directed multigraph encoding works this way only up to $n = 6$. For larger n, the value is instead established by the hand proof in Appendix A. At $n = 7$, the certifier ran out of time without giving an answer. The generation enumerator's degree-capped classification at $n = 7$ (Remark A.45) is verified only for extremisers with maximum total degree at most 8, not for all graphs. These results do not close the structural gaps that remain open. The next section lists them. Three of those gaps cannot be solved by computation alone. The decomposition in Conjecture A.28 would prove Conjecture 3.1. Its backward-arc clause is only one part of that decomposition. The directed vertex problem for $m \geq 3$ does not follow from Whitney's inequality and the arc case. The undirected vertex-disjoint problem for $m \geq 6$ has been open for half a century, and the program can explore it but not solve it.

3.5 Open problems

Table 3.3 separates what this thesis proves from what it leaves open, one row per frontier. Each row states the strongest thing known here and the next step, so that a reader who wants to continue knows which statement to attack.

The map is filled in where the arguments reached, and the blanks are marked where they did not. The program does not answer every question. It records which questions are left and the form each of them now takes.

Problem	Known here	Remaining target
Directed simple arc, $m \geq 3$	$\ell_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$ (Theorem 1.11)	Prove the decomposition of Conjecture A.28, or otherwise determine the linear coefficient.
Directed simple vertex, $m \geq 3$	$k_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$ (Theorem 1.12)	Decide whether its exact value equals the arc value, given that an arc upper bound does not transfer.
Directed multigraph classification	Exact value for all n, m and quadratic-branch uniqueness for $n \geq \max(8, 2m)$	Classify the range $n < 2m$ and the remaining linear-branch extremisers.
Undirected simple vertex, $m \geq 6$	Block formulation and $m/2 < c_m \leq 2(m-1)$, where $c_m = \lim_n k_m(n)/(n-1)$ (Theorem A.13)	Determine c_m, whose likely next structure is triconnected rather than merely blockwise.
Hypergraph vertex, $m \geq 4$	Exact at $m = 2$ and, in the stated range, $m = 3$ (Theorems 1.9 and 1.10)	Locate the first failure for $r \geq 3$ and replace the rank-two decomposition used at $m = 3$.
Directed hypergraphs	For fixed r, m, leading term $(m-1)n^2/(4(r-1))$ for all orientations and both separations	Determine finite exact values and whether forward and general agree once $n > r$.

Table 3.3. The remaining problems, separated from the results already proved in this thesis.

Appendix A

Full Proofs and Computational Audit

This appendix carries every proof in the thesis. The body states the results and says in a few sentences what makes each one work, and the argument itself is here, together with the supporting lemmas and propositions that no chapter states in its own right, and with figures beside the arguments that need them.

How to read this appendix. It runs in four parts, and they are not of equal weight. PART I is classical machinery, Theorems A.1 and A.2 and Mader's theorem, and may be skipped by a reader who trusts the cited results. PARTS II and III carry the original arguments, and a reader with time for three of them should read Theorem 1.13 (the directed multigraph value, closed for every n and m by deleting a reachability skeleton rather than a vertex), Lemma A.58 with Theorem 1.10 (the incidence rank bound proved through Tutte's triconnected decomposition, which settles the hypergraph vertex problem at $m = 3$), and Lemma A.33 with the three theorems it drives, Theorems 1.11, 1.12 and 1.14, which pin six of the sixteen variants asymptotically with no hypothesis on the shape of the extremiser. PART IV holds one self-contained extension, the search's connectivity bookkeeping, and the computational audit. Every construction the body refers to by name is defined here, the directed ones in Section A.3. A heading carrying the badge [AI] means that the proof under it was written by an AI model and is not verified line by line by the author, as the Contribution Statement sets out. In this appendix a heading without the badge is a cited classical result, the Gomory–Hu double count of Section A.2 which proves Theorems 1.2 and 1.3 out of Lemmas A.5 to A.7, or an argument the author has read and checked.

Part I. Classical machinery

The two theorems every later argument runs on, and Mader's theorem proved in full through the Gomory–Hu tree. Cited results, reproved here so the thesis stands on its own.

A.1 Menger, Gomory–Hu, and the degree bound

We use two classical results in their capacitated form. Unit capacities represent simple graphs, integer capacities represent multiplicities, and cut capacity is the total capacity crossing the cut.

Theorem A.1 (Menger and max-flow/min-cut [9, 13])**.** *In any capacitated digraph, and for distinct vertices u, v, the maximum flow from u to v equals the minimum capacity of a cut separating u*

from v. This is the max-flow/min-cut theorem. With integer capacities, integral flow decomposition recovers Menger's path–separator theorem. At unit capacities it gives the combinatorial statement we cite throughout: the maximum number of pairwise edge-disjoint u–v paths equals the minimum number of edges whose removal separates them,

$$\lambda_G(u,v) = \min\{\, |C| : C \subseteq E(G),\ G - C \text{ has no } u\text{–}v \text{ path} \,\}.$$

Undirected graphs become symmetric networks, multigraph multiplicities become capacities, hypergraphs use the gate network of Figure 2.3b, and vertex-disjoint routes use the vertex splitting of Figure 2.3a. Thus one maximum-flow theorem supports every checker variant.

Theorem A.2 (Gomory–Hu [4])**.** *Every* undirected *capacitated graph G has a weighted tree T on the same vertex set, its* Gomory–Hu tree*, such that for all u, v the value $\lambda_G(u,v)$ equals the minimum edge weight on the u–v path in T, and each tree edge e corresponds to a minimum cut of G whose capacity is the weight of e. All $\binom{n}{2}$ pairwise values are thereby stored in the $n-1$ weights of a single tree.*

The theorem applies to undirected simple graphs, multigraphs, and hyperedge cuts (Theorem A.51), but not to asymmetric directed cuts or vertex connectivity. Its standard submodular-cut construction is proved in [4]. Here we use only the displayed path-minimum identity.

We also use the immediate endpoint bound.

Lemma A.3 (Degree bound)**.** *For every pair of vertices of a graph G, $\lambda_G(u,v) \leq \min(\deg u, \deg v)$. In a digraph D, $\lambda_D(u,v) \leq \min(d^+(u), d^-(v))$.*

Proof. Every edge-disjoint u–v path uses a distinct edge at u, so there are at most $\deg u$ of them, and likewise at most $\deg v$. In the directed case each arc-disjoint path leaves u by a distinct out-arc and enters v by a distinct in-arc. □

Chapter 2 leans on two monotonicity facts to keep the search's connectivity checks cheap. Both follow directly from max-flow / min-cut.

Proposition A.4 (Monotonicity of the binding connectivity)**.** *Let G be any graph, digraph, hypergraph, or directed hypergraph model used in this thesis, simple or with multiplicities, and let G' be obtained from G by changing one edge, arc, or hyperedge by a single copy. Then*

(1) (monotonicity) *removing the unit cannot raise $\lambda^{\max}$, and adding it cannot lower $\lambda^{\max}$, and*

(2) (unit step) *adding the unit raises $\lambda^{\max}$ by at most one.*

Both statements hold verbatim for the vertex measure $\kappa^{\max}$.

Proof. Write G^+ for G with one edge, arc, or hyperedge copy added, and fix a pair (s,t), ordered in a directed model. Removing a copy only removes routes, so neither local connectivity can increase under removal. This proves the monotonicity part after taking the maximum over all pairs.

For the unit step in edge connectivity, take a largest edge-, arc-, or hyperedge-disjoint family of s–t routes in G^+. At most one member of the family uses the new copy. Discard that member if it exists. Every remaining route lies in G, so

$$\lambda_G(s,t) \le \lambda_{G^+}(s,t) \le \lambda_G(s,t) + 1.$$

The vertex measure has the same unit step, but it is safer to see it directly than through a split-network cut. In a graph or digraph, two internally vertex-disjoint routes cannot both use the new adjacency. If it is internal to both routes, they share its endpoints internally. If it meets s or t, they share its other endpoint internally, except for the unique one-step s–t route. In the hypergraph convention of Section A.8, the routes are also hyperedge-copy-disjoint, so again at most one uses the new copy. Deleting that route leaves a family in G and gives

$$\kappa_G(s,t) \le \kappa_{G^+}(s,t) \le \kappa_G(s,t) + 1.$$

Taking the maximum over all pairs proves the two asserted unit-step bounds. If the new copy is parallel to an existing adjacency under the reduced convention of Section 1.3, it changes $\kappa^{\max}$ by zero, which is already covered by the inequality. □

A.2 Mader's theorem in full

Let T be a Gomory–Hu tree of G and let $\mathrm{dist}_T(e)$ count the tree edges on the path between the endpoints of $e \in E(G)$, as Figure A.1 illustrates. The upper bound is the following double count. Simplicity enters only because at most $n-1$ graph edges can join adjacent vertices of T.

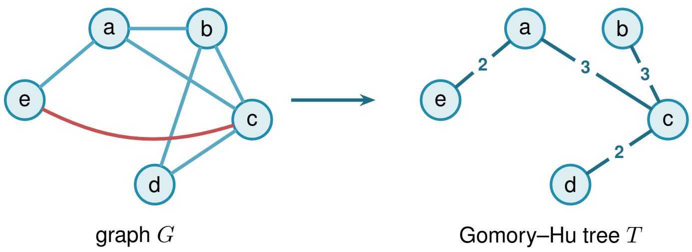


Figure A.1. A graph G and a Gomory–Hu tree T on the same vertices. Each graph edge crosses $\mathrm{dist}_T(e)$ tree cuts, and at most $n-1$ edges of a simple graph have tree distance one.

Lemma A.5 (Distance-one edges)**.** *At most $n-1$ edges of a simple graph G satisfy $\mathrm{dist}_T(e) = 1$.*

Proof. An edge $e \in E(G)$ has $\mathrm{dist}_T(e) = 1$ if and only if its endpoints are adjacent in T. The tree T has exactly $n-1$ edges, and since G is simple each pair of adjacent tree vertices supports at most one edge of G. □

Lemma A.6 (Sum of tree distances)**.** $\displaystyle\sum_{e \in E(G)} \mathrm{dist}_T(e) \;\ge\; 2|E(G)| - (n-1).$

Proof. Partition $E(G)$ into $E_1 = \{e : \mathrm{dist}_T(e) = 1\}$ and $E_{\geq 2} = \{e : \mathrm{dist}_T(e) \geq 2\}$. Then

$$\sum_{e \in E(G)} \mathrm{dist}_T(e) \;\geq\; |E_1| + 2\,|E_{\geq 2}| \;=\; 2|E(G)| - |E_1| \;\geq\; 2|E(G)| - (n-1),$$

using Lemma A.5 in the last step. □

Lemma A.7 (Double-counting identity)**.** *Let G carry* unit capacities*, so that each weight $c(t)$ of a Gomory–Hu tree T is the plain* number *of edges of G crossing the cut that t induces. Then*

$$\sum_{t \in E(T)} c(t) \;=\; \sum_{e \in E(G)} \mathrm{dist}_T(e).$$

This is the form used for Mader's theorem, where G is a simple graph and so unit-capacitated by default.

Proof. By Theorem A.2 each tree edge t induces a cut of G of capacity $c(t)$, so $c(t) = \sum_{e \in E(G)} \mathbf{1}_{e \text{ crosses } t}$. Exchanging the order of summation,

$$\sum_{t \in E(T)} c(t) = \sum_{e \in E(G)} \sum_{t \in E(T)} \mathbf{1}_{e \text{ crosses } t} = \sum_{e \in E(G)} \mathrm{dist}_T(e),$$

where the last equality holds because an edge $\{u, v\}$ crosses exactly those tree cuts induced by the tree edges on the unique u–v path in T. □

Proof of Theorem 1.2, upper bound. Let G be a simple graph on n vertices with $\lambda_G^{\max} \leq m-1$ and let T be a Gomory–Hu tree of G. Each tree edge $t = \{x, y\}$ has weight $c(t) = \lambda_G(x, y) \leq m - 1$, so summing over the $n-1$ tree edges, $\sum_t c(t) \leq (m-1)(n-1)$. Combining with Lemmas A.6 and A.7,

$$2|E(G)| - (n-1) \;\leq\; \sum_{e \in E(G)} \mathrm{dist}_T(e) \;=\; \sum_{t \in E(T)} c(t) \;\leq\; (m-1)(n-1).$$

Rearranging gives $2|E(G)| \leq m(n-1)$, and since $|E(G)|$ is an integer, $|E(G)| \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$. □

The factor two comes from charging all but at most $n-1$ edges to at least two tree cuts. Parallel edges destroy that refinement, leaving the multigraph bound $(m-1)(n-1)$ of Theorem 1.3.

Proof of Theorem 1.3. For the lower bound, take a spanning tree of K_n, that is, a connected cycle-free subgraph that touches all n vertices and so has exactly $n-1$ edges, and give every tree edge multiplicity $m-1$, meaning $m-1$ parallel copies. The result has $(m-1)(n-1)$ edges. Any two vertices are joined by a unique path in the tree, of some length $k \geq 1$. To separate them one must cut some edge of that path, and the cheapest choice is the lightest tree edge on it, which still carries $m-1$ copies, so $\lambda(u, v) = m-1$ for every pair and $\lambda^{\max} = m-1$. The graph sits right at the cap.

For the upper bound, suppose $\lambda_G^{\max} \le m-1$ and build a Gomory–Hu tree T of G (Theorem A.2), the same $(n-1)$-edge summary of all connectivities used for Mader's theorem. Each tree edge e stands for a minimum cut of G, of capacity $c(e) = \lambda_G(u,v) \le m-1$ for the pair (u,v) it represents. Now charge each edge of G to a tree edge: an edge $\{x,y\}$ crosses the cut of the lightest tree edge on the tree path from x to y, so it is counted inside that tree edge's capacity. Summing the $n-1$ capacities therefore counts every edge of G at least once, which already gives $|E(G)| \le \sum_{e\in T} c(e) \le (m-1)(n-1)$. Unlike Mader's simple-graph count there is no second charge to halve, because parallel edges are now allowed and nothing forces an edge to be counted twice. The tree-of-multiplicity-$(m-1)$ construction above meets this bound exactly, which is why the answer is the clean product $(m-1)(n-1)$ and not a floor. $\square$

Construction A.8 (Hub-and-spokes graph)**.** For $n \ge m \ge 2$, let $H_{m,n}$ have vertex set $\{h, v_1, \ldots, v_{n-1}\}$ with (i) all $n-1$ *hub edges* $\{h, v_i\}$, and (ii) a *spoke graph* on $\{v_1, \ldots, v_{n-1}\}$ with $\left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor$ edges in which every v_i has spoke degree at most $m-2$. Such a spoke graph exists by Lemma A.9. This is the general extremal construction. If $(m-1) \mid (n-1)$, take the spoke graph to be disjoint copies of K_{m-1}, and adjoining h turns them into copies of K_m with one shared vertex, the shared-clique construction.

Lemma A.9 (Near-regular graphs exist)**.** *For $N \ge 1$ and $0 \le r \le N-1$ there is a graph on N vertices with $\left\lfloor \frac{rN}{2} \right\rfloor$ edges and maximum degree at most r.*

Proof. If rN is even we build an r-regular circulant graph on vertex set $\{0, \ldots, N-1\}$: for even r connect each i to $i \pm j \pmod N$ for $j = 1, \ldots, r/2$, and for odd r (then N is even) use $j = 1, \ldots, (r-1)/2$ together with the diameter distance $N/2$, which contributes degree one. Both are r-regular with $rN/2$ edges. If rN is odd, then r and N are both odd: take the $(r-1)$-regular circulant with distances $1, \ldots, (r-1)/2$ and add the near-perfect matching that pairs vertex i with $i + (N+1)/2 \pmod N$ for $i = 0, \ldots, (N-3)/2$, which covers all vertices but one. The total is $\frac{(r-1)N}{2} + \frac{N-1}{2} = \frac{rN-1}{2} = \left\lfloor \frac{rN}{2} \right\rfloor$ edges, with one vertex of degree $r-1$ and all others of degree r. $\square$

Theorem A.10 (Extremal graphs exist)**.** *For all $n \ge m \ge 2$ the graph $H_{m,n}$ has $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ edges and $\lambda^{\max} \le m-1$.*

Proof. The edge count is $(n-1) + \left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, since $(m-2)(n-1)$ and $m(n-1)$ have the same parity. For the connectivity, any two distinct vertices include at least one spoke vertex v_i, whose total degree is at most $1 + (m-2) = m-1$. Lemma A.3 then gives $\lambda(u,v) \le m-1$. $\square$

The upper bound above proves Theorem 1.2, and it uses no relation between n and m: the Gomory–Hu tree has $n-1$ edges whatever m is. Theorem A.10 adds that the bound is attained once $n \ge m$, which is where that hypothesis is needed and where it belongs. For $n < m$, the complete graph is itself feasible and the bound is slack except at $(n,m) = (2,3)$, where the floor creates the isolated equality noted in the main text. Equality then holds systematically from $n = m$ onward. The proof also reveals which graphs are extremal: equality forces both inequalities in the chain to be tight.

Theorem A.11 (Characterisation of equality)**.** *Let G be simple with $\lambda_G^{\max} \le m-1$ and $|E(G)| = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and let T be a Gomory–Hu tree of G. If $m(n-1)$ is even, then every tree edge has weight exactly $m-1$, every tree edge is also an edge of G, and every edge of G joins vertices at tree distance at most 2. If $m(n-1)$ is odd, there is exactly one unit of slack, located in exactly one of three places: one tree edge has weight $m-2$, or one tree edge is not an edge of G, or one edge of G joins vertices at tree distance 3. Everything else is tight.*

Proof. Write the chain of the upper-bound proof as three non-negative integer slacks, one for each inequality it uses:

$$S_1 = (m-1)(n-1) - \sum_t c(t), \qquad S_2 = \sum_{e:\, \mathrm{dist}_T(e) \ge 2} \big(\mathrm{dist}_T(e) - 2\big), \qquad S_3 = (n-1) - |E_1|.$$

In words, S_1 counts tree weights below $m-1$, S_2 counts edges beyond tree distance two, and S_3 counts tree edges that support no edge of G. Retracing the proof gives two expressions for the same quantity,

$$\sum_e \mathrm{dist}_T(e) \;=\; 2|E| - |E_1| + S_2 \qquad \text{and} \qquad \sum_e \mathrm{dist}_T(e) \;=\; \sum_t c(t) \;=\; (m-1)(n-1) - S_1.$$

Substituting $|E_1| = (n-1) - S_3$ into the first and equating it with the second,

$$\begin{aligned} 2|E| - (n-1) + S_2 + S_3 &\;=\; (m-1)(n-1) - S_1, \\ \text{so} \qquad 2|E| &\;=\; m(n-1) - (S_1 + S_2 + S_3). \end{aligned}$$

The whole characterisation reads off that last identity. If $m(n-1)$ is even, equality $|E| = m(n-1)/2$ forces $S_1 = S_2 = S_3 = 0$. If $m(n-1)$ is odd the floor absorbs exactly one unit, so $S_1 + S_2 + S_3 = 1$ and exactly one of the three slacks equals one. □

For $m = 2$ the characterisation says the extremal graphs are exactly the spanning trees: every tree edge must carry weight one and lie in G, so G contains its own Gomory–Hu tree and has only $n-1$ edges to spend. For $m = 4$ and $n \equiv 1 \pmod 3$ it is consistent with the trees of K_4 blocks glued at shared cut vertices, the case of Construction A.8 that the search of Chapter 2 rediscovers.

Remark A.12 (The counting convention, and the audit it forces)*.* This thesis defines $\ell_m(n)$ and $k_m(n)$ as the *largest* number of edges a graph can carry while *no* pair is joined by m independent routes. Sørensen and Thomassen, and the Erdős Problems database [10] after them, state the same results the other way round, as the *smallest* number of edges that *forces* such a pair. Those two quantities are adjacent integers by construction: if k is the largest feasible edge count then no graph on n vertices with $k+1$ edges is feasible, so $k+1$ is exactly the forcing threshold.

What follows is the bookkeeping that convention forces on every constant quoted from the literature.

The shift is visible in every entry the database and this thesis both state. It gives $k_2(n) = n$ where a tree has $n-1$ edges, $k_3(2n) = 3n-1$ against $\lfloor 3(2n-1)/2 \rfloor = 3n-2$ here, $k_4(n) = 2n-1$ against $2n-2$, $\ell_5(2n) = 5n-2$ against $5n-3$, $\ell_6(n) = 3n-2$ [18] against $3n-3$, and it writes the Bollobás-Erdős conjecture as $1 + \binom{m}{2}n$ *on* $1 + (m-1)n$ *vertices* against $\binom{m}{2}n$ here. Every one is exactly one apart, and the values in this thesis are the ones an exhaustive search returns: $k_3(4) = 4$, $k_3(5) = 6$ and $k_4(5) = 8$ were each confirmed by walking every graph of that size.

The conjecture is the one entry that must keep its vertex count attached, and an earlier draft of this thesis lost it. On $N = 1 + (m-1)n$ vertices the conjectured value is $\binom{m}{2}\frac{N-1}{m-1} = \frac{m(N-1)}{2}$, which is the rate $c_m = m/2$ of Theorem A.13. Read instead as a count on n vertices, $\binom{m}{2}n$ overstates it by a factor of $m-1$, giving $3n$ at $m = 3$ where the true value is about $3n/2$.

A second trap is worth naming, because an earlier draft fell into that one too. The comparison $k_5 > \ell_5$ is invariant under the shift, since *both* functions move by one, so no internal check can detect a half-converted statement. What does detect it is evaluating the two sides in different conventions, which manufactures a spurious divergence one size early. Reading $\lfloor 8 \cdot 13/3 \rfloor - 3 = 31$ against this thesis's $\ell_5(13) = 30$ appears to separate the two problems at $n = 13$, but the 31 is a forcing count and the 30 is an avoiding count. Done consistently, $k_5(13) = 30 = \ell_5(13)$ and the separation begins at $n = 14$, which is also what [8] proves directly by showing $f_5(n) = \lfloor 5(n-1)/2 \rfloor + 1$ throughout $6 \le n \le 13$.

The range likewise comes from the paper rather than from the database, which records only the clean tail $n \ge 13$. Theorem 4 there states $f_5(n) = \lfloor 8n/3 \rfloor - 3$ for every $n \ge 6$ apart from $n = 7$ and $n = 12$, and supplies $f_5(7) = 16$ and $f_5(12) = 28$ separately. Its closing line then covers the whole small range at once, $f_5(n) = \lfloor 5(n-1)/2 \rfloor + 1$ for $6 \le n \le 13$. That is why Theorem 1.5 is stated in two ranges and needs no exceptions: the two sizes the paper sets aside are the only points of $6 \le n \le 13$ at which the large-n formula misses the small-n one, low by one at $n = 7$ and high by one at $n = 12$.

A.2.1 The vertex problem is a block problem

The *vertex* problem $k_m(n)$ has been open since 1974, and this short section records its block structure. It changes the object being searched: from graphs on n vertices, with n growing, to 2-connected graphs alone.

Theorem A.13 (The vertex problem is a knapsack over blocks, [Al]). *For $b \ge 2$ let*

$$h_m(b) \;:=\; \max\{\, |E(B)| \;:\; B \text{ 2-connected or a single edge},\ |V(B)| = b,\ \kappa_B^{\max} \le m-1 \,\},$$

with $h_m(b) := -\infty$ when no such B exists, as happens for every $b \ge 3$ at $m = 2$, where a cycle *already carries two internally disjoint routes. Then for all $m \geq 2$ and $n \geq 2$,*

$$k_m(n) \;=\; \max\Big\{ \sum_i h_m(b_i) \;:\; b_i \geq 2,\; \sum_i (b_i - 1) \leq n - 1 \Big\}.$$

Consequently $k_m(n) \geq k_m(n_1) + k_m(n_2)$ whenever $n = n_1 + n_2 - 1$, and

$$c_m \;:=\; \lim_{n\to\infty} \frac{k_m(n)}{n-1} \;=\; \sup_{b\geq 2} \frac{h_m(b)}{b-1}$$

exists, the limit being approached from below. The supremum is finite: Remark A.14 gives $k_m(n) < 2(m-1)n$, so $c_m \leq 2(m-1)$.

Proof. Let G be feasible. Adjacent vertices u, v lie in a common block B, and no u-v path leaves B: leaving means crossing a cut vertex, and returning means crossing that same cut vertex a second time, which no path does. So $\kappa_G(u,v) = \kappa_B(u,v)$, every block inherits feasibility, and since every edge lies in exactly one block, $|E(G)| = \sum_B |E(B)| \leq \sum_B h_m(|V(B)|)$. Within one component the block sizes satisfy $\sum_B(|V(B)| - 1) =$ (order of the component) $- 1$, and summing over components loses one further vertex apiece, so $\sum_B(|V(B)| - 1) \leq n - 1$. That is $\leq$.

For $\geq$, take blocks $B_1, \ldots, B_t$ with $\sum(b_i - 1) \leq n - 1$, pick one vertex, and attach all the B_i to it, pairwise disjoint otherwise. The result has $1 + \sum(b_i - 1) \leq n$ vertices. Add isolated vertices if necessary to bring its order to exactly n. Its nontrivial blocks are exactly the B_i, so pairs inside a block keep their κ and pairs split between two blocks are separated by the shared vertex and have $\kappa \leq 1 \leq m - 1$. It is feasible with $\sum_i |E(B_i)|$ edges.

Superadditivity is the special case of gluing two extremal graphs at a single vertex. Writing $a_n := k_m(n)$, the sequence (a_{n+1}) is superadditive in $n - 1$, so Fekete's lemma gives $a_n/(n-1) \to \sup_n a_n/(n-1)$, and $a_n/(n-1) \geq h_m(b)/(b-1)$ whenever $(b-1) \mid (n-1)$, while $a_n \leq \max_b h_m(b)/(b-1) \cdot (n-1)$ from the display. So the limit is $\sup_b h_m(b)/(b-1)$. □

Three things follow at once.

At $m = 3$ it *proves* Theorem 1.4's value. A block with $\kappa^{\max} \leq 2$ is an edge or a cycle, since a block that is neither contains a theta subgraph and a theta gives three internally disjoint paths. So $h_3(2) = 1$ and $h_3(b) = b$ for $b \geq 3$, the rate $b/(b-1)$ is largest at $b = 3$, and the knapsack packs triangles: $k_3(n) = 3\lfloor (n-1)/2 \rfloor + ((n-1) \bmod 2) = \left\lfloor \frac{3(n-1)}{2} \right\rfloor$.

At $m = 4$ the picture is $h_4(b) = 2(b-1)$ for every $b \geq 4$, so the rate $h_4(b)/(b-1)$ reaches 2 at $b = 4$ and stays there, against 1 at $b = 2$ and $3/2$ at $b = 3$. The knapsack therefore spends its whole budget on blocks of size at least four as soon as one fits, giving $k_4(n) = 2(n-1)$ for $n \geq 4$. Below that only the small blocks are available and the formula overshoots, $k_4(2) = 1$ against its 2 and $k_4(3) = 3$ against its 4, which is where the hypothesis $n \geq m$ of Theorem 1.4 earns its place. This one is a repackaging rather than a second proof. The upper bound $h_4(b) \leq k_4(b) = 2(b-1)$ is Theorem 1.4 itself read on a block, so it cannot turn round and reprove it, and nothing as elementary as the theta argument above is on offer for the blocks with $\kappa^{\max} \leq 3$. What the reformulation adds is the extremal block, one the appendix has already built: the hub-and-spokes graph $H_{4,b}$ of Construction A.8, a hub joined to every other vertex with a cycle on the rest. Unlike the shared-clique graph it is a single block, and it meets $2(b-1)$ at every size. An exhaustive sweep of the 2-connected graphs confirms that it is optimal, and the unique optimum, at every b from 4 to 8.

At every m it identifies the Bollobás–Erdős construction, as many copies of K_m sharing one vertex as fit, as the case $b = m$: $h_m(m) = \binom{m}{2}$ and the rate is exactly $m/2$. Their conjecture is precisely the assertion $c_m = m/2$, that no block beats K_m. Exhaustive computation over all 2-connected graphs on at most nine vertices, of which there are 194066 at the top size alone, confirms that none does, for every $m \leq 8$: the rate $m/2$ is attained and never exceeded. That sweep runs from its own standalone script, `research_notes/scripts/simple_vertex_blocks.py`, rather than from the main program, so it corroborates the checker instead of sharing code with it. Since the rate of a bouquet of blocks is a weighted average of the rates of its blocks, a counterexample must contain a single 2-connected block that beats $m/2$ on its own, and by the computation such a block needs at least ten vertices.

The reason the computation does not reach further is structural rather than a matter of processor time. The conjecture is false. Leonard disproved it first, at $m = 5$, with an explicit graph on 57 vertices [19], and Mader showed that the difference $k_m(n) - mn/2$ is unbounded for $m \geq 6$ [11]. What replaces it, and what the rest of this section leans on, is a lower bound of Sørensen and Thomassen [8]: for each fixed m,

$$c_m \;\geq\; \frac{m(m-1)-2}{2m-3} \;=\; \frac{m}{2} + \frac{1}{4} + O(1/m),$$

so the conjectured rate is wrong, but only by a constant. That display alone refutes the conjecture for every $m \geq 5$, without a second reading of either earlier paper: subtracting gives

$$2\big(m(m-1)-2\big) - m(2m-3) \;=\; m-4,$$

which is positive exactly when $m > 4$, so the Sørensen–Thomassen rate exceeds $m/2$ from $m = 5$ onwards.

Their witness is built by a recursion, described here in the language of this section, since it says why the blocks above cannot find it. Start from $G^0 = K_m$ minus an edge, and form G^j from two fresh copies of G^0 together with G^{j-1}, identifying one vertex of each with one vertex of the next, the three identifications arranged in a *cycle*. Each step adds $2m-3$ vertices and $m(m-1)-2$ edges, which is the rate above. The arrangement is cyclic, and three graphs glued in a cycle at three vertices have no cut vertex at all, so G^j is 2-connected and is a *single block*. Theorem A.13 therefore sees it as one indivisible object of size $m + (2m-3)j$ and gets no purchase on it. What the graph does have is 2-cuts, one pair of identification vertices separating each copy from the rest, so the decomposition that would see it is the triconnected one, along 2-cuts rather than cut vertices, which is the same Tutte/SPQR machinery Lemma A.58 already uses on incidence graphs. That is the concrete next step this reformulation points at, and it is pointing somewhere the answer is already known: Sørensen and Thomassen prove in the same paper that the Bollobás–Erdős bound *does* hold for 3-connected graphs [8]. So the rate $m/2$ survives exactly one refinement past the block decomposition, and what a triconnected version of Theorem A.13 would have to supply is not a new bound on the rigid pieces but the bookkeeping that assembles them, which is where the recursion above hides.

It also explains the sizes. The recursion beats the rate $m/2$ only once $j > 2/(m-4)$, so the smallest member of the family that outruns a bouquet of K_m's has 26 vertices at $m = 5$, 24 at $m = 6$ and 18 at $m = 7$, against the nine that exhaustive enumeration reaches. The construction was rebuilt and checked here at $m = 5, 6, 7$ and $j \leq 3$: the vertex and edge counts match the formulas exactly, every member is 2-connected, and every member has $\kappa^{\max} = m - 1$, so each is feasible and none contains the forbidden m internally disjoint paths.

Remark A.14 (A degeneracy question, and what it would give). A general density bound is $k_m(n) < 2(m-1)n$: Mader's density theorem [20] would otherwise produce an m-connected subgraph. Feasibility says much more than that. It gives, for instance, that any two adjacent vertices have at most $m - 2$ common neighbours, since each common neighbour is a route of length two beside the edge.

The natural strengthening is that a feasible graph always has a vertex of degree at most $m - 1$. Since feasibility passes to induced subgraphs, that would make every feasible graph $(m-1)$-degenerate and give

$$k_m(n) \;\leq\; (m-1)n - \binom{m}{2},$$

halving the constant in the only general upper bound known. It holds for $m = 3$, where it is classical: a graph of minimum degree at least three contains a subdivision of K_4, and two branch vertices of that subdivision are joined by three internally disjoint paths. Exhaustive search over all graphs with minimum degree at least m finds no feasible one for $m \leq 6$ and $n \leq 10$. It is stated here as a question rather than a conjecture, since the sizes reached are well below those at which the Bollobás–Erdős conjecture itself first fails.

Part II. The directed results

The three constructions the body names but does not define, the exact value at $m = 2$, the structural propositions that bound the frontier at $m \geq 3$, and the directed multigraph problem closed for every n and every m.

A.3 The directed constructions

Three shapes carry every directed lower bound in this thesis. Chapter 1 shows the first two side by side in Figure 1.7 and records what they reach. Here each is defined precisely and its connectivity checked.

Construction A.15 (Directed hub)**.** For $n \geq m \geq 2$, take a hub h and label the other $N = n-1$ vertices $0, \dots, N-1$. Add the $2N$ hub arcs $h \to i$ and $i \to h$, and on the non-hub vertices the $(m-2)N$ ring arcs $i \to i+j \bmod N$ for $j = 1, \dots, m-2$. The total is $m(n-1)$ arcs and every non-hub vertex has $d^+ = d^- = m-1$.

Every ordered pair has a non-hub member, whose relevant degree is exactly $m-1$ and, since $n \geq m$, at most the hub's own $n-1$, with equality only at $n = m$. That member is therefore the binding one and the degree bound gives $\lambda^{\max} \leq m-1$. At $m = 2$ the ring is empty and what remains is the *bidirected star*, in which two spokes can only reach each other through the hub.

Construction A.16 (One-directional bipartite digraph)**.** Split the vertices into parts A and B of sizes $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$, and include every arc from A to B and none the other way. The total is $\lfloor n^2/4 \rfloor$ arcs.

Nothing leaves B and nothing enters A, so no route has a second step and every pair is joined by at most the one direct arc, giving $\lambda^{\max} = 1$.

Construction A.17 (Augmented bipartite digraph)**.** For $n \geq m \geq 2$, set $|B| = \lceil (n+m-2)/2 \rceil$ and $|A| = n - |B|$, include every arc from A to B, and inside B give each b_j the $m-2$ in-arcs from its cyclic predecessors $b_{j-1}, \dots, b_{j-(m-2)}$. The total is $\lfloor (n+m-2)^2/4 \rfloor$ arcs.

The only in-arcs of a $b \in B$ that an $a \in A$ can reach are the direct arc $a \to b$ and the $m-2$ two-step detours through b's in-neighbours inside B, so such a pair carries exactly $m-1$ routes and no more, while pairs inside B are held down by the in-degree $m-2$ and pairs from B to A or inside A have no routes at all.

At $m = 3$, $n = 10$ this is the digraph that refutes the natural first guess $\max(m(n-1), \lfloor n^2/4 \rfloor)$. Here $|B| = \lceil 11/2 \rceil = 6$ and $|A| = 4$, so its 24 arcs from A to B carry a six-cycle inside B, one arc per b_j since $m-2 = 1$, giving every b_j one extra in-arc, so each pair a_i, b_j has the direct arc and one detour and $\lambda^{\max} = 2$. That is $24 + 6 = 30$ arcs against the guess of 27, and the correction is the shift of the partition away from a balanced split of n, which the guess assumes, towards a balanced split of $n+m-2$. At this size that quantity is 11 and odd, so $|B| = 6$ and $|B| = 5$ tie, the latter giving $25 + 5 = 30$ as well.

A.4 The directed arc value at $m = 2$

Three short lemmas feed the induction. The first shows a hub forces a linear bound for every m, and is the proof behind the hub branch of Conjecture 3.1. The three lemmas below carry no badge, while the induction of Theorem 1.6 that they support keeps one at its statement, and its finite base cases $2 \leq n \leq 7$ were verified by the author's program, through the exhaustive search of Chapter 2 whose completed range is recorded in Table A.1.

Lemma A.18 (Two-hop lemma)**.** *Let D be a simple digraph with $\lambda_D^{\max} \leq m-1$ containing a vertex r with $d^+(r) = n-1$. Then every $v \neq r$ has at most $m-2$ in-arcs from $V \setminus \{r, v\}$.*

Proof. If $u_1, \ldots, u_{m-1}$ were distinct in-neighbours of v other than r, that is, distinct vertices each sending an arc to v, the paths $r \to v$ and $r \to u_i \to v$ for $i = 1, \ldots, m-1$ would be m arc-disjoint routes, since the u_i are distinct and $d^+(r) = n-1$ supplies every starting arc. That contradicts $\lambda_D(r, v) \leq m-1$. □

Theorem A.19 (Hub upper bound)**.** *If a simple digraph D with $\lambda_D^{\max} \leq m-1$ has a vertex r with $d^+(r) = n-1$, then $|A(D)| \leq m(n-1)$.*

Proof. Split the arcs into those leaving r (exactly $n-1$), those entering r (at most $n-1$), and the internal arcs avoiding r. By Lemma A.18 each of the $n-1$ other vertices receives at most $m-2$ internal arcs, so there are at most $(m-2)(n-1)$ internal arcs, and the total is at most $m(n-1)$. □

Lemma A.20 (No transitive triangle)**.** *A simple digraph with $\lambda^{\max} \leq 1$ contains no three vertices x, y, z with all of $(x, y), (y, z), (x, z)$ present, since $x \to z$ and $x \to y \to z$ would be two arc-disjoint routes.*

Lemma A.21 (Deletion monotonicity)**.** *Write $D - v_0$ for the digraph obtained from D by deleting the vertex v_0 together with all arcs incident to it. For any digraph D and vertex v_0,*

$$\lambda^{\max}(D - v_0) \leq \lambda^{\max}(D) \qquad \text{and} \qquad \kappa^{\max}(D - v_0) \leq \kappa^{\max}(D),$$

since arc-disjoint paths in $D - v_0$ are still arc-disjoint paths in D, and internally vertex-disjoint paths in $D - v_0$ are still internally vertex-disjoint paths in D. The two statements are proved by the same one-line observation because deleting a vertex only removes routes, never creates any, and it cannot make two surviving routes share anything they did not share before.

In words: deleting a vertex can never manufacture independent routes. Every route surviving in $D - v_0$ was already a route in D, so no pair comes out of the deletion better connected than it went in. This single guarantee is what every vertex-deletion induction in this section rests on, and it holds for both separations alike.

Proof of Theorem 1.6. Write $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$. A direct computation shows $2(n-1) \geq \lfloor n^2/4 \rfloor$ exactly for $n \leq 7$, with equality at $n = 7$, so $M(n) = 2(n-1)$ for $n \leq 7$ and $M(n) = \lfloor n^2/4 \rfloor$ for $n \geq 7$. The lower bound $\ell_2^{\mathrm{dir}}(n) \geq M(n)$ is witnessed by the directed hub construction (Construction A.15 at $m = 2$, the bidirected star) and the one-directional bipartite construction (Construction A.16).

For the upper bound we show by strong induction on n that $\lambda_D^{\max} \leq 1$ forces $|A(D)| \leq M(n)$.

Base cases $2 \leq n \leq 7$. All six are settled uniformly by the pruned exhaustive search of Chapter 2, which walks every simple digraph with $\lambda^{\max} \leq 1$ on up to seven vertices and reports the maxima $2, 4, 6, 8, 10, 12$, matching $M(n)$ at each size. The transcript is recorded in Table A.1, and the cut-counting certifier independently confirms the values within its reach through the $m = 2$ reduction of Corollary A.42. The two smallest are also immediate by hand, as a check on the machine rather than as a separate argument. At $n = 2$ there are at most 2 arcs. At $n = 3$ a fifth arc would leave at most one arc of the complete bidirected triangle missing, so three of the remaining arcs form a transitive triangle, which Lemma A.20 forbids. Unlike $n = 2, 3$, the cases $n = 4, 5, 6, 7$ are not immediate by hand, which is why the case-check is handed to the program.

Inductive step $n \geq 8$. Let D be a simple digraph on n vertices with $\lambda_D^{\max} \leq 1$ and put $a := |A(D)|$. Choose v_0 minimising the total degree $d(v) = d^+(v) + d^-(v)$. Since $\sum_v d(v) = 2a$, averaging gives $d(v_0) \leq 2a/n$. By Lemma A.21 the digraph $D - v_0$ on $n-1 \geq 7$ vertices satisfies the induction hypothesis, so $a - d(v_0) \leq M(n-1) = \left\lfloor \frac{(n-1)^2}{4} \right\rfloor$. Combining,

$$a \;\leq\; \frac{2a}{n} + \left\lfloor \frac{(n-1)^2}{4} \right\rfloor \qquad \Longrightarrow \qquad a \;\leq\; \frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor .$$

It remains to check that this right-hand side never exceeds $\lfloor n^2/4 \rfloor$, and the two parities behave differently enough to treat them in turn.

For even $n = 2k$ with $k \geq 4$, first simplify $\lfloor (n-1)^2/4 \rfloor$. Expanding $(2k-1)^2 = 4k^2 - 4k + 1$ and dividing by 4 gives $k^2 - k + \frac{1}{4}$, whose floor is $k(k-1)$. The bound then reads

$$a \;\leq\; \frac{2k}{2k-2}\, k(k-1) \;=\; \frac{k}{k-1}\, k(k-1) \;=\; k^2,$$

and since $\lfloor n^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$ as well, the bound lands exactly on the target with no slack left over.

For odd $n = 2k+1$ with $k \geq 4$, the same simplification gives $\lfloor (n-1)^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$, so the bound is $a \leq \frac{2k+1}{2k-1} k^2$. To compare this fraction with the target, split off its integer part by dividing $(2k+1)k^2$ by $2k-1$. The remainder is exactly k, because $(2k-1)(k^2+k) = 2k^3 + k^2 - k$ and $(2k+1)k^2 = 2k^3 + k^2$ differ by k, so

$$\frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor \;=\; \frac{(2k+1)k^2}{2k-1} \;=\; k(k+1) + \frac{k}{2k-1}.$$

The leftover fraction $\frac{k}{2k-1}$ lies strictly between 0 and 1 for every $k \geq 2$ (the induction here is already past the base cases, so $k \geq 4$), so the bound puts a at most an integer $k(k+1)$ plus a positive amount below one. Since a is itself an integer it cannot reach the next integer up, so $a \leq k(k+1)$. The target $\lfloor n^2/4 \rfloor = \lfloor (2k+1)^2/4 \rfloor = k^2 + k = k(k+1)$ agrees, so for odd n it is integrality, not the raw arithmetic, that closes the last fraction of a unit. In both parities $a \leq \lfloor n^2/4 \rfloor \leq M(n)$, completing the induction. □

Three remarks on the shape of this proof. First, the minimum-degree deletion closes the step on its own, and no separate hub case is needed, because the averaging bound holds whether or not D contains a hub, although Theorem A.19 shows directly that a hub digraph never exceeds the linear branch. Second, for even n the bound is exactly tight against the one-directional bipartite construction, which is $\frac{n}{2}$-regular in total degree, so every step of the chain is achieved by the conjectured extremiser, and for odd n the slack $\frac{k}{2k-1} < 1$ is absorbed by integrality. Third, the proof's length comes from carrying the two regimes of $M(n)$ through the arithmetic separately, the same bookkeeping Chapter 2 hands to the machine at the base cases.

Remark A.22 (Which way Whitney runs). Whitney's inequality is easy to misread: the tempting direction is the wrong one. From $\kappa_D(u,v) \le \lambda_D(u,v)$ for every pair it follows that $\kappa_D^{\max} \le \lambda_D^{\max}$, hence that

$$\{D : \lambda_D^{\max} \le m-1\} \subseteq \{D : \kappa_D^{\max} \le m-1\}.$$

Every digraph feasible for the *arc* problem is feasible for the *vertex* problem, and not the other way round. The vertex-feasible family is the larger of the two, so the inequality between the extremal numbers it hands over for free is

$$k_m^{\mathrm{dir}}(n) \;\ge\; \ell_m^{\mathrm{dir}}(n),$$

and an upper bound for the arc problem says nothing at all about the vertex problem. The gap is not vacuous: two routes that share an interior vertex but no arc are counted by λ and not by κ, so there really are digraphs in the difference of the two families. Whitney therefore gives one thing, that an arc construction is a vertex *lower* bound, and the vertex upper bound has to be proved separately. The proof below does that, and Chapter 3 treats the two separations the same way for $m \ge 3$, where the vertex problem stays open even though the arc conjecture is stated.

Proof of Theorem 1.7. Write $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ as before.

Lower bound. The directed hub and the one-directional bipartite construction both have $\lambda^{\max} = 1$, so by Remark A.22 both have $\kappa^{\max} \le 1$ and are feasible for the vertex problem. Hence $k_2^{\mathrm{dir}}(n) \ge M(n)$.

Upper bound. We show by strong induction on n that $\kappa_D^{\max} \le 1$ forces $|A(D)| \le M(n)$. The induction is the one that proved Theorem 1.6, and every step of it survives the change of separation, because the only two properties of the cap that argument uses are that it is inherited by vertex deletion and that it holds at the base sizes.

The base cases $2 \le n \le 7$ are settled by the same pruned exhaustive search of Chapter 2, run with the vertex feasibility test in place of the arc one. It walks every simple digraph with $\kappa^{\max} \le 1$ on up to seven vertices and reports the maxima $2, 4, 6, 8, 10, 12$, matching $M(n)$ at each size. Table A.1 records this run beside the arc one.

For the inductive step $n \ge 8$, let D be a simple digraph on n vertices with $\kappa_D^{\max} \le 1$ and put $a := |A(D)|$. Choose v_0 of minimum total degree, so $d(v_0) \le 2a/n$ by averaging. By Lemma A.21, which holds for κ exactly as it does for λ, the digraph $D - v_0$ on $n-1 \ge 7$ vertices again satisfies $\kappa^{\max} \le 1$, so the induction hypothesis gives $a - d(v_0) \le M(n-1) = \lfloor (n-1)^2/4 \rfloor$. From here the two parity computations are verbatim those of Theorem 1.6: they use only the two displayed inequalities and the integrality of a, no property of the separation. They yield $a \le \lfloor n^2/4 \rfloor \le M(n)$, completing the induction.

Hence $k_2^{\mathrm{dir}}(n) = M(n) = \ell_2^{\mathrm{dir}}(n)$. The two problems have the same answer at $m = 2$, but this is a computed coincidence of the two searches meeting at every base size, not a formal consequence of Whitney's inequality. $\square$

Remark A.23 (The directed vertex problem at $m \geq 3$). The proof above transfers the $m = 2$ induction but nothing more, and for $m \geq 3$ the vertex problem is separate: the arc conjecture, and the decomposition of Conjecture A.28 that would prove it, both concern the smaller family and leave $k_m^{\mathrm{dir}}(n)$ untouched. What can be said is computational. Running the exhaustive search of Chapter 2 under both feasibility tests at $m = 3$ gives

n	2	3	4	5	6
$\ell_3^{\mathrm{dir}}(n)$	2	6	9	12	15
$k_3^{\mathrm{dir}}(n)$	2	6	9	12	15

with every entry proved complete, so the two separations agree at every size the exhaustion reaches, and from $n = 3$ onwards both match the conjectured $\max(m(n-1), \lfloor (n+m-2)^2/4 \rfloor)$. The table stops at $n = 6$ because that is where the exhaustion stops: at $n = 7$ neither sweep closes inside an hour and a half, and each returns the construction's 18 as a bare lower bound. At $n = 2$ the conjecture's formula gives 3 where only 2 arcs exist at all, the usual degeneracy of the formula below $n = m$. Agreement over five sizes is evidence and not a proof, and the natural next question is structural rather than numerical: to separate the two problems one needs a digraph where some pair carries strictly more arc-disjoint routes than internally disjoint ones *and* where that slack pays for extra arcs. The two demands pull against each other, which is perhaps why no small example does both.

Lemma A.24 (The augmented bipartite digraph is feasible)**.** *For $m \geq 2$ and $n \geq m$, the digraph of Construction A.17 satisfies $\lambda^{\max} = m - 1$ and has $\lfloor (n+m-2)^2/4 \rfloor$ arcs. Hence $\ell_m^{\mathrm{dir}}(n) \geq \lfloor (n+m-2)^2/4 \rfloor$.*

Proof. The arc count is $|B|\,(|A| + m - 2)$, one arc from each of the $|A|$ vertices of A plus $m - 2$ internal in-arcs, for each of the $|B|$ vertices of B. Now $|A| + m - 2 = \lfloor (n+m-2)/2 \rfloor$ and $|B| = \lceil (n+m-2)/2 \rceil$, and the product of the two halves of an integer N is $\lfloor N^2/4 \rfloor$, which gives the stated total at $N = n + m - 2$.

For the connectivity, fix $a \in A$ and $b \in B$. Two kinds of route run from a to b: the direct arc, and the two-step detours $a \to b' \to b$ that pass through an in-neighbour b' of b inside B. There are $m - 2$ detours, so $m - 1$ routes in all, and no two of them share an arc. They are also all there are, because the only in-arcs of b that a can reach are exactly those $m - 1$ arcs. Deleting them therefore cuts every route from a to b. Pairs inside B are limited by the in-degree $m - 2$, pairs inside A and pairs from B to A have no routes at all, so $\lambda^{\max} = m - 1$. $\square$

A.5 Structural propositions on the directed frontier

These are the proved ingredients of the reduction discussed in Chapter 3. The section ends with what the reduction still assumes, stated as Conjecture A.28, and with the one quadratic bound that assumes nothing, Proposition A.29.

Proposition A.25 (Mutually unreachable out-neighbourhoods)**.** *Let D be a simple digraph with*

$\lambda_D^{\max} \leq 1$. *For any vertex u and distinct out-neighbours v, w of u, there is no directed path from v to w in $D - u$.*

Proof. If P were a directed v–w path avoiding u, then $u \to w$ and $u \to v \xrightarrow{P} w$ would be two arc-disjoint u-to-w routes: the first uses only the arc (u, w), the second starts with (u, v) and continues along P, none of whose arcs touch u. This contradicts $\lambda_D(u, w) \leq 1$. The restriction to $D - u$ is not cosmetic, and Figure A.2 draws both the argument and the reason for the restriction: in the digraph with arcs (u, v), (v, u), (u, w) every local connectivity is 1, yet $v \to u \to w$ reaches w from v through u. $\square$

(a) If a directed $v \rightsquigarrow w$ path (red) existed in $D - u$, then $u \to w$ and $u \to v \rightsquigarrow w$ would be two arc-disjoint u–w routes, against $\lambda(u, w) \leq 1$.

(b) The counterexample $\{(u, v), (v, u), (u, w)\}$: $\lambda^{\max} = 1$, yet w *is* reachable from v only *through* u.

Figure A.2. Why Proposition A.25 must exclude paths through u. **(a)** For out-neighbours v, w of u in a digraph with $\lambda^{\max} \leq 1$, no $v \rightsquigarrow w$ path can avoid u: such a path would combine with the lone arc $u \to w$ into two arc-disjoint routes. **(b)** The restriction to $D - u$ is essential. In the three-vertex digraph $\{(u, v), (v, u), (u, w)\}$ we have $\lambda^{\max} = 1$, yet v reaches w via $v \to u \to w$, a route that reuses the vertex u, which the proposition allows.

Proposition A.26 (Disjoint second out-neighbourhoods)**.** *Let D be a simple digraph with $\lambda_D^{\max} \leq 1$, let $u \in V$, and let $v_1 \neq v_2$ be out-neighbours of u. Then $(N^+(v_1) \setminus \{u\}) \cap (N^+(v_2) \setminus \{u\}) = \emptyset$.*

Proof. A common out-neighbour $w \neq u$ would give the two arc-disjoint routes $u \to v_1 \to w$ and $u \to v_2 \to w$, contradicting $\lambda_D(u, w) \leq 1$. $\square$

Proposition A.27 (Two-hop budget on bipartite partitions, [AI])**.** *Let D be a simple digraph with $\lambda_D^{\max} \leq m - 1$. If V admits a partition $A \cup B$ with $A \neq \emptyset$ and every arc from A to B is present, then every $b \in B$ has in-degree at most $m - 2$ from inside B, and consequently the number of arcs inside B is at most $(m - 2)|B|$.*

In words: once every arc from A to B is present, a vertex of B can afford very few in-arcs from inside B. Each such arc creates a fresh two-step detour on top of the direct arc that every vertex of A already has, and the cap allows only $m - 2$ of them.

Proof. Fix $a \in A$ and $b \in B$, and let $b'_1, \ldots, b'_d$ be the in-neighbours of b inside B, where $d = \deg_B^-(b)$. The direct arc $a \to b$ together with the two-step detours $a \to b'_i \to b$ are $1 + d$ arc-disjoint a-to-b routes, all present because $A \to B$ is complete. Feasibility forces $1 + d \leq m - 1$, so $d \leq m - 2$, and summing over B bounds the internal arcs by $(m - 2)|B|$. $\square$

We now combine Proposition A.27 with the best choice of partition. Suppose every arc of D either runs from A to B or stays inside B: the partition $A \to B$ is complete, no arc lies inside A, and crucially *no arc runs from B back to A*. Write $b = |B|$, so that $|A| = n - b$. The arcs then come in exactly two kinds. The arcs crossing the partition number $|A|\,|B| = (n-b)\,b$, one for each ordered A–B pair. The arcs inside B number at most $(m-2)\,b$: since no route between two vertices of B ever needs to leave B (there is no arc back to A to leave by), Proposition A.27 applies to B on its own and caps the in-degree of each $b \in B$ from inside B at $m-2$. There are no other arcs, so, adding the two kinds,

$$|A(D)| \;\leq\; (n-b)\,b \;+\; (m-2)\,b \;=\; b\,\bigl(n+m-2-b\bigr).$$

The right-hand side is a downward parabola in b. Writing $s = n+m-2$, it is $b(s-b) = -b^2 + sb$, maximised at $b = s/2$, so over integers the best partition takes $b = \lceil s/2 \rceil$ (equivalently $\lfloor s/2 \rfloor$). That value lies in the admissible range $1 \leq b \leq n-1$, which $A \neq \emptyset$ forces, exactly when $m \leq n$, and $n \geq m$ is the range Conjecture 3.1 is stated on. The maximum value is

$$\max_b \; b\,(s-b) \;=\; \left\lfloor \frac{s^2}{4} \right\rfloor \;=\; \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor .$$

This is the quadratic branch of Conjecture 3.1, reached uniformly in m. The count is often summarised as resting on the absence of backward arcs alone, and that summary is too narrow. Four things were granted at once: that a partition $V = A \cup B$ exists at all, that every arc from A to B is present, that no arc lies inside A, and that no arc runs from B back to A. Only the last of the four is the backward-arc condition. The first three describe the shape of the extremiser, and nothing proved so far forces an arbitrary extremal digraph to have that shape. The gap is therefore a structural one:

Conjecture A.28 (The missing decomposition)**.** *Let n be large enough that the quadratic branch of Conjecture 3.1 strictly exceeds the linear one, that is, $\lfloor (n+m-2)^2/4 \rfloor > m(n-1)$. Then every extremal digraph D on n vertices with $\lambda_D^{\max} \leq m-1$ admits a partition $V(D) = A \cup B$ into two non-empty parts such that every arc from A to B is present, no arc lies inside A, and no arc runs from B to A.*

The hypothesis on n has to be there, because the obvious phrasing is too narrow. On the linear branch the extremisers are *not* just the hub. At $m = 2$, every bidirected spanning tree is feasible and has exactly $2(n-1)$ arcs, since between two vertices the only route follows the unique tree path, so the whole extremal set on that branch is the set of bidirected trees. An exhaustive classification confirms it: at $n = 5$ and $n = 6$ there are exactly 3 and 6 extremal digraphs up to isomorphism, which are precisely the numbers of trees on 5 and 6 vertices, and none of them admits the decomposition. Excluding “the hub” would leave all the other trees behind, so the hypothesis is on the size rather than on the shape.

Given Conjecture A.28 the count above closes Conjecture 3.1 for $m \geq 3$. The backward-arc clause is the part where the natural delete-and-compensate exchange fails to be monotone, which is why it is the clause usually named, but the partition itself is not free either, and Chapter 3 states the position that way. Testing the conjecture by exhaustion is harder than it looks: the crossover sits at $n = 8$ for $m = 2$ and at $n = 9$ for $m = 3$, so every size an exhaustive sweep can reach at $m = 3$ still lies on the linear branch, where the statement says nothing. That is the practical reason this conjecture has resisted a computational probe rather than a proof.

What can be proved without any of the four is the leading constant, and the argument is short enough to give in full. It also says something the conjecture does not: after deleting a lower-order number of arcs, every feasible digraph becomes a subgraph of a one-directional bipartite digraph.

Proposition A.29 (Unconditional quadratic bound and stability, [AI])**.** *Let D be a simple digraph on $n \geq 2$ vertices with $\lambda_D^{\max} \leq m - 1$. Then*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; \sqrt{m}\, n^{3/2}.$$

Moreover D can be made one-directional bipartite, meaning every remaining arc runs from one part to the other and none returns, by deleting at most $\sqrt{m}\, n^{3/2}$ arcs. Consequently, for each fixed m,

$$\ell_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; O_m\!\left(n^{3/2}\right),$$

so the leading constant $\frac{1}{4}$ of Conjecture 3.1 holds unconditionally, with no hypothesis on the structure of the extremiser.

Proof. Step 1, a budget on two-step routes. Fix an ordered pair (u, v) with $u \neq v$, and let $p_2(u, v)$ count the vertices x with both $u \to x$ and $x \to v$ present. Such an x is automatically distinct from u and from v, since D has no loops, so each one gives a genuine two-step route $u \to x \to v$. Two of these routes with different midpoints $x \neq x'$ share no arc, since the first uses (u, x) and (x, v) and the second uses (u, x') and (x', v), and neither uses the arc (u, v), which has no midpoint at all. So the direct arc, when present, together with all the two-step routes, is a family of pairwise arc-disjoint u-to-v routes, and feasibility caps its size:

$$\mathbf{1}_{(u,v)\in A(D)} + p_2(u, v) \;\leq\; \lambda_D(u, v) \;\leq\; m - 1.$$

Summing over all $n(n-1)$ ordered pairs gives $|A(D)| + \sum_{u \neq v} p_2(u, v) \leq (m - 1)\, n(n - 1)$.

Step 2, from pairs to degrees. Write Δ for the number of digons of D, meaning the pairs of vertices carrying an arc in each direction. Three small facts turn Step 1 into a statement about degrees alone.

(2a) Walks of length two, counted by their midpoint.

$$\sum_{u,\, v \in V} p_2(u, v) \;=\; \sum_{x \in V} d^-(x)\, d^+(x),$$

where the left-hand sum runs over *all* ordered pairs, the diagonal $u = v$ included. Fixing a vertex x, a walk $u \to x \to v$ is nothing more than an independent choice of an in-neighbour u of x and an out-neighbour v of x, so x is the midpoint of exactly $d^-(x)\, d^+(x)$ of them.

(2b) What the diagonal counts. A diagonal term $p_2(u,u)$ counts the closed walks $u \to x \to u$, and each digon contributes two such walks, one based at each of its endpoints. Hence

$$\sum_{u \in V} p_2(u,u) \;=\; 2\Delta.$$

(2c) Digons are few. Distinct digons use disjoint pairs of arcs, so $2\Delta \le |A(D)|$.

Splitting the all-pairs sum of (2a) along its diagonal and feeding in Step 1 now gives the inequality the rest of the argument runs on:

$$\begin{aligned}
\sum_{x \in V} d^+(x)\, d^-(x) \;&=\; \sum_{u \neq v} p_2(u,v) \;+\; 2\Delta && \text{by (2a) and (2b),}\\
&\le\; \big[(m-1)\, n(n-1) - |A(D)|\big] \;+\; 2\Delta && \text{by Step 1,}\\
&\le\; \big[(m-1)\, n(n-1) - |A(D)|\big] \;+\; |A(D)| && \text{by (2c),}\\
&=\; (m-1)\, n(n-1).
\end{aligned}$$

The last two lines absorb the digon term exactly against the slack Step 1 left behind, so nothing is given away in passing from pairs to degrees. In words, a vertex with a large in-degree *and* a large out-degree is expensive, because it manufactures two-step routes for many pairs at once, and the budget for those routes is only quadratic in n.

Step 3, the smaller half-degree is small on average. Since $\min(a,b) \le \sqrt{ab}$ for non-negative a, b, and by the Cauchy-Schwarz inequality in the form $\sum_x \sqrt{a_x} \le \sqrt{n \sum_x a_x}$,

$$\begin{aligned}
\sum_{x \in V} \min\big(d^+(x), d^-(x)\big) \;&\le\; \sum_{x \in V} \sqrt{d^+(x)\, d^-(x)} \;\le\; \sqrt{n \sum_{x \in V} d^+(x)\, d^-(x)}\\
&\le\; \sqrt{n \cdot (m-1)\, n(n-1)} \;=\; n\sqrt{(m-1)(n-1)} \;\le\; \sqrt{m}\, n^{3/2}.
\end{aligned}$$

Step 4, deleting the smaller half of every vertex. Call x *source-like* if $d^+(x) \ge d^-(x)$ and *sink-like* otherwise, and write S and T for the two classes, so $V = S \cup T$. Delete every in-arc of a source-like vertex and every out-arc of a sink-like vertex. A source-like x loses $d^-(x) = \min(d^+(x), d^-(x))$ arcs and a sink-like x loses $d^+(x) = \min(d^+(x), d^-(x))$ arcs, so the number of deleted arcs is at most $\sum_x \min(d^+(x), d^-(x))$, which Step 3 bounds by $\sqrt{m}\, n^{3/2}$. An arc (a,b) survives only if a is not sink-like and b is not source-like, that is, only if $a \in S$ and $b \in T$. So what remains runs entirely from S to T, which is the one-directional bipartite shape, and it has at most $|S|\,|T| \le \lfloor n^2/4 \rfloor$ arcs. Adding the two counts gives the bound.

The asymptotic statement follows because the augmented bipartite construction (Construction A.17) already gives $\ell_m^{\mathrm{dir}}(n) \ge \lfloor (n+m-2)^2/4 \rfloor = n^2/4 + O_m(n)$, and $n^{3/2}$ dominates n. ☐

The error term $n^{3/2}$ is larger than the $(m-2)n/2$ that Conjecture 3.1 predicts, so the proposition as it stands does not separate the conjectured formula from its near neighbours. It can be sharpened to the right order, and the extra idea is small enough to be worth giving, because it explains where the $\sqrt{n}$ was being wasted.

Step 3 is lossy exactly when every vertex carries a middling degree. The Cauchy-Schwarz step is tight only if $d^+(x)d^-(x)$ is about $(m-1)n$ at every vertex, which forces every degree down to about $\sqrt{mn}$. But a digraph with $\Theta(n^2)$ arcs has average degree $\Theta(n)$, and at a vertex of large degree the constraint $d^+(x)d^-(x) \le (m-1)n$ bites much harder: with $d^+ + d^-$ of order n, the smaller of the two must be of order m rather than $\sqrt{mn}$. The dense case and the tight case of Step 3 are therefore incompatible, and the way to exploit that is a dichotomy on the minimum degree, with vertex deletion to handle the sparse side.

Lemma A.30 (Small side of a high-degree vertex)**.** *For any vertex x of a digraph with total degree $d(x) = d^+(x) + d^-(x) > 0$,*

$$\min\bigl(d^+(x), d^-(x)\bigr) \;\le\; \frac{2\,d^+(x)\,d^-(x)}{d(x)}.$$

Proof. Write $\mu = \min(d^+(x), d^-(x))$ and $M = \max(d^+(x), d^-(x))$, so $\mu + M = d(x)$ and $\mu M = d^+(x)d^-(x)$. Since $\mu \le M$ we have $M \ge d(x)/2$, hence $d^+(x)d^-(x) = \mu M \ge \mu\, d(x)/2$, which rearranges to the claim. ☐

Theorem 1.11, stated in Chapter 1, says in words that a feasible simple digraph is never more than a linear amount denser than the balanced bipartite wall, which holds $\lfloor n^2/4 \rfloor$ arcs. The wall's quadratic term is therefore the leading term, and what remains in question is how large the linear correction is. The proof below assumes nothing about the shape of D.

Proof of Theorem 1.11. Induction on n. For $n = 2$ a simple digraph has at most 2 arcs and the right-hand side is $1 + 4(m-1) \ge 5$, so the base holds. Let $n \ge 3$ and split on the minimum total degree.

Case 1: some vertex v has $d(v) < n/2$. Since $d(v)$ is an integer, $d(v) \le \lfloor n/2 \rfloor$. By Lemma A.21 the digraph $D - v$ on $n-1$ vertices is still feasible, so the induction hypothesis applies to it and

$$|A(D)| \;\le\; \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 4(m-1)(n-2) + \left\lfloor \frac{n}{2} \right\rfloor \;=\; \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-2),$$

using Lemma A.34 for the last step, and this is below the claimed bound.

Case 2: every vertex has $d(x) \ge n/2$. By Lemma A.30 and Step 2 of Proposition A.29,

$$\sum_{x\in V} \min\bigl(d^+(x), d^-(x)\bigr) \;\le\; \sum_{x\in V} \frac{2\,d^+(x)\,d^-(x)}{d(x)} \;\le\; \frac{4}{n}\sum_{x\in V} d^+(x)\,d^-(x) \;\le\; \frac{4}{n}\,(m-1)\,n(n-1),$$

which is $4(m-1)(n-1)$. Step 4 of Proposition A.29 deletes at most that many arcs and leaves at most $\lfloor n^2/4 \rfloor$, so the bound holds in this case too.

For the two-sided asymptotic statement, Construction A.17 gives

$$\ell_m^{\mathrm{dir}}(n) \;\geq\; \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor \;=\; \frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1),$$

so the second-order term is linear in n from both sides. □

What is left open is therefore a constant factor in one coefficient, not a shape. Conjecture 3.1 says the second-order coefficient is exactly $(m-2)/2$, and the theorem pins it between that and $4(m-1)$, a factor of $8(m-1)/(m-2)$: sixteen at $m=3$, and tending to eight as m grows. Closing that gap is where the structure of Conjecture A.28 lives, and Remark A.31 rules out the most obvious way to try.

Remark A.31 (Case 2's coefficient needs a new inequality). One might hope to cut Case 2's constant using the observation that in the conjectured extremiser, one whole side has $\min(d^+, d^-) = 0$, so summing Lemma A.30's worst case at every vertex looks wasteful. That observation alone does not cut the coefficient. The value four is the best universal coefficient obtainable from the two facts used in Case 2, the budget $\sum_x d^+(x)\, d^-(x) \leq (m-1)n(n-1)$ of Step 2 in Proposition A.29 and the floor $d(x) \geq n/2$.

Think of the budget as money and of $t_x = \min(d^+(x), d^-(x))$ as what each vertex buys with it. At a vertex with $d(x) = s \geq n/2$, one also has $s \geq 2t$. Hence the cheapest possible cost for a given t is the two-branch function

$$c(t) = \min_{s \geq \max(n/2, 2t)} t(s-t) = \begin{cases} t(n/2-t), & 0 \leq t \leq n/4, \\ t^2, & t \geq n/4. \end{cases}$$

On the first branch the value per unit cost is $t/c(t) = 1/(n/2-t)$, which increases to $4/n$, whereas on the second it is $1/t$, which decreases from $4/n$. Thus $t \leq (4/n)c(t)$ for every t, with equality in the continuous relaxation at $t = n/4$. Summing and using the shared budget $Q = (m-1)n(n-1)$ gives

$$\sum_{x \in V} t_x \leq \frac{4}{n} \sum_{x \in V} d^+(x) d^-(x) \leq \frac{4Q}{n} = 4(m-1)(n-1).$$

Conversely, fix m and take n large. Assign $t = n/4$ to as many abstract vertices as the budget permits, assign one further $t \in [0, n/4]$ so that its continuous cost absorbs the remainder, and assign zero to the rest. This uses at most $16(m-1)+1 \leq n$ vertices. Since the remainder is within $O_m(n)$ of the full cost $n^2/16$, the final t is $n/4 - O_m(\sqrt{n})$. The resulting total is $4(m-1)n - O_m(\sqrt{n})$, matching the coefficient four asymptotically. Therefore no smaller universal coefficient follows from these two inequalities alone.

The crude per-vertex bound therefore loses nothing within this relaxation, and the reason is that its two sources of slack close at that very point. Lemma A.30 is an equality only when $d^+(x) = d^-(x)$, and the floor substitution $d(x) \to n/2$ is an equality only when $d(x) = n/2$. Both hold at once exactly when $d^+(x) = d^-(x) = n/4$, which is the profile the relaxed optimum concentrates on. Cutting the constant therefore needs a second and independent inequality, one that constrains interference among the handful of expensive near-balanced vertices themselves rather than a sharper reading of the one already in hand.

A.5.1 What that proof actually used

Neither the proposition nor the theorem above ever used feasibility twice. Each used it once, in Step 1, to cap a single explicit family of routes, and everything after that is arithmetic on in-degrees and out-degrees. Promoting that one cap from a deduction to a hypothesis costs nothing here and buys two further results below, one of them for a problem Chapter 3 lists as untouched by the arc case.

Definition A.32 (Two-step budget)**.** Let D be a simple digraph and let $C \geq 1$. Call D *C-budgeted* if every ordered pair (u, v) of distinct vertices satisfies

$$\mathbf{1}_{(u,v)\in A(D)} + p_2(u, v) \;\leq\; C,$$

where $p_2(u, v)$ counts, as in Proposition A.29, the vertices $x \notin \{u, v\}$ with both (u, x) and (x, v) present.

Lemma A.33 (The counting core, [AI])**.** *Every C-budgeted simple digraph D on $n \geq 2$ vertices satisfies*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4C(n-1).$$

Proof. Read the proofs of Proposition A.29 and Theorem 1.11 again with C written wherever $m-1$ stood. Step 1 of the proposition is now the hypothesis rather than a deduction, and it was the only place feasibility ever entered, since Steps 2, 3 and 4 manipulate in-degrees and out-degrees and nothing else. The theorem's induction needs one further point, that the hypothesis is inherited by induced subdigraphs, which Lemma A.21 supplied there by way of feasibility. Here it is immediate: deleting a vertex only deletes arcs and removes candidate midpoints, so neither term of the budget can rise. With those two observations the induction runs verbatim and delivers the stated bound. □

The first application is the directed *vertex* problem. Theorem 1.7 settles it at $m = 2$, and for $m \geq 3$ this thesis has been careful to record that it does not follow from the arc case: Whitney's $\kappa \leq \lambda$ makes the vertex-feasible family the larger of the two, so an arc upper bound restricts nothing there. That remains true of the *bound*. What does carry across is the route-counting, because the family Step 1 exhibits happens to be disjoint in the stronger sense as well, and once that is noticed the argument never needs the arc value at all.

Proof of Theorem 1.12. Fix an ordered pair (u, v) of distinct vertices. The direct arc (u, v), when it is present, is a route with no interior vertex at all, and a two-step route $u \to x \to v$ has interior exactly $\{x\}$, one vertex, distinct from both u and v. Distinct midpoints give disjoint interiors, and the direct arc's interior is disjoint from all of them for want of any element, so these $\mathbf{1}_{(u,v)\in A(D)} +$

$p_2(u, v)$ routes are pairwise internally vertex-disjoint and not merely arc-disjoint. Feasibility caps the family at $\kappa_D(u, v) \le m - 1$, so D is $(m - 1)$-budgeted and Lemma A.33 applies.

For the lower bound, the augmented bipartite digraph of Construction A.17 has $\lambda^{\max} \le m-1$, hence $\kappa^{\max} \le m - 1$ by Whitney, so for $n \ge m$ it witnesses $k_m^{\mathrm{dir}}(n) \ge \lfloor (n + m - 2)^2/4 \rfloor$. For fixed $m \ge 3$ this is $n^2/4 + \Theta_m(n)$, and at $m = 2$ Theorem 1.7 gives the sharper $n^2/4 + O(1)$ asymptotic. □

The vertex problem at $m \ge 3$ is therefore open in the same narrow sense as the arc problem rather than in the wide one. Its leading constant is $\frac{1}{4}$ unconditionally, the term after it is linear in n, and since $\ell_m^{\mathrm{dir}}(n) \le k_m^{\mathrm{dir}}(n)$ with both now inside $n^2/4 + \Theta_m(n)$, the two values agree to first order. What is left is the coefficient of the linear term, the same quantity Conjecture A.28 is needed for on the arc side, together with the conjecture that the two values coincide exactly. What did and did not transfer is stated precisely here. The upper bound of Theorem 1.11 did not: it is proved over the smaller family and says nothing here. The route family behind it did, and that is a statement about a construction rather than about a bound, which is why Whitney's inequality does not block it.

A.6 The directed multigraph problem, solved in full

An optimisation that scales m out of the problem entirely gives $L_m^{\mathrm{dir}}(n) = 2(n - 1)(m - 1)$ for $n \le 6$ (Theorem A.49, recorded in Section A.6.4 below), and the section above, on the simple digraph at $m = 2$, closes *every* n by repeatedly deleting one vertex of least degree and averaging. The same argument almost settles the multigraph case at every m. Dividing every multiplicity by $m-1$, as in the scaling step of Chapter 2, turns a feasible multigraph into a weighting with values in $[0, 1]$ whose pairwise maximum flows are all at most one. Deleting a vertex of least weighted degree from that weighting then drives the same averaging argument through cleanly on the linear branch, and on the quadratic branch as well whenever n is even. What breaks it is a single fractional remainder at odd n: a leftover of size $\frac{k}{2k-1}$ that an integer count could round away but a genuinely fractional weight cannot. Finding a specific vertex that is provably light enough to absorb that remainder turns out to resist every direct attempt.

This section abandons that attempt and proves the full value a different way. Instead of deleting one vertex at a time, it deletes one whole *sparse subgraph* at a time, chosen so that the deletion is guaranteed to lower every pair's connectivity by at least one in a single stroke. That change removes the fractional step altogether, along with the restriction to a particular m or a particular parity of n.

Lemma A.34 (Arithmetic identity)**.** *For all* $n \ge 2$, $\left\lfloor \frac{(n-1)^2}{4} \right\rfloor + \lfloor n/2 \rfloor = \left\lfloor \frac{n^2}{4} \right\rfloor$.

Proof. For $n = 2k$ the left side is $k(k - 1) + k = k^2$, and for $n = 2k + 1$ it is $k^2 + k = k(k + 1)$. Both match the right side. □

A.6.1 The lower bound: thickening

The lower bound rests on one observation: multiplying every arc's multiplicity by a fixed integer multiplies every cut's capacity, and hence every pairwise connectivity, by that same integer.

Lemma A.35 (Thickening)**.** *Let D_0 be a directed multigraph and $k \geq 1$ an integer. Write $k \cdot D_0$ for the multigraph obtained by multiplying every multiplicity $\mu(u, v)$ by k. Then $\lambda_{k \cdot D_0}(u, v) = k \cdot \lambda_{D_0}(u, v)$ for every ordered pair $u \neq v$, and $|A(k \cdot D_0)| = k\,|A(D_0)|$.*

Proof. By Theorem A.1, $\lambda_{D_0}(u, v)$ equals the minimum capacity, over all u–v cuts, of the total multiplicity crossing that cut. Multiplying every multiplicity by k multiplies the capacity of *every* cut by k at once, so it multiplies the minimum over all cuts by k as well, which is precisely $\lambda_{k \cdot D_0}(u, v) = k \cdot \lambda_{D_0}(u, v)$. The arc count scales by k because every one of its summands, each multiplicity, does. □

Construction A.36 (Thickened tree and thickened bipartite digraph)**.** For $m \geq 2$, take any spanning tree T of K_n and give every edge multiplicity $m - 1$ in each direction. Between two vertices, T offers only its one tree path, so this is $m-1$ copies of the $m = 2$ bidirected tree of Theorem 1.6, feasible at $\lambda^{\max} = 1$ there, and Lemma A.35 with $k = m - 1$ gives $\lambda^{\max} = m - 1$ here and an arc count of $2(n-1)(m-1)$. Alternatively, take the one-directional complete bipartite digraph of Construction A.16 (balanced parts A, B with $|A| = \lfloor n/2 \rfloor$, $|B| = \lceil n/2 \rceil$, feasible at $\lambda^{\max} = 1$) and give every arc multiplicity $m - 1$: the same lemma gives $\lambda^{\max} = m - 1$ and $(m-1)\left\lfloor n^2/4 \right\rfloor$ arcs.

Taking whichever of the two constructions is larger gives $L_m^{\mathrm{dir}}(n) \geq (m-1)M(n)$ for every n and m, which is the lower bound half of Theorem 1.13. The rest of this section proves the matching upper bound.

A.6.2 The upper bound: a reachability skeleton

Consider first what a single vertex deletion gives at $m = 2$: removing a vertex can only remove routes passing through it, so no pair's connectivity ever rises, and the $m = 2$ induction rests on that guarantee together with an averaging bound on how much one well-chosen vertex carries. The multigraph problem at a general m needs those same two ingredients: a deletion that is guaranteed to lower every connectivity, paired with a bound on how much it removes. There, though, the natural analogue of "one vertex" breaks down. Once multiplicities are free to sit anywhere between 0 and $m - 1$, no single vertex is guaranteed to carry a controllable share of the arcs, which is exactly the fractional remainder above. The fix is to stop insisting on a vertex and delete a whole *subgraph* instead: one general enough to always exist, sparse enough to bound, and specific enough that removing it is guaranteed to lower every pairwise connectivity by at least one.

Definition A.37 (Reachability skeleton)**.** Let D be a directed multigraph. A spanning subgraph $R \subseteq D$, using at most one copy of each arc it uses (no parallel copies, even if D has them), is a *reachability skeleton* of D if, for every pair of vertices $u \neq v$, u reaches v by some directed path in R exactly when u reaches v by some directed path in D.

A reachability skeleton keeps every yes-or-no fact about who can reach whom and discards everything else, in particular every parallel copy of every arc. It is a caricature of D: thin, but faithful on the one question connectivity actually depends on. The next two lemmas turn that faithfulness into the whole induction. The first says a reachability skeleton always exists and is never too big. The second says deleting it always lowers every connectivity by at least one.

Lemma A.38 (A sparse reachability skeleton always exists, [AI]). *Every loopless directed multigraph D on $n \geq 2$ vertices has a reachability skeleton R with $|A(R)| \leq M(n)$.*

Proof. Building R takes three steps: handle the traffic *inside* each strongly connected piece of D, handle the traffic *between* pieces, then count. A *strongly connected component* (an SCC) is a maximal set of vertices any two of which reach each other, and every directed multigraph partitions uniquely into its SCCs, since mutual reachability is an equivalence relation. Figure A.3 carries one small running example through all three steps, and every object named below appears in it concretely.

Step 1: inside each component. Fix one SCC of order $n_i \geq 1$ and pick any vertex r inside it as a root. Since r reaches every vertex of the component, a standard fact about digraphs gives a spanning *out-arborescence*: a tree of $n_i - 1$ arcs of D, every one pointing away from r, along which r alone reaches every other vertex of the component (grow it one hop at a time, always available since the component is strongly connected). Symmetrically, because every vertex of the component reaches r, there is a spanning *in-arborescence* of $n_i - 1$ arcs, every one pointing toward r, along which every vertex reaches r. Taking both trees together, any two vertices u, v of the component are joined by the route $u \to r \to v$, first the in-arborescence, then the out-arborescence, so the $2(n_i - 1)$ arcs of the two trees alone reconstruct *every* reachability relation inside this one component.

Step 2: between components. Contract each SCC to a single point and, between two contracted points, keep one arc whenever D has at least one arc from the first component to the second (dropping any further parallel arcs between the same two components at this stage). Call the result Q_0, a digraph on q points, one per component. Two different SCCs can never reach each other in both directions, since that would make them one strongly connected component after all, so Q_0 has no directed cycle: it is a DAG (a directed acyclic graph). Reachability between two different components of D is, by this very construction, exactly reachability between the two corresponding points of Q_0. Thin Q_0 down to an *inclusion-minimal* spanning subdigraph $Q \subseteq Q_0$ that still has the same reachability relation, deleting one arc at a time for as long as some arc is redundant, which must stop since Q_0 has only finitely many arcs.

The reason to thin Q_0 down to Q is that Q's *underlying undirected graph*, meaning Q with the arrows erased, has no triangle. Suppose three points x, y, z of Q were pairwise joined by some arc of Q. Since $Q \subseteq Q_0$ and Q_0 has no cycle, Q has none either, so between any two of x, y, z the arc runs in only one direction, never both. Three points pairwise joined with a consistent direction on each pair admit only one pattern, up to relabelling: a linear order, say $x \to y$, $y \to z$, and $x \to z$. But then $x \to z$ is redundant, since x already reaches z via y, so deleting it would leave the reachability relation of Q unchanged, contradicting that Q is inclusion-minimal. So no three points of Q can be pairwise joined at all: that is triangle-freeness of the underlying graph.

A triangle-free graph on q vertices has at most $\lfloor q^2/4 \rfloor$ edges, by Mantel's theorem [21], the $r = 2$ case of the more general theorem of Turán [22]. Applied to Q,

$$|A(Q)| \;\le\; \left\lfloor \frac{q^2}{4} \right\rfloor.$$

Step 3: assembling R and counting. Build R from the two arborescences of Step 1 inside every SCC, together with one witness arc of D for every arc of Q (some arc of D realises it, by the very definition of Q_0). By Step 1, every pair inside one component reaches each other in R exactly as in D. By Step 2, and because within-component reachability is already secured, every pair in two different components reaches each other in R exactly as in D too: to travel from u to v across the components Q's path visits, reach each witness arc's tail inside its own component (Step 1), cross the witness arc, and repeat. Since $R \subseteq D$ throughout, R can never certify a reachability D does not already have, so R is a genuine reachability skeleton. Writing $n_1, \ldots, n_q$ for the SCC orders, so $\sum_i n_i = n$, its arc count is at most the $2(n_i - 1)$ arborescence arcs of each component plus one witness per arc of Q. The first count is an upper bound rather than an equality, since the in- and out-arborescence of one component may share an arc, which only helps. So

$$|A(R)| \;\le\; \sum_{i=1}^{q} 2(n_i - 1) \;+\; |A(Q)| \;\le\; 2(n-q) \;+\; \left\lfloor \frac{q^2}{4} \right\rfloor \;=:\; f(q).$$

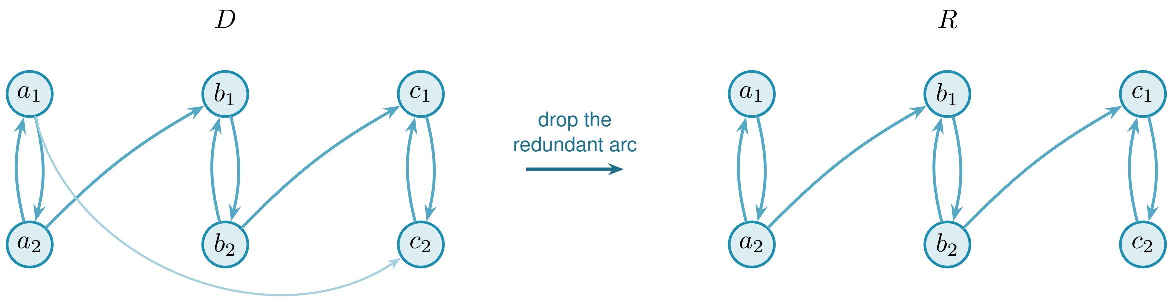
It remains to bound $f(q)$ over the range $1 \le q \le n$ that the number of components can actually take. Its increments are

$$f(q+1) - f(q) \;=\; -2 \;+\; \left\lfloor \frac{q+1}{2} \right\rfloor,$$

using the identity $\lfloor (q+1)^2/4 \rfloor - \lfloor q^2/4 \rfloor \;=\; \lfloor (q+1)/2 \rfloor$, checked on the two parities of q (at $q = 2j$ both sides are j, at $q = 2j+1$ both are $j+1$). Since $\lfloor (q+1)/2 \rfloor$ never decreases as q grows, neither do the increments of f, which is to say f is convex. A convex function on an interval attains its largest value at one of the two ends, because a value at an interior point is a weighted average of the two ends or less. Hence

$$f(q) \;\le\; \max\bigl(f(1),\, f(n)\bigr) \qquad \text{for } 1 \le q \le n.$$

At $q = 1$ the whole digraph is one component, so $f(1) = 2(n-1) + 0$. At $q = n$ every component is a single vertex, so $f(n) = 0 + \lfloor n^2/4 \rfloor$. Hence $f(q) \le \max\bigl(2(n-1), \lfloor n^2/4 \rfloor\bigr) = M(n)$ for every q in range, which is the lemma. □



redundant: a_1 already reaches c_2 via a_2, b_1, b_2, c_1

Figure A.3. A small instance of Lemma A.38. In D (left) three bidirected pairs $A = \{a_1, a_2\}$, $B = \{b_1, b_2\}$, $C = \{c_1, c_2\}$ are the strongly connected components, joined by $a_2 \to b_1$, $b_2 \to c_1$, and $a_1 \to c_2$. Contracting each pair to a point gives a condensation Q_0 on three points, carrying all three arcs $A \to B$, $B \to C$ and $A \to C$. That is exactly the transitively-closed triangle the proof rules out of the thinned Q. The arc $A \to C$ is redundant, since A already reaches C through B, so the transitive reduction Q keeps only $A \to B$ and $B \to C$. The underlying graph of Q is then a path on three points, triangle-free as claimed. The reachability skeleton R (right) reflects exactly this: every SCC keeps both of its internal arcs (an in- and an out-arborescence on two vertices is just the pair itself), while of the three component-crossing arcs of D only the two witnesses of Q's two arcs survive. The direct arc $a_1 \to c_2$ is dropped, yet a_1 still reaches c_2 in R: first $a_1 \to a_2$, then $a_2 \to b_1$, then $b_1 \to b_2$, then $b_2 \to c_1$, then $c_1 \to c_2$.

Lemma A.39 (Deleting a skeleton lowers every connectivity by one, [AI]). *Let D be a directed multigraph with $\lambda^{\max}(D) \le k$ for some $k \ge 1$, and let R be a reachability skeleton of D. Write $D - R$ for the multigraph obtained by deleting, from each multiplicity $\mu(u, v)$, the one copy (if any) that R uses on that pair. Then $\lambda^{\max}(D - R) \le k - 1$.*

In words: subtracting one copy of each skeleton arc costs every reachable pair at least one route (a single pair can lose more, if some of its own routes happened to run through the deleted arcs too). The skeleton preserves reachability, so however many arc-disjoint routes survive the deletion, one further route can always be run entirely inside the arcs that were deleted, and that extra route is what the original cap has to pay for.

Proof. Fix an ordered pair $u \ne v$ and suppose $D - R$ carries $t \ge 1$ pairwise arc-disjoint u–v paths. These use arcs of $D - R$ and hence of D, since $D - R \subseteq D$, so u reaches v in D. Because R is a reachability skeleton, u then also reaches v in R: some directed path P from u to v uses only arcs of R. Every arc of R has been subtracted out of $D - R$, so P shares no arc with any of the t paths already found inside $D - R$. Together, P and those t paths are $t + 1$ pairwise arc-disjoint u–v paths in D, so $t + 1 \le \lambda_D(u, v) \le k$, that is, $t \le k - 1$. Pairs with $t = 0$ trivially satisfy $t \le k - 1$ once $k \ge 1$, so this holds for every pair, giving $\lambda^{\max}(D - R) \le k - 1$. ☐

Proof of Theorem 1.13. The lower bound is Construction A.36. For the upper bound, fix n and prove by induction on $k = m - 1 \ge 0$ that every directed multigraph D on n vertices with $\lambda^{\max}(D) \le k$ has $|A(D)| \le k\, M(n)$. At $k = 0$, every pair has $\lambda_D(u, v) \le 0$, so no arc (u, v) can even be present, since a single arc is already one route, and $|A(D)| = 0 = 0 \cdot M(n)$.

For $k \ge 1$, take D with $\lambda^{\max}(D) \le k$ and let R be a reachability skeleton, which exists with $|A(R)| \le M(n)$ by Lemma A.38. By Lemma A.39, $D - R$ has $\lambda^{\max}(D - R) \le k - 1$, and it is again a directed multigraph on the same n vertices, so the induction hypothesis gives $|A(D - R)| \le (k-1)M(n)$. Since R uses at most one copy of each arc it touches, the arcs of D split exactly into those of R and those left in $D - R$, so

$$|A(D)| \;=\; |A(R)| \;+\; |A(D-R)| \;\le\; M(n) \;+\; (k-1)M(n) \;=\; k\,M(n).$$

Taking $k = m - 1$ gives $L_m^{\mathrm{dir}}(n) \le (m-1)M(n)$, matching the lower bound. $\square$

Nothing above depended on m or on the parity of n: the same three-step construction of R and the same one-line deletion bound run identically at every level. In particular, the induction closes exactly the finite computation the rest of this appendix once needed help to settle.

Corollary A.40 ([AI]). $L_3^{\mathrm{dir}}(7) = 24$.

Proof. $M(7) = \max(12, 12) = 12$, so Theorem 1.13 at $m = 3$, $n = 7$ gives $L_3^{\mathrm{dir}}(7) = 2 \cdot 12 = 24$. Concretely, the induction peels two reachability skeletons off any 7-vertex witness in turn, each of at most $M(7) = 12$ arcs by Lemma A.38, first lowering $\lambda^{\max}$ from 2 to 1 and then from 1 to 0: $24 = 12 + 12$ spends the whole budget twice over, with nothing left for a third pass. This is a proof by hand of the statement Chapters 2 and 3 once recorded as an outstanding computation. That computation was the mixed-integer program `prove_integral_arc_bound(7, 3, 25)`, whose `INFEASIBLE` verdict would have settled the same fact, and which never returned one: it reached its time limit on the bundled CBC solver, and a fourteen-hour uncapped enumeration was stopped without a verdict. It was never run on a stronger solver, and it no longer needs to be. The machine route to this fact was abandoned unfinished and the hand proof above replaced it, rather than the two confirming each other. $\square$

Corollary A.41 (The fractional relaxation is integral). *For every $n \ge 2$, the m-free optimum $M^*(n)$ of Chapter 2 equals $M(n)$, and it is attained by a $\{0,1\}$ weight matrix. Consequently the scaling identity*

$$L_m^{\mathrm{dir}}(n) \;=\; (m-1)\,M^*(n)$$

holds for every n and every $m \ge 2$, with no hypothesis on the optimum.

In words: allowing fractional weights does not raise the optimum. The relaxed optimum is already attained by a multigraph with weights 0 and 1, so the scaling identity that carries one computed number to every m at once needs no integrality hypothesis attached to it.

Proof. For $M^*(n) \ge M(n)$, both constructions of Construction A.36 at multiplicity one are $\{0,1\}$ matrices with all pairwise maximum flows equal to one, of total weight $2(n-1)$ and $\lfloor n^2/4 \rfloor$ respectively.

For the reverse, note first that the optimum is attained at a *rational* matrix. A weight matrix is feasible exactly when, for each ordered pair, some cut separating that pair has capacity at most one, so the feasible region is the union, over the finitely many ways of choosing one cut per pair, of the bounded rational polytopes cut out by those choices together with $0 \le w \le 1$. A linear objective over a bounded rational polytope attains its maximum at a rational vertex, and $M^*(n)$ is the largest of finitely many such maxima.

So let w be a feasible rational optimum and let q be a common denominator of its entries. Then $q\,w$ has integer entries, and multiplying every weight by q multiplies every cut capacity by q and hence every maximum flow by q, exactly as in Lemma A.35. So $q\,w$ is a directed multigraph with $\lambda^{\max} \le q$, that is, feasible at $m-1=q$, and Theorem 1.13 bounds its arc count by $q\,M(n)$. Dividing by q gives $M^*(n) = \sum_{u \ne v} w(u,v) \le M(n)$.

The identity follows because Theorem 1.13 gives $L^{\mathrm{dir}}_m(n) = (m-1)M(n)$. □

This settles the integrality question Chapter 2 raises when it introduces $M^*(n)$, namely whether the heaviest fractional weight matrix is always a $\{0,1\}$ one. It is, and the argument runs backwards from the usual direction. Nothing here inspects the fractional problem at all. An integral theorem, proved for every m at once, is strong enough to be applied to the scaled-up matrix $q\,w$, and the fractional statement falls out of the integral one rather than the other way round. That is only possible because Theorem 1.13 holds at every m, however large, so no denominator is out of reach.

Corollary A.42 (The $m=2$ case collapses to the simple digraph)**.** *At $m=2$ (so $k=1$), Theorem 1.13 reads $L^{\mathrm{dir}}_2(n) = M(n)$, matching Theorem 1.6 exactly, and this is not a coincidence. A directed multigraph with $\lambda^{\max} \le 1$ can carry no parallel arcs at all, since two parallel copies of one arc are already two arc-disjoint routes between their endpoints, so at $m=2$ the multigraph problem and the simple digraph problem are literally the same problem. Theorem 1.13 recovers Theorem 1.6 as its $k=1$ case, rather than needing it as a separate input.*

Theorem 1.13 settles the value at every n and m, but a value is not a full description of the extremal digraphs, and on the linear branch a companion fact about *which* multigraphs attain it follows from the same argument. Call a directed multigraph *everywhere-saturated* at level m when every ordered pair of distinct vertices has local connectivity exactly $m-1$. The doubled bidirected spanning trees at multiplicity $m-1$ form an extremal family on the linear branch and are everywhere-saturated: the $m-1$ routes between two vertices run along the doubled tree path, and cutting one doubled edge of that path is a cut of $m-1$ arcs.

Lemma A.43 (Saturated attachment, [AI])**.** *Let $m \ge 2$ and let D_0 be a directed multigraph on at least two vertices with $\lambda_{D_0}(x,y) = m-1$ for every ordered pair of distinct vertices. Let D be obtained from D_0 by adding one new vertex v together with any multiset of arcs incident to v. If $\lambda^{\max}_D \le m-1$, then $d(v) \le 2(m-1)$. If $d(v) = 2(m-1)$, then every arc at v joins v to one single vertex u, with $\mu(v,u) = \mu(u,v) = m-1$, and D is again everywhere-saturated.*

Proof. No mixed pair. Suppose v has an in-arc from u and an out-arc to w with $u \ne w$. By Menger and saturation D_0 carries $m-1$ pairwise arc-disjoint u-to-w routes, and the route $u \to v \to w$ uses only arcs incident to v, so it is arc-disjoint from all of them and $\lambda_D(u,w) \ge m$, a contradiction. Hence v is a pure source, a pure sink, or every arc at v touches one single vertex u.

A pure source has $d(v) \le m-1$*.* Let $t = \sum_x \mu(v,x)$ and fix u with $\mu(v,u) \ge 1$. Any cut $(S,\bar S)$ with $v \in S$ and $u \in \bar S$ has capacity $\sum_{x\in\bar S}\mu(v,x) + e_{D_0}(S\setminus\{v\},\bar S)$. There are two kinds. If $S=\{v\}$ the capacity is exactly t. If S also contains a vertex x of D_0, then $S\setminus\{v\}$ and $\bar S$ separate x from u inside D_0, so the second term is at least $\lambda_{D_0}(x,u) = m-1$, while the first is at least $\mu(v,u)$. Every cut therefore has capacity at least $\min\bigl(t,\ \mu(v,u)+m-1\bigr)$, and by max-flow min-cut so does $\lambda_D(v,u)$, which feasibility caps at $m-1$. Since $\mu(v,u) \ge 1$, the second argument of that minimum exceeds $m-1$, so the minimum is t and $t \le m-1$.

A pure sink has $d(v) \le m-1$*.* Reverse every arc: the reverse of an everywhere-saturated multigraph is everywhere-saturated, a pure sink becomes a pure source, and the previous step applies.

A single partner. If every arc at v touches one vertex u, then parallel arcs are already that many arc-disjoint one-step routes, so $\mu(v,u) \le m-1$ and $\mu(u,v) \le m-1$, giving $d(v) \le 2(m-1)$.

Equality. Since $2(m-1) > m-1$, a degree of $2(m-1)$ rules out the pure cases, so v has a single partner at full multiplicity both ways. The result is feasible and again everywhere-saturated: a route through v must enter and leave through u, so it adds nothing to any flow between old vertices. For every old $x \ne u$, $\lambda_D(x,v) = \min\bigl(\lambda_{D_0}(x,u), \mu(u,v)\bigr) = m-1$. For $x=u$, the $m-1$ parallel arcs $u \to v$ give the same conclusion directly. Symmetrically, $\lambda_D(v,x) = m-1$ for every old x. $\square$

Remark A.44 (The linear branch grows one leaf at a time)*.* The equality case of Lemma A.43 attaches a new leaf to an arbitrary vertex at full multiplicity, and that is the only attachment reaching $2(m-1)$. Iterating from a single doubled edge, itself everywhere-saturated on two vertices, generates all doubled bidirected spanning trees. Thus this extremal family is closed under maximal attachment and grows one leaf at a time. The lemma does not assert that every linear-branch extremiser admits such a leaf-deletion sequence. The program confirms the lemma exhaustively at $m=3$: over every doubled tree on five and six vertices and every one of the 3^{2n} possible attachment patterns, the densest feasible attachment has degree exactly $4 = 2(m-1)$, and the degree-4 patterns are precisely the single-partner attachments at full multiplicity, one per choice of partner.

Remark A.45 (The degree-capped 24-arc extremisers at $n=7$)*.* Lemma A.43 proves rigidity for maximal one-vertex attachments to an everywhere-saturated graph, but it does not classify every extremiser on the linear branch. At $n=7$, $m=3$, the two branches tie at 24 arcs, and every doubled bidirected tree is already an extremiser, giving eleven tree isomorphism classes before the bipartite examples are counted. The separate machine run discussed here answered only a degree-capped question relevant to extension arguments. The tool was the sound generation enumerator of Chapter 2, in the call `enumerate_extremal_directed_multigraphs_via_generation(7, 3, 24, max_degree=8)`, which prunes only with proved bounds and is therefore complete within the imposed maximum-total-degree cap. It ran for about seven hours across ten processor cores and returned exactly three isomorphism classes satisfying that cap. Two of them are the bipartite shapes $2\cdot B_{3,4}$ and $2\cdot B_{4,3}$. The third is the doubled bidirected path P_7, the one non-star tree whose maximum degree (2) is compatible with an 8-regular graph one vertex up. Each of the three was re-verified by the exact checker. This is a restricted classification, not a classification of all 24-arc extremisers. The reachability-skeleton induction proves the value 24 without it, and the unrestricted extremal family at this tie size is substantially larger.

A.6.3 Extremal classification on the quadratic branch

The classification above is one finite data point, and the linear branch is settled by Lemma A.43, so what is missing is the quadratic branch at a general n. The reachability-skeleton proof looks unpromising for this, since it is designed never to need to know which multigraph it is taking apart. But that indifference is only in the direction of the *bound*. Run backwards, from the assumption that the bound is met exactly, the same construction is unexpectedly rigid: it pins down the number of strongly connected components, then the shape of the skeleton, and finally the multigraph.

Two consequences of equality do the work, and both come from reading the count of Lemma A.38 as an equation rather than an inequality.

Lemma A.46 (What equality forces, [AI])**.** *Let $n \geq 8$, so that $\lfloor n^2/4 \rfloor > 2(n-1)$ and $M(n) = \lfloor n^2/4 \rfloor$, and let D be a directed multigraph on n vertices with $\lambda_D^{\max} \leq m-1$ and $|A(D)| = (m-1)\lfloor n^2/4 \rfloor$. Then*

(1) D is acyclic, and

(2) D has a reachability skeleton R, inclusion-minimal for reachability in the sense that no arc of it is redundant, whose underlying undirected graph is exactly the balanced complete bipartite graph $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$, so R is an acyclic orientation of it.

Proof. The proof of Theorem 1.13 peels reachability skeletons $R_1, \ldots, R_{m-1}$, each deletion dropping every pairwise connectivity by at least one, so that after $m-1$ peels no arc survives and $|A(D)| = \sum_i |A(R_i)|$. Each $|A(R_i)| \leq M(n)$, so equality in the hypothesis forces $|A(R_i)| = M(n)$ for every i, and in particular for $R := R_1$.

Now read the count of Lemma A.38 backwards. It gives $|A(R)| \leq f(q) := 2(n-q) + \lfloor q^2/4 \rfloor$, where q is the number of strongly connected components of D, so $f(q) \geq \lfloor n^2/4 \rfloor = f(n)$. The increments $f(q+1) - f(q) = -2 + \lfloor (q+1)/2 \rfloor$ are non-decreasing, so f is convex on $q \in \{1, \ldots, n\}$. Write $d_i := f(i+1) - f(i)$. Suppose toward a contradiction that $q < n$. If $q = 1$ then $f(q) = f(1) = 2(n-1)$, and the hypothesis $n \geq 8$ gives $\lfloor n^2/4 \rfloor > 2(n-1)$, so $f(q) < f(n)$, contradicting $f(q) \geq f(n)$. So $1 < q < n$. Since $f(1) < f(n) \leq f(q)$, the sum $d_1 + \cdots + d_{q-1} = f(q) - f(1)$ is positive, and since the d_i are non-decreasing this forces the last term d_{q-1} to be positive too (otherwise every term up to it would be at most $d_{q-1} \leq 0$ and the sum could not be positive). Non-decreasing again gives $d_q, \ldots, d_{n-1} \geq d_{q-1} > 0$, so $f(n) - f(q) = d_q + \cdots + d_{n-1} > 0$, that is $f(n) > f(q)$, contradicting $f(q) \geq f(n)$. Both cases are impossible, so $q = n$: every strongly connected component is a single vertex, which is (1).

With $q = n$ the two arborescences of Step 1 of that proof are empty, so R consists of the witness arcs of Q alone and $|A(Q)| = \lfloor n^2/4 \rfloor$. The underlying undirected graph of Q is triangle-free, proved there from the inclusion-minimality of Q. A triangle-free graph on n vertices with $\lfloor n^2/4 \rfloor$ edges meets Mantel's bound [21] with equality, and the extremal graph for Mantel's theorem is unique: it is $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$. That is (2), and R is acyclic because D is. Its inclusion-minimality is the minimality of Q, the two objects coinciding once $q = n$ leaves the arborescences empty, and it is what Lemma A.47 uses next. □

The next lemma is where the skeleton's minimality, which had only been used to make it small, is used to make it *shallow*. It costs nothing and it is what makes the theorem work.

Lemma A.47 (A minimal bipartite skeleton has no long path, [AI])**.** *Let R be as in Lemma A.46, with parts A and B. Then R contains no directed path with three arcs.*

Proof. Suppose $u \to v \to w \to x$ were such a path. A path in a bipartite graph alternates sides, so u and x lie on opposite sides, and since R's underlying graph is the *complete* bipartite graph, R contains an arc between them. It cannot be $x \to u$, which would close a directed cycle against the path, and D is acyclic. So $u \to x$ lies in R. But then deleting $u \to x$ from R changes no reachability, since u still reaches x along the path, contradicting the inclusion-minimality of Q. □

Theorem A.48 (Extremal classification on the quadratic branch, [AI])**.** *Let $m \geq 2$ and $n \geq \max(8, 2m)$, so that $M(n) = \lfloor n^2/4 \rfloor$ and n is on the quadratic branch. If D is a directed multigraph on n vertices with $\lambda_D^{\max} \leq m - 1$ and $|A(D)| = L_m^{\mathrm{dir}}(n) = (m-1) \lfloor n^2/4 \rfloor$, then*

$$D \in \left\{ (m-1) \cdot B_{\lceil n/2 \rceil,\, \lfloor n/2 \rfloor},\ (m-1) \cdot B_{\lfloor n/2 \rfloor,\, \lceil n/2 \rceil} \right\},$$

where $B_{a,b}$ has all arcs from its part of size a to its part of size b, and every displayed arc has multiplicity exactly $m - 1$. Thus there is one isomorphism class when n is even and two when n is odd, interchanged by reversing every arc.

In words: past $n = \max(8, 2m)$ the underlying bipartite shape is forced. Split the vertices as evenly as possible, choose either part as the source, run every arc from it to the other part, and repeat each arc $m - 1$ times. When n is odd the two choices give two non-isomorphic digraphs, because the source has different size, and they are reversals of one another.

Proof. Take R, A and B from Lemma A.46, and write $\alpha = |A| \geq \beta = |B| = \lfloor n/2 \rfloor$.

Step 1: R has no directed path with two arcs. Suppose $a \to b \to a'$ with $a, a' \in A$ and $b \in B$. If a had an incoming arc $b_0 \to a$ in R, then $b_0 \to a \to b \to a'$ would be a path with three arcs, which Lemma A.47 forbids. So a has no incoming arc in R, and since R's underlying graph is complete bipartite, $a \to b''$ for *every* $b'' \in B$. Symmetrically a' has no outgoing arc, so $b'' \to a'$ for every $b'' \in B$. The β routes $a \to b'' \to a'$, one through each $b'' \in B$, use pairwise distinct arcs, and each of their arcs is present in D because $R \subseteq D$. Hence $\lambda_D(a, a') \geq \beta = \lfloor n/2 \rfloor \geq m$, against feasibility. The case of a path $b \to a \to b'$ with middle vertex in A is identical with the sides exchanged and gives $\lambda_D(b, b') \geq \alpha \geq \beta \geq m$, equally impossible. This is the one step that uses $n \geq 2m$.

Step 2: R is one-directional. By Step 1 no vertex of R has both an incoming and an outgoing arc, so every vertex is a source or a sink. Fix $a \in A$. It is adjacent in R to every vertex of B, so it is not isolated. If a is a source then every $b \in B$ has an incoming arc, hence is a sink, hence has no outgoing arc, so every arc at every b runs into it: all arcs of R run from A to B. If instead a is a sink the same argument gives all arcs from B to A. Thus R is either $B_{\alpha,\beta}$ or $B_{\beta,\alpha}$, according to which part is the source. The remaining steps apply identically to either orientation, so from now on use A for the source part and B for the sink part without imposing an order on their sizes.

Step 3: D has no arc outside R. R has the same reachability as D, and in a one-directional bipartite digraph the only ordered pairs joined by a directed path are the pairs $(a, b) \in A \times B$, each by the single arc $a \to b$. So if D had an arc (u, x), then u reaches x in R, forcing $(u, x) \in A \times B$, and R already contains that arc. Hence D and R use exactly the same $\alpha\beta = \lfloor n^2/4 \rfloor$ ordered pairs.

Step 4: every multiplicity is $m - 1$. A pair joined by $\mu(a, b)$ parallel arcs has $\lambda_D(a, b) \geq \mu(a, b)$, so feasibility gives $\mu(a, b) \leq m - 1$ on each of the $\lfloor n^2/4 \rfloor$ pairs. Their total is $|A(D)| = (m - 1)\lfloor n^2/4 \rfloor$, so every one of them attains the cap. Hence D is the multiplicity-$(m - 1)$ one-directional complete bipartite digraph on these two parts, and their product $\lfloor n^2/4 \rfloor$ forces their sizes to be $\lceil n/2 \rceil$ and $\lfloor n/2 \rfloor$. Either size may be the source size, giving exactly the two displayed possibilities (which coincide up to isomorphism when n is even). $\square$

Three remarks. First, on where the hypothesis $n \geq 2m$ is used and what it costs. It enters once, in Step 1, where a two-arc path in the skeleton is converted into $\lfloor n/2 \rfloor$ arc-disjoint routes between one pair, and feasibility kills that only when $\lfloor n/2 \rfloor$ reaches m. For $n < 2m$ the theorem is not proved, and it is not known to fail either. Nothing in the argument suggests a boundary there: the step is a lower bound on one pair's connectivity, and losing it removes a contradiction rather than supplying a counterexample. Second, a degree fact falls out of the same equality, and it is what makes a direct search for extremisers finite in practice. Deleting a vertex leaves at most $(m - 1)M(n - 1)$ arcs, so every vertex of an extremiser carries degree at least $(m - 1)\lfloor n/2 \rfloor$. For even n the degree sum $2(m - 1)\lfloor n^2/4 \rfloor = n \cdot (m - 1)(n/2)$ leaves no slack at all, so every vertex has degree exactly $(m - 1)n/2$: an extremiser at even n is regular. Third, on the shape of the argument. Theorem 1.13 is indifferent to which multigraph it dismantles, and Chapter 3 records that indifference as the reason the value says nothing about uniqueness. That is true of the proof read forwards. Read backwards from equality, the same skeleton is a rigid object, and the two facts that had been mere bookkeeping, that f is maximised at one end and that a minimal skeleton is triangle-free, become a determination of q and an application of the equality case of Mantel's theorem.

A.6.4 The historical m-free model

Chapter 2 reduces the directed multigraph problem at every m to one m-free optimisation, $M^*(n)$, by dividing multiplicities by $m - 1$. This subsection writes out the model that computed it, the theorem it settled and the record of what was run. Corollary A.41 above has since replaced it, so nothing below is load-bearing.

By max-flow/min-cut, $\mathrm{maxflow}(s,t) \le 1$ exactly when some s–t cut has capacity at most one. The optimisation therefore chooses one cut per ordered pair. A binary label x_u^{st} records its side, and p_{uv}^{st} counts $w(u,v)$ only when $u \to v$ crosses from source to sink (Figure A.4). This describes the exact feasible region. The archived wrapper retained an `OPTIMAL` status and primal witness but no independently replayable matching bound, so the runs are finite solver checks rather than formal certificates.

With scaled weights w, binary side labels x, and non-negative crossing weights p, the model is

$$
\begin{array}{lll}
\text{maximise} & \displaystyle\sum_{u\neq v} w(u,v) & \\
\text{subject to} & x_s^{st} = 1, \quad x_t^{st} = 0 & \text{for all } (s,t), \\
& p_{uv}^{st} \ge w(u,v) + x_u^{st} - x_v^{st} - 1 & \text{for all } (s,t),\ (u,v), \\
& \displaystyle\sum_{u\neq v} p_{uv}^{st} \le 1 & \text{for all } (s,t), \\
& w(s,t) + \displaystyle\sum_{x} \min\bigl(w(s,x), w(x,t)\bigr) \le 1 & \text{for all } (s,t), \\
& d(v_0) \le d(v_1) \le \cdots \le d(v_{n-1}), & \text{with } d(v) = \sum_{u\neq v}\bigl(w(u,v)+w(v,u)\bigr), \\
& w \in [0,1], \quad x \in \{0,1\}. &
\end{array}
$$

The first line is what makes the labelling an s–t cut rather than an arbitrary bipartition, and it is not optional. Without it the all-zero labelling $x \equiv 0$ is admissible, no ordered pair (u,v) ever crosses, every p_{uv}^{st} rests at 0, and the cut constraint below is satisfied by every weight matrix whatsoever. Pinning the source side of s and the sink side of t removes that escape and leaves the remaining $n-2$ labels free, which is precisely the choice of an s–t cut. The implementation supplies these two values as constants rather than as variables (`program/erdos915_unified.py`, in `_cut_counting_model`), which is the same restriction imposed one step earlier.

Because $w \le 1$, the second constraint forces $p_{uv}^{st} \ge w(u,v)$ only in the crossing case $(x_u^{st}, x_v^{st}) = (1,0)$. In the other three cases its right-hand side is non-positive. The third constraint then caps the chosen cut at one. The remaining two lines are redundant accelerators: the first counts the direct arc and all two-step detours, and the second removes equivalent vertex relabellings.

The $\min$ in the two-step line is written as a minimum for readability only. A linear program cannot take one directly. It is linearised in the usual way, by an auxiliary variable $z^{sxt} \in [0,1]$ held below both arguments, $z^{sxt} \le w(s,x)$ and $z^{sxt} \le w(x,t)$, together with a binary selector b^{sxt} and the two big-M inequalities $z^{sxt} \ge w(s,x) - (1-b^{sxt})$ and $z^{sxt} \ge w(x,t) - b^{sxt}$, which force z^{sxt} up to whichever argument is the smaller. The displayed $\min$ is then the value of z^{sxt}.

Theorem A.49 (Directed multigraph value, small n)**.** *For $n \le 6$, $M^*(n) = 2(n-1)$, and the*

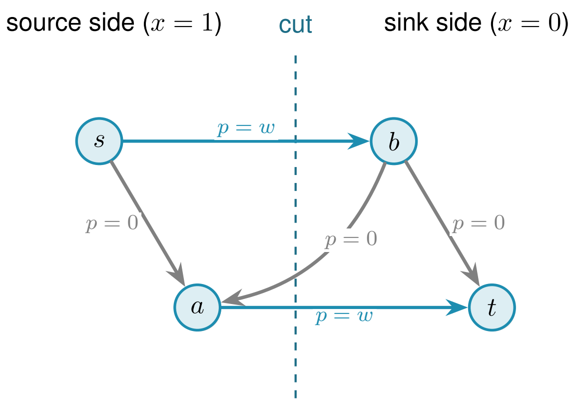


Figure A.4. One chosen s–t cut. Only the two thick source-to-sink arcs contribute crossing weight $p = w$, while same-side and backward arcs have $p = 0$.

optimum is attained by a $\{0, 1\}$ weight matrix. Hence for all $m \geq 2$, $L_m^{\mathrm{dir}}(n) = 2(n-1)(m-1)$, attained by the double star.

Historical runs gave $M^*(n) = 2(n-1)$ with status `OPTIMAL` for $3 \leq n \leq 6$, and $n = 7$ did not close. Their archived output lacks a separately checkable bound or gap, and no theorem now depends on it: Appendix A proves the value for all n, m by hand and Section A.11 retains the solver record.

Part III. The hypergraph results

The edge bound through a hypergraph Gomory–Hu tree, the vertex problem at $m = 2$ and $m = 3$ by way of an incidence rank bound, and the directed hypergraph leading term.

A.7 The hypergraph bounds

Connectivity in a hypergraph admits several inequivalent definitions, surveyed with their cut algorithms by Chekuri and Xu [23] and by Dewar, Pike and Proos [24]. This thesis fixes on the Berge notion throughout, which is the one the checker of Chapter 2 measures and the one the bounds below are stated for.

The body's hypergraph claims rest on two theorems: a Menger theorem for Berge routes, which also certifies the helper-point checker of Chapter 2, and a hypergraph Gomory–Hu theorem, which is the classical construction of Gomory and Hu [4] run on a symmetric submodular cut function rather than on the edge-cut function, as set out by Hanifehnezhad and Dolati [25].

Theorem A.50 (Menger for multihypergraphs, [AI])**.** *For a multihypergraph $\mathcal{H}$ and vertices u, v, regard parallel hyperedge copies as distinct. The maximum number of hyperedge-copy-disjoint Berge u–v paths equals the minimum number of individual copies whose deletion separates u from v. Equivalently, if a cut C is specified as a set of distinct hyperedge types, its cost is*

$\sum_{e \in C} \mu(e)$*, because every copy of each selected type must be deleted. A simple hypergraph is the case where every multiplicity is one.*

Proof. Build the helper-point network of Figure 2.3b: for each distinct hyperedge e a helper arc $e^- \to e^+$ whose capacity is the multiplicity $\mu(e)$, which is one for a simple hypergraph, and arcs $x \to e^-$ and $e^+ \to x$ of unbounded capacity for each member $x \in e$. The proof matches the number of disjoint Berge paths to the size of a minimum cut by translating between Berge-path language and flow language in each direction: a family of hyperedge-copy-disjoint Berge paths becomes a feasible integral flow, and conversely such a flow becomes a family of Berge paths, as the next two paragraphs spell out in turn.

Berge paths become flow. A Berge path $u, e_1, v_1, \ldots, e_k, v$ becomes the directed walk $u \to e_1^- \to e_1^+ \to v_1 \to \cdots \to v$, crossing one helper arc per hyperedge it uses, and carrying one unit of flow. The e_i along a single Berge path are distinct, and a hyperedge-disjoint family rides no copy twice, so at most $\mu(e)$ of its paths cross the helper arc of e. Each helper arc therefore carries at most its own capacity and the family is a feasible flow of value k. Only at multiplicity one are these paths pairwise arc-disjoint, and the argument never needs them to be.

Flow becomes Berge paths. Conversely, because every capacity is an integer, a maximum flow may be taken integral, and an integral flow of value k decomposes into k unit paths. Reading a helper crossing $\cdots x \to e^- \to e^+ \to y \cdots$ back as the single hyperedge step x, e, y, that is, *suppressing* the two helper nodes, turns each unit path into a Berge path. At most $\mu(e)$ of them cross the helper arc of e, since that is its capacity, so each may be given a copy of e of its own and no copy serves two routes. The k Berge paths are therefore hyperedge-disjoint.

Cuts. Finally, the member arcs are uncapped, so any cut of finite capacity uses helper arcs only, and a set of helper arcs separates u from v in the network exactly when the corresponding hyperedge types separate them in $\mathcal{H}$. Its capacity is $\sum \mu(e)$ over those types, which is the number of copies removed. Conversely, deleting only some copies of a type while leaving another copy changes no reachability, so a minimum copy cut may be taken to delete either all copies of a type or none. Thus network-cut capacity and the minimum number of deleted copies are the same count. The integral max-flow/min-cut theorem equates the largest feasible integral flow value with this smallest cut capacity, which is precisely the claim. □

Theorem A.51 (Hypergraph Gomory–Hu, [AI])**.** *Every r-uniform multihypergraph $\mathcal{H}$ on vertex set V has a weighted tree T^* on V, its* hypergraph Gomory–Hu tree*, such that $\lambda_{\mathcal{H}}(u, v)$ equals the minimum weight on the u–v tree path. Each tree edge t induces a vertex partition $(S_t, V \setminus S_t)$ with*

$$c_{\mathcal{H}}(S_t) = c^*(t), \qquad c_{\mathcal{H}}(S) := \sum_{e \in \delta_{\mathcal{H}}(S)} \mu(e),$$

where $\delta_{\mathcal{H}}(S)$ is the set of distinct hyperedge types incident to both sides. Thus $c_{\mathcal{H}}(S)$ counts crossing copies with multiplicity. For a simple hypergraph it is just $|\delta_{\mathcal{H}}(S)|$.

This is the exact hypergraph echo of Theorem A.2, the same tree-of-cuts summary of all pairwise connectivities, with the single change that the capacity of a cut now counts the hyperedge copies straddling it, $c_{\mathcal{H}}(S)$, in place of the edges. Gomory and Hu's construction [4] needs nothing about the cut function beyond it being symmetric and submodular [25], and the weighted hyperedge-cut function $c_{\mathcal{H}}(\cdot)$ has both properties exactly as the ordinary edge-cut function does, so the identical uncrossing construction builds this tree too. We use only the two properties in the statement, the path-minimum reading of $\lambda_{\mathcal{H}}$ and the identity $c_{\mathcal{H}}(S_t) = c^*(t)$.

Proof of Proposition 1.8. Upper bound. Let $\mathcal{H}$ be r-uniform with $\lambda^{\max}_{\mathcal{H}} \leq m-1$, and let T^* be its hypergraph Gomory–Hu tree (Theorem A.51), a weighted tree on the same n vertices with weights c^*. The plan is to charge each hyperedge copy to tree edges and count the charges two ways, exactly as in the Mader proof, with the one change that a hyperedge spans r vertices rather than two.

Step 1: each hyperedge copy crosses at least $r-1$ tree cuts. Fix a copy of a hyperedge e, a set of r vertices, and let $\mathrm{ST}(e)$ be its *smallest connected subtree*, the minimal subtree of T^* that contains all r of them (its Steiner tree). A tree spanning at least r vertices has at least $r-1$ edges, so $\mathrm{ST}(e)$ has at least $r-1$ of them. Each such edge t is a genuine cut for e: deleting t from T^* splits the vertices into the two sides $(S_t, V \setminus S_t)$, and because t lies on $\mathrm{ST}(e)$ the hyperedge e has at least one vertex on each side. So e touches both sides. Therefore

$$\bigl|\{\, t \in E(T^*) : e \text{ straddles } S_t \,\}\bigr| \;\geq\; r-1 \qquad \text{for every hyperedge copy } e.$$

Step 2: count the incidences two ways. Sum that inequality over all hyperedge copies, then exchange the order of summation. This is legitimate because both of the middle sums below simply count the same pairs (e,t) in which the copy e straddles the cut of t, once grouped by the hyperedge copy and once by the tree edge:

$$(r-1)\,|E(\mathcal{H})| \;\leq\; \sum_{e \text{ (with copies)}} \bigl|\{\, t : e \text{ straddles } S_t \,\}\bigr| \;=\; \sum_t c_{\mathcal{H}}(S_t).$$

Step 3: read off the bound. By the Gomory–Hu property each tree edge has $c_{\mathcal{H}}(S_t) = c^*(t)$. Moreover $c^*(t)$ is itself a local connectivity $\lambda_{\mathcal{H}}$ of t's two endpoints: since those endpoints are adjacent in T^*, the u–v tree path between them is the single edge t, so the path-minimum reading of Theorem A.51 gives $\lambda_{\mathcal{H}}$ of that pair exactly $c^*(t)$, hence at most $m-1$ by feasibility. Summing over the $n-1$ tree edges, $\sum_t c_{\mathcal{H}}(S_t) = \sum_t c^*(t) \leq (m-1)(n-1)$. Chaining Steps 2 and 3,

$$(r-1)\,|E(\mathcal{H})| \;\leq\; (m-1)(n-1),$$

and dividing by $r-1$, then rounding down because $|E(\mathcal{H})|$ is an integer, gives $|E(\mathcal{H})| \leq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$. *Lower bound.* When $(r-1) \mid (n-1)$, take a hub h and split the other $n-1$ vertices into groups $G_1, G_2, \ldots$ of size $r-1$ each. Form the star hypertree whose hyperedges are $\{h\} \cup G_i$, one per group, and give each hyperedge multiplicity $m-1$. Every non-hub vertex lies in exactly $m-1$ hyperedges, which form a cut isolating it, so by Theorem A.50 every pair has Berge connectivity at most $m-1$, while the count is $\frac{(m-1)(n-1)}{r-1}$. For general n under the hypothesis $m-1 \leq \binom{n-2}{r-2}$, Theorem A.52 below attains the floor exactly, without even needing repeated hyperedges. $\square$

The multiplicities in the lower bound can in fact be avoided: for $r \geq 3$, *simple* r-uniform hypergraphs already attain the same bound, in sharp contrast to the graph case $r = 2$, where simplicity costs the factor that separates Mader's $\lfloor m(n-1)/2 \rfloor$ from the multigraph value $(m-1)(n-1)$.

Theorem A.52 (Simplicity is free for $r \geq 3$, [AI])**.** *Let $r \geq 3$, $n \geq r$, and $2 \leq m$ with $m-1 \leq \binom{n-2}{r-2}$. Then the maximum number of hyperedges in a* simple *r-uniform hypergraph on n vertices with Berge edge-connectivity at most $m-1$ is exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$.*

Proof. The upper bound is Proposition 1.8, since a simple hypergraph is a multihypergraph. For the lower bound, fix a hub h and let $V' = V \setminus \{h\}$. By Lemma A.55 applied with rank $r-1$ and degree cap $d = m-1$ on the $n-1$ vertices of V', there is an $(r-1)$-uniform simple hypergraph $\mathcal{G}^*$ on V' with maximum degree at most $m-1$ and exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. Adding h to each hyperedge produces distinct r-sets, so the result $\mathcal{H}$ is simple, and every non-hub vertex w lies in $\deg_{\mathcal{G}^*}(w) \leq m-1$ hyperedges, which form a cut isolating w. By Theorem A.50, $\lambda_{\mathcal{H}}^{\max} \leq m-1$. $\square$

Remark A.53 (The hypothesis is not decorative)*.* Outside this range the displayed value can fail for simple hypergraphs, even though it stays a valid upper bound (Proposition 1.8) and, when $(r-1) \mid (n-1)$, stays exactly attained by a multihypergraph. The smallest case already shows it. At $n = r = m = 3$ there is only one 3-subset of a 3-element vertex set, so a simple hypergraph carries at most one hyperedge. The hypothesis needs $m-1 \leq \binom{n-2}{r-2} = \binom{1}{1} = 1$, and $m-1 = 2$ misses it, so the formula's value $\lfloor 2 \cdot 2/2 \rfloor = 2$ is not attained by any simple hypergraph there. It is attained by the multihypergraph star, two parallel copies of that one triple. Theorem 1.10 carries the same hypothesis at $m-1 = 2$, where it reduces to $2 < r < n$, for the same reason. Table 3.1 splits the simple and multi hypergraph rows on exactly this distinction, and the program enumerates the two models separately for the same reason, stated where each formula is used.

The sparse hypergraph the construction needs is a corollary of a classical partition theorem, which we state before using it.

Theorem A.54 (Baranyai [26])**.** *If r divides n, the $\binom{n}{r}$ distinct r-subsets of an n-element set can be partitioned into $\binom{n-1}{r-1}$ classes, each a* perfect matching*, meaning a family of n/r disjoint r-sets that together cover every vertex exactly once. More generally, for any prescribed class sizes $a_1 + \cdots + a_t = \binom{n}{r}$, and with no divisibility assumed, the r-subsets can be partitioned into classes of exactly those sizes so that in the class of size a_i every vertex lies in either $\lfloor ra_i/n \rfloor$ or $\lceil ra_i/n \rceil$ of the sets, as evenly balanced as the size permits.*

The first statement is the memorable one. When r divides n the complete r-uniform hypergraph, the one holding every r-set, comes apart into $\binom{n-1}{r-1}$ perfect matchings, the exact hypergraph analogue of splitting a regular bipartite graph into perfect matchings by König's theorem. The proof is an induction that keeps the partial partition as balanced as possible at every step, an integer-flow argument in the spirit of Hall's marriage theorem rather than an explicit construction, and its full form is in Baranyai [26]. We need only the general equitable form, and only its conclusion about degrees.

Lemma A.55 (Sparse uniform hypergraphs exist)**.** *Let $r \geq 1$, $n \geq r$, and $0 \leq d \leq \binom{n-1}{r-1}$. There exists an r-uniform simple hypergraph on n vertices with maximum degree at most d and exactly $\left\lfloor \frac{dn}{r} \right\rfloor$ hyperedges.*

Proof. For $r = 1$ the hypotheses force $d \in \{0, 1\}$, and the empty family or the family of all n singleton edges, respectively, has the required size and degree. Assume henceforth that $r \geq 2$. Put $e := \left\lfloor \frac{dn}{r} \right\rfloor \leq \frac{n}{r}\binom{n-1}{r-1} = \binom{n}{r}$. Apply the equitable form of Theorem A.54 with two classes, of sizes e and $\binom{n}{r} - e$. Within each class every vertex is covered either $\lfloor re/n \rfloor$ or $\lceil re/n \rceil$ times, so the class of size e is a simple r-uniform hypergraph with $e = \left\lfloor \frac{dn}{r} \right\rfloor$ hyperedges and maximum degree at most $\lceil re/n \rceil \leq \lceil d \rceil = d$, the last step because $re/n \leq r(dn/r)/n = d$ and d is an integer. ☐

A.8 The hypergraph vertex problem

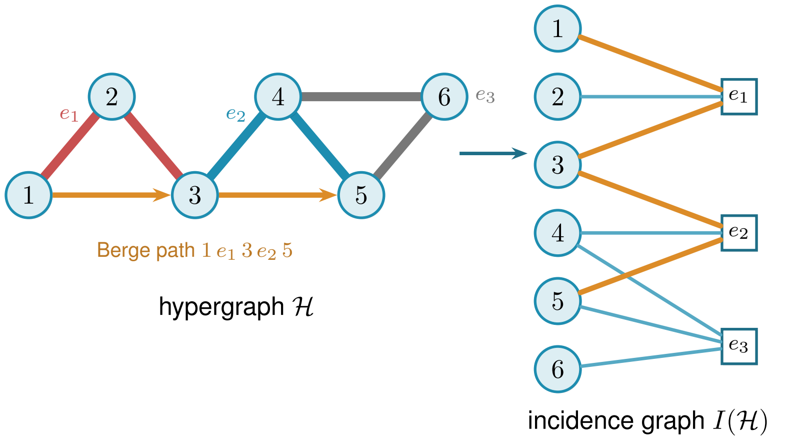
The whole vertex problem translates into a graph problem through one observation, the one the rest of this section rests on and which Figure A.5 draws in full. Write $I(\mathcal{H})$ for the bipartite *incidence graph* of a hypergraph $\mathcal{H}$: one node per vertex, one node per hyperedge, an edge when the vertex lies in the hyperedge. A Berge path with distinct hyperedges is exactly a path of $I(\mathcal{H})$ between two vertex-class nodes. The interior of such a path consists of both kinds of node, the hyperedges it rides and the vertices where it changes hyperedge, so two paths of $I(\mathcal{H})$ are internally disjoint in the ordinary graph sense exactly when the two Berge routes share neither a hyperedge nor an intermediate vertex. That hybrid is what $\kappa_{\mathcal{H}}$ means throughout, and both halves are needed: dropping the hyperedge half would let two routes ride the same hyperedge, which a single Berge edge cannot serve twice. Hence

$$\kappa_{\mathcal{H}}(u, v) \;=\; \kappa_{I(\mathcal{H})}(u, v),$$

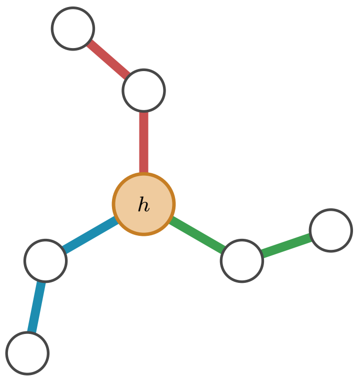
the maximum number of internally disjoint u–v paths in $I(\mathcal{H})$, and the hypergraph vertex problem asks for the maximum number of degree-r nodes on one side of a bipartite graph whose other side holds n nodes, subject to no two of those n nodes being joined by m internally disjoint paths. Figure A.5a traces a single Berge path through both pictures, and Figure A.5b shows the star hypertree of Theorem 1.9, whose incidence graph is a tree.

Theorem 1.9, stated in Chapter 1, follows at once from the incidence translation.

Proof of Theorem 1.9. We claim $\kappa_{\mathcal{H}}^{\max} \leq 1$ holds if and only if $I(\mathcal{H})$ is a forest. If $I(\mathcal{H})$ contains a cycle $v_1, e_1, v_2, e_2, \ldots, v_\ell, e_\ell, v_1$ (it alternates classes, so $\ell \geq 2$, all v_i and e_i distinct), then v_1, e_1, v_2 and $v_1, e_\ell, v_\ell, e_{\ell-1}, \ldots, e_2, v_2$ are two Berge v_1–v_2 paths with disjoint hyperedge sets



(a) A 3-uniform hypergraph $\mathcal{H}$ (left) and its bipartite incidence graph $I(\mathcal{H})$ (right): circles for vertices, hollow squares for hyperedges, an edge when the vertex lies in the hyperedge. The Berge path $1\,e_1\,3\,e_2\,5$ is traced in orange in both pictures. In $\mathcal{H}$ it rides the red line e_1 and the blue line e_2, changing at the shared vertex 3, while the grey line e_3 lies off it. In $I(\mathcal{H})$ it is the ordinary path $1 - e_1 - 3 - e_2 - 5$ alternating sides.



(b) A star hypertree: every hyperedge $\{h\} \cup G_i$ shares only the hub h (orange), and the other $r - 1 = 2$ vertices of each block sit in a single hyperedge. Its incidence graph is a tree, so $\kappa^{\max} \leq 1$.

Figure A.5. The translation the vertex problem rests on. Internally vertex-disjoint Berge paths in $\mathcal{H}$ correspond exactly to internally disjoint paths in $I(\mathcal{H})$, so $\kappa_{\mathcal{H}}(u, v) = \kappa_{I(\mathcal{H})}(u, v)$. **(top)** A hypergraph and its incidence graph with one Berge path drawn in both. **(bottom)** The star hypertree of Theorem 1.9, whose incidence graph is a tree.

and disjoint interior vertex sets, so $\kappa(v_1, v_2) \geq 2$. (At $\ell = 2$ this is the repeated pair: two hyperedges sharing two vertices.) Conversely, two such paths between u and v concatenate to a cycle of $I(\mathcal{H})$. This proves the claim. A forest on $n + q$ nodes has at most $n + q - 1$ edges, while $I(\mathcal{H})$ has exactly rq edges, so $rq \leq n + q - 1$, i.e. $q \leq (n-1)/(r-1)$, and q is an integer. The star hypertree has $\lfloor (n-1)/(r-1) \rfloor$ hyperedges and a forest as incidence graph (one tree together with any unused isolated vertices) because each block vertex lies in one hyperedge and every cycle would need two hyperedges sharing two vertices, so the floor is attained. □

Proposition A.56 (General lower bound)**.** *For $r \geq 3$, $m \geq 2$ and $m - 1 \leq \binom{n-2}{r-2}$,*

$$k_m^{(r)}(n) \;\geq\; \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

attained by a simple hypergraph.

Proof. The simple construction of Theorem A.52 has Berge edge-connectivity at most $m - 1$, and Whitney's inequality $\kappa \leq \lambda$ holds pairwise (an internally vertex-disjoint family is in particular hyperedge-disjoint under our definition), so the same hypergraph is feasible for the vertex problem. □

The simple model needs $m-1$ distinct $(r-1)$-sets to hang off the hub, which is what $m-1 \leq \binom{n-2}{r-2}$ asks for. A multihypergraph does not, because it may repeat one hyperedge instead, and that closes a different range of cases.

Proposition A.57 (The multihypergraph lower bound)**.** *For $r \geq 2$, $m \geq 2$ and $(r-1) \mid (n-1)$,*

$$K_m^{(r)}(n) \;\geq\; \frac{(m-1)(n-1)}{r-1},$$

attained by the star hypertree of Figure 1.8 with every hyperedge taken at multiplicity $m - 1$. At $m = 3$ this meets the upper bound of Theorem 1.10, so

$$K_3^{(r)}(n) \;=\; \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor \qquad \text{whenever } (r-1) \mid (n-1).$$

For $r \geq 3$, that range neither contains nor is contained in the simple model's $2 < r < n$, so the two models close different cases.

Proof. Take the hub h, split the other $n - 1$ vertices into blocks $G_1, \ldots, G_q$ of size $r - 1$ with $q = (n-1)/(r-1)$, and give each hyperedge $\{h\} \cup G_i$ multiplicity $m - 1$. The count is $(m-1)q$, which is the displayed value.

For the connectivity, take $u \neq w$. If both lie in one block G_i, or if one of them is h and the other lies in G_i, the $m - 1$ copies of $\{h\} \cup G_i$ are $m - 1$ one-step Berge routes between them. A one-step route has no interior vertex, so no two of them share one, and they ride different copies, so the family is internally vertex-disjoint and $\kappa(u, w) \geq m - 1$. Those same $m - 1$ copies are also every hyperedge containing the block vertex among u, w, so deleting them separates the pair and Theorem A.50 caps $\kappa(u, w)$ at $m - 1$.

If $u \in G_i$ and $w \in G_j$ with $i \neq j$, then two different blocks meet only at h, so every u–w Berge route has h in its interior. Two internally vertex-disjoint routes cannot both use h there, so $\kappa(u, w) \leq 1$. Hence $\kappa^{\max} = m - 1$ and the hypergraph is feasible.

At $m = 3$ the count is $2(n-1)/(r-1)$, which is $\lfloor 2(n-1)/(r-1) \rfloor$ under the divisibility hypothesis, and Theorem 1.10 bounds every multihypergraph by that same quantity. □

At $m = 3$ the upper-bound side closes completely, and equality follows in the construction range stated below. The engine is a rank bound for graphs in which one side of a partition must stay below local 3-connectivity. The hypergraph theorem then falls out of the incidence translation. Throughout, $\mathrm{rank}(G) = |E(G)| - |V(G)| + 1$ denotes the cycle rank of a connected multigraph, that is, the number of independent cycles it carries. A tree has cycle rank 0, and each edge added beyond a spanning tree closes exactly one new cycle and raises the rank by one, so bounding the rank is the same as bounding how many independent cycles the graph can hold.

Primer: global connectivity and the pieces of a graph

The lemma below speaks of a graph being 2-connected or 3-connected and of taking it apart at small separators. These are the global cousins of the local $\kappa(u, v)$ from Chapter 1, so we fix the words once. A graph on more than k vertices is *k-connected* if it stays connected after the removal of any $k - 1$ vertices. By Menger in its global form [27, Ch. 3] this is the same as asking that every pair of vertices be joined by at least k internally vertex-disjoint paths, so k-connected means $\kappa(u, v) \geq k$ for all pairs at once, the uniform version of the local measure. Thus 2-connected means no single vertex disconnects the graph, and 3-connected means no two vertices do.

Three named obstructions go with this. A *cut vertex* is a vertex whose removal disconnects the graph, so a connected graph on at least three vertices is 2-connected exactly when it has no cut vertex. A *separating pair* is a set of two vertices whose removal disconnects the graph: it is the obstruction that a 2-connected graph must lack to be 3-connected. Finally a *block* is a maximal piece with no cut vertex of its own, hence either a single bridge edge or a maximal 2-connected subgraph. Two blocks overlap in at most one vertex, always a cut vertex, and the blocks and cut vertices hang together in a tree, the *block-cut tree*, with one node per block and one per cut vertex. Step 1 of the proof is precisely an induction over that tree.

Lemma A.58 (Incidence rank lemma, [AI]). *Let G be a connected multigraph with vertex set $X \cup Z$ such that (i) Z is independent, (ii) no edge incident to Z is parallel to another edge, (iii) every $z \in Z$ has degree at least 2, and (iv) $\kappa_G(x, x') \leq 2$ for all distinct $x, x' \in X$. Then* $\mathrm{rank}(G) \leq |X| - 1$.

In words: this is the engine behind the $m = 3$ hypergraph result, stated for a bipartite graph rather than for a hypergraph. Read X as the vertices of a hypergraph and Z as its hyperedges, so that G is the incidence graph and $\operatorname{rank}(G)$, the number of independent cycles it carries, is the quantity the hyperedge count is eventually read off from. The four hypotheses say in turn that the picture really is an incidence graph (i), that no hyperedge repeats a vertex (ii), that no hyperedge dangles as a leaf (iii), and that the hypergraph is feasible at $m = 3$ (iv). The conclusion caps the independent cycles by the number of vertices.

Proof. Induction on $|V|+|E|$. The proof peels G down to a core that cannot exist. Step 1 handles a cut vertex by arguing block by block. Steps 2 and 3 remove a degree-two Z-vertex or X-vertex without changing the bound. What survives is 2-connected with minimum degree at least three, and Step 4 shows that such a graph already violates the connectivity cap (iv), so it never arises. *Base* $|V| \leq 2$. A single vertex has degree 0, too little for the degree-at-least-2 floor (iii) imposes on a Z-vertex, so it cannot lie in Z and must lie in X, giving $0 \leq |X| - 1$. On two vertices the same reasoning applies to either one: a vertex in Z could reach only the other vertex, and (ii) forbids that single edge from being parallel to another, so it would have degree at most 1, again short of (iii). Hence (i)–(iii) force both vertices into X. Parallel edges between them are internally disjoint paths (each has empty interior, so no two can share an interior vertex), so (iv) caps the multiplicity at 2 and $\operatorname{rank} \leq 1 = |X| - 1$.

Step 1: a cut vertex. Suppose G has a cut vertex, so it is not yet 2-connected. Break it into its *blocks* $B_1, \ldots, B_c$ with $c \geq 2$, a block being a maximal piece with no cut vertex of its own, that is, either a single bridge edge or a maximal 2-connected subgraph. Two blocks meet in at most one vertex, and any shared vertex is a cut vertex of G. No cycle can pass through two different blocks: since two blocks share at most one vertex, a cycle straddling both would have to visit that shared vertex twice, which a simple cycle cannot do. So the cycle rank adds up over blocks, $\operatorname{rank}(G) = \sum_i \operatorname{rank}(B_i)$, and it is enough to bound each $\operatorname{rank}(B_i)$ by $|X \cap B_i| - 1$ and sum.

Each block inherits the hypotheses. A bridge block is one edge, so its rank is 0, and by (i) at least one endpoint lies in X, giving $0 \leq |X \cap B| - 1$. Every other block is 2-connected, hence has minimum degree at least 2, so (i) to (iv) hold inside it and the induction hypothesis gives $\operatorname{rank}(B_i) \leq |X \cap B_i| - 1$.

Adding these requires care, because a cut vertex is shared by several blocks and would be counted several times. Write $b(v)$ for the number of blocks containing v, so $b(v) = 1$ except at cut vertices. Counting each X-vertex once for every block it lies in,

$$\sum_i |X \cap B_i| = |X| + \sum_{x \in X \text{ cut}} (b(x) - 1).$$

The number of blocks satisfies the block–cut tree identity $c = 1 + \sum_{v \text{ cut}}(b(v) - 1)$. This follows from counting the edges of the block-cut tree two ways. That tree has c block-nodes and one node per cut vertex, so being a tree its edge count is one less than its total number of nodes, $c + \#\{\text{cut vertices}\} - 1$. The same edge count also equals $\sum_{v \text{ cut}} b(v)$, since each cut vertex v is joined in the tree to exactly the $b(v)$ blocks that contain it. Equating the two expressions for the edge count and solving for c gives the identity. Subtracting this count of blocks from the previous display and separating the cut vertices into their X- and Z-parts leaves

$$\operatorname{rank}(G) = \sum_i \operatorname{rank}(B_i) \le \sum_i (|X \cap B_i| - 1) = |X| - 1 - \sum_{z \in Z \text{ cut}} (b(z) - 1) \le |X| - 1,$$

where the last inequality holds because each $b(z) - 1 \ge 0$, so the Z-cut sum only pushes the bound further below $|X| - 1$.

Step 2: a Z-vertex of degree two. From now on G is 2-connected, every cut vertex having been handled. Suppose some $z_0 \in Z$ has degree exactly 2. Both its neighbours lie in X by (i), and they are distinct, since two edges to the same vertex would be parallel at a Z-vertex, which (ii) forbids. *Suppress* z_0 by deleting it and joining its two neighbours x, x' with one new edge. This loses two edges and one vertex and gains one edge, so $|E|$ and $|V|$ each drop by one and $\operatorname{rank} = |E| - |V| + 1$ is unchanged. The new edge runs X to X, so $|X|$ is unchanged and (i), (ii), (iii) still hold for the remaining Z-vertices. Replacing the path x, z_0, x' by a direct edge alters no route among the surviving vertices, so every $\kappa(x_1, x_2)$ is preserved and (iv) holds. The graph is strictly smaller, so the induction hypothesis applies and returns the bound for G.

Step 3: an X-vertex of degree two. Now G is 2-connected and, Step 2 being used up, every Z-vertex has degree at least 3. Suppose some $x_0 \in X$ has degree 2, and delete it. Deleting one vertex from a 2-connected graph leaves it connected. Stripping a degree-2 vertex removes two edges and one vertex, so rank drops by exactly one. Each neighbour of x_0 loses one edge, and a neighbour in Z only falls from degree at least 3 to degree at least 2, so (iii) survives, while deleting a vertex cannot raise a connectivity, so (iv) survives. Here $|X|$ drops by one, and $|X| \ge 2$ to begin with, for if $|X| \le 1$ then by (i) and (ii) every z would send all its edges to one X-vertex without repeats, hence have degree at most 1, against (iii). The smaller graph satisfies the hypotheses, so induction gives $\operatorname{rank}(G) - 1 \le (|X| - 1) - 1$, that is $\operatorname{rank}(G) \le |X| - 1$.

Step 4: the remaining case is impossible. Now G is 2-connected with $|V| \ge 3$ and every degree at least 3. First, G is simple: a parallel class lies inside X by (ii), and if $\mu(x, x') \ge 2$ then a third internally disjoint route exists: walk from x to any third vertex w inside $G - x'$, then from w to x' inside $G - x$, both walks available because deleting a single vertex from a 2-connected graph leaves it connected, exactly as already used in Step 3. The concatenation visits x only first and x' only last, so it contains an x–x' path with at least one interior vertex, giving $\kappa(x, x') \ge 3$, against (iv). If G is 3-connected, Menger puts every pair at $\kappa \ge 3$, so two vertices of X would violate the cap $\kappa(x, x') \le 2$ that (iv) imposes on every pair drawn from X, forcing $|X| \le 1$, and then every z has degree at most 1 by (i), absurd.

Otherwise G is 2-connected but not 3-connected. Tutte's triconnected decomposition [28], whose algorithmic form is the SPQR tree of Hopcroft and Tarjan [29], expresses it as a tree of cycle, bond, and 3-connected pieces joined along virtual edges. For a leaf piece L its unique virtual edge $\{a, b\}$ represents a real a–b path through the rest of G, internally outside L. Vertices of L away from that edge keep their degree in G. These are the only properties of the decomposition used below.

Now G itself is none of the three piece types, since it is not a single cycle (its degrees are at least three), not a bond (it is simple), and not 3-connected (the case just handled), so the tree has at least two leaves. Let L be one, with unique virtual edge $\{a, b\}$. If L is a cycle, a vertex of L off $\{a, b\}$ has degree 2 in G, absurd. If L is a bond, it holds at least two real parallel edges, absurd in a simple graph. If L is 3-connected, then any two of its vertices are joined by three internally disjoint paths inside L, at most one through the virtual edge, which the far-side path replaces: so $\kappa_G(u, v) \geq 3$ for *all* $u, v \in V(L)$, forcing $|X \cap V(L)| \leq 1$ by (iv). But $|V(L)| \geq 4$ leaves a Z-vertex $z^* \notin \{a, b\}$. Its neighbours lie in $X \cap V(L)$ by (i), at most one vertex, reached by at most one edge since G is simple, giving degree at most 1, which is absurd. □

The rank bound now yields Theorem 1.10, stated in Chapter 1. In words: at the route limit $m = 3$ a hyperedge network can hold no more hyperedges under the stricter separation than the displayed edge bound. When $2 < r < n$, that count is reached without repeating a hyperedge, so the two separations agree in this range exactly as they did at $m = 2$.

Proof of Theorem 1.10. Apply Lemma A.58 to each connected component of the incidence graph $I(\mathcal{H})$, with X the vertex class and Z the hyperedge class: Z is independent, incidences are simple even when hyperedges repeat, hyperedge-nodes have degree $r \geq 2$, components without an X-node are impossible, and $\kappa_I(u, v) = \kappa_{\mathcal{H}}(u, v) \leq 2$ for vertex pairs. Sum $\mathrm{rank} \leq |X_c| - 1$ over the C components. The incidence graph $I(\mathcal{H})$ has rq edges (each of the q hyperedges contributes r incidences) and $n + q$ nodes, so $\sum_c \mathrm{rank} = rq - (n + q) + C$, while $\sum_c (|X_c| - 1) = n - C$. The inequality reads $rq - (n + q) + C \leq n - C$, hence $q(r - 1) \leq 2(n - 1) + 2(1 - C) \leq 2(n - 1)$, the last step using $C \geq 1$. The lower bound is Proposition A.56 at $m = 3$. □

Remark A.59 (Edges as gates at $r = 2$, and the boundary of the method)*.* At $r = 2$ the theorem speaks of multigraphs under the hyperedge-as-gate convention, in which parallel edges are distinct one-step routes, giving the multigraph value $2(n - 1)$ of Theorem 1.3, complementing the body's convention for graphs, where parallel adjacencies count once. The proof of Lemma A.58 leans on the triconnected decomposition exactly where $\kappa \leq 2$ lives. For $m = 4$, the analogous question asks for a rank bound relative to $\kappa \leq 3$, where no comparably clean decomposition exists. Thus the proof is confined to $m \leq 3$, and no general $m = 4$ formula is claimed.

A.9 The directed hypergraph

The remaining four of the sixteen variants orient the Berge edge. They are grouped here rather than with the undirected hypergraph bounds above because the questions they raise are different ones: not which decomposition controls the rank of an incidence graph, but how much a single edge with one tail and many heads can carry, and whether the way it is oriented changes the answer. The first result gives a pair of bounds a factor of four apart, and the last closes that factor for the model the program measures.

Proposition A.60 (Directed hypergraphs: first bounds, [AI])**.** *Let $n \geq r \geq 2$ and $m \geq 2$. Model a directed r-uniform hyperedge as a tail vertex together with $r - 1$ head vertices, a Berge route entering at the tail and leaving at a head. If every ordered pair of vertices has at most $m - 1$*

hyperedge-disjoint directed Berge routes, then

$$|E(\mathcal{H})| \;\le\; \frac{(m-1)\,n(n-1)}{r-1},$$

and, for every $1 \le \alpha \le n-r+1$, there is a feasible directed multihypergraph with

$$\alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor$$

hyperedges. This construction is simple whenever $m-1 \le \binom{n-\alpha-1}{r-2}$. Consequently, for fixed r and m, the balanced choice $\alpha = \lfloor n/2 \rfloor$ is available for all sufficiently large n and proves that the extremal value is $\Theta_{r,m}(n^2)$.

Proof. Upper bound: a hyperedge with tail u serves the $r-1$ ordered pairs (u,h), h a head. If some ordered pair (u,v) were served by m distinct hyperedges, those would be m hyperedge-disjoint one-step routes, so each ordered pair is served at most $m-1$ times. Summing that cap over all $n(n-1)$ ordered pairs counts every hyperedge once for each of its $r-1$ tail-head pairs, so the sum is $(r-1)|E|$ on one side and at most $(m-1)n(n-1)$ on the other, giving $(r-1)|E| \le (m-1)n(n-1)$.

For the lower bound, split V into tails A, $|A| = \alpha$, and heads B, where $b := |B| = n-\alpha \ge r-1$. Put $k := r-1$, $d := m-1$ and $C := \binom{b-1}{k-1}$, the degree of a vertex in the complete k-uniform hypergraph on B. Write $d = qC + s$ with $0 \le s < C$. Take q copies of that complete hypergraph and, by Lemma A.55, a simple k-uniform hypergraph on B of maximum degree at most s and with $\lfloor sb/k \rfloor$ edges. Their multiset union $\mathcal{G}^*$ has maximum degree at most $qC + s = d$ and

$$q\binom{b}{k} + \left\lfloor \frac{sb}{k} \right\rfloor = q\frac{bC}{k} + \left\lfloor \frac{sb}{k} \right\rfloor = \left\lfloor \frac{db}{k} \right\rfloor$$

hyperedges. Give every tail $a \in A$ the hyperedges (a,H) for $H \in \mathcal{G}^*$, with the same multiplicities. No vertex of B is a tail, so every route is a single step and $\lambda(a,b) = \deg_{\mathcal{G}^*}(b) \le m-1$, while pairs inside A or starting in B have no routes. If $d \le C$, the auxiliary hypergraph can instead be taken directly from Lemma A.55, so the resulting directed hypergraph is simple. $\square$

The forward model is one of three ways to orient a Berge edge. In general a directed r-uniform hyperedge is a split of its r vertices into a non-empty *tail set* T and a non-empty *head set* H, a route entering at a tail and leaving at a head. The *forward* model is $|T| = 1$, the *backward* model is $|H| = 1$, and the *general* model allows any split (genuinely mixed splits, both parts at least two, exist for $r \ge 4$). The next two results place all three. The first is that backward needs no separate study, and the second is that the general model, despite admitting more hyperedges on a single vertex set, obeys the very same first-order bound as forward.

Lemma A.61 (Backward is the reverse of forward, [AI])**.** *Reversing every hyperarc, $(T,H) \mapsto (H,T)$, is an edge-count-preserving bijection between the forward directed r-uniform hypergraphs and the backward ones on the same vertices, and the reverse $\mathcal{H}^R$ satisfies $\lambda_{\mathcal{H}}(u,v) =$*

$\lambda_{\mathcal{H}^R}(v,u)$ and $\kappa_{\mathcal{H}}(u,v) = \kappa_{\mathcal{H}^R}(v,u)$. Hence $\lambda^{\max}$ and $\kappa^{\max}$ are invariant under reversal, and the backward edge and vertex extremal numbers equal the forward ones, for every r, m and n.

In words: the backward model is the forward model seen in a mirror. Turning every hyperarc around swaps the tails with the heads and reverses every route, which leaves the number of independent routes between a pair untouched. The two models have identical extremal numbers, so nothing below needs to treat backward as a separate case.

Proof. Reversal sends a backward hyperarc ($|T| = r-1$, $|H| = 1$) to a forward one ($|T| = 1$) and is its own inverse, so it is an edge-count-preserving bijection between the two families. Let $u = x_0, e_1, x_1, \ldots, e_k, x_k = v$ be a directed Berge route in $\mathcal{H}$, where x_{i-1} is a tail and x_i a head of e_i. The reversed edge $e_i^R = (H_i, T_i)$ has x_i as a tail and x_{i-1} as a head, so $v = x_k, e_k^R, x_{k-1}, \ldots, e_1^R, x_0 = u$ is a directed Berge route from v to u in $\mathcal{H}^R$ on the same edges and the same internal vertices. This is a bijection between u–v routes in $\mathcal{H}$ and v–u routes in $\mathcal{H}^R$ that preserves which edges and which internal vertices any two routes share, so it carries an edge-disjoint (respectively internally vertex-disjoint) family to one of equal size. Therefore $\lambda_{\mathcal{H}}(u,v) = \lambda_{\mathcal{H}^R}(v,u)$ and $\kappa_{\mathcal{H}}(u,v) = \kappa_{\mathcal{H}^R}(v,u)$, and maximising over ordered pairs gives $\lambda^{\max}(\mathcal{H}) = \lambda^{\max}(\mathcal{H}^R)$ and the same for κ. Since reversal is a bijection preserving feasibility and edge count, the two models share every extremal number. □

Proposition A.62 (The general model obeys the forward bound, [AI])**.** *Let a directed r-uniform hyperedge be any split into a non-empty tail set T and head set H. If every ordered pair has at most $m-1$ hyperedge-disjoint, or at most $m-1$ internally vertex-disjoint, directed Berge routes, then*

$$|E(\mathcal{H})| \;\le\; \frac{(m-1)\,n(n-1)}{r-1},$$

the same bound as the forward model of Proposition A.60. The forward construction of that proposition is itself a set of general hyperedges, so the general value obeys the same upper bound and inherits the same quadratic lower bound, and in particular is $\Theta_{r,m}(n^2)$ for fixed r and m.

Proof. Fix an ordered pair (u,v). A hyperedge (T,H) with $u \in T$ and $v \in H$ carries the one-step route $u \to v$ that enters at the tail u and leaves at the head v. Distinct such hyperedges give routes that share no edge and no internal vertex, so if k hyperedges had u in their tail set and v in their head set then $\lambda(u,v) \ge k$ and $\kappa(u,v) \ge k$, so feasibility forces $k \le m-1$. Each hyperedge (T,H) is counted by exactly the $|T|\,|H|$ ordered pairs (u,v) with $u \in T$ and $v \in H$, so summing the per-pair bound over all pairs gives $\sum_{(T,H)} |T|\,|H| \le (m-1)\,n(n-1)$. As $|T|+|H| = r$ with both parts at least one, $|T|\,|H| \ge 1 \cdot (r-1) = r-1$, whence $(r-1)|E| \le (m-1)n(n-1)$. The forward family of Proposition A.60 consists of general hyperedges, so it witnesses the same $\approx (m-1)n^2/(4(r-1))$ lower bound. □

The two models are therefore trapped between the same pair of bounds, which differ by a factor of 4. That is enough to fix the *order* of both values and not enough to fix either leading constant, so it does not follow that the forward and general models agree asymptotically, only that any disagreement between them lives inside that factor. The rest of this section closes that gap for the forward model, which leaves the comparison between the two models as the question that survives.

The two bounds are not equally tight, and the simple digraph case indicates which end of the gap the slack sits at. The upper bound charges each hyperedge to the ordered tail-head pairs it occupies and then allows all $n(n-1)$ ordered pairs to be used to capacity. The construction uses only the $\alpha(n-\alpha) \le n^2/4$ pairs that run from the tail class to the head class, and leaves the pairs inside each class, and those running backwards, entirely idle. The factor of four is exactly that difference, n^2 against $n^2/4$, and it is the same difference that appears in the simple directed problem, where Construction A.17 also uses only a one-directional bipartite pattern. There, Theorem 1.11 now proves that the bipartite pattern is right to first order, that no arrangement recovers the idle pairs. If that phenomenon is a feature of directedness rather than of graphs specifically, then it is the upper bound here that should come down by a factor of four, not the construction that should climb, and that is what happens.

The obstacle to running the argument of Theorem 1.11 here is real and worth naming before it is disposed of, because the way around it is not to remove it. There, two-step routes through distinct midpoints were automatically arc-disjoint. Here a single hyperedge carries $r-1$ heads at once, so a pair of two-step routes with different midpoints may enter through the very same hyperedge, and the family of two-step routes is not edge-disjoint at all. The resolution is to accept the loss. A packing can always be extracted from that family at a cost of a factor $r-1$, and that factor turns out to be free, because of where it lands. The route count does not bound the hyperedges directly. It bounds the arcs of an auxiliary digraph, and there the factor $r-1$ enters only the *linear* error term of Lemma A.33, never the $n^2/4$ leading term. Meanwhile a second, independent factor of $r-1$ is gained when the auxiliary digraph is exchanged back for hyperedges, and that one does divide the leading term. The two factors are not the same factor, and only one of them is paid where it matters.

Definition A.63 (One-step shadow)**.** Let $\mathcal{H}$ be a forward directed r-uniform hypergraph on V. Its *one-step shadow* is the simple digraph $R(\mathcal{H})$ on V with an arc (u, v) exactly when some hyperedge of $\mathcal{H}$ has tail u and v among its heads. Write $\operatorname{cod}(u, v)$ for the number of hyperedges realising that arc.

Theorem 1.14, stated in Chapter 1, says in words that the bipartite construction of Proposition A.60 is asymptotically the best there is. Its count already matches the leading term $\frac{m-1}{r-1}\lfloor n^2/4 \rfloor$, so the factor of four that used to separate the known bounds is gone and the remaining uncertainty sits entirely in a term linear in n. The bound covers both separations at once.

Proof of Theorem 1.14. Write $R = R(\mathcal{H})$ for the one-step shadow and note it has no loops, since a hyperedge's tail is not one of its heads.

Step 1: hyperedges against shadow arcs. Suppose $\operatorname{cod}(u, v) = k$. Those k hyperedges each give a one-step Berge route from u to v, and the routes use one hyperedge apiece and different ones, so they are pairwise hyperedge-disjoint, while having no interior vertex at all they are internally vertex-disjoint as well. Hence $k \le \kappa_{\mathcal{H}}(u,v) \le m-1$. Every hyperedge is counted by exactly its $r-1$ tail-head pairs, so summing over the arcs of R,

$$(r-1)\,|E(\mathcal{H})| \;=\; \sum_{(u,v)\in A(R)} \operatorname{cod}(u,v) \;\le\; (m-1)\,|A(R)|.$$

This is the count of Proposition A.60 stopped one step earlier. That proof then bounded $|A(R)|$ by the trivial $n(n-1)$, which is exactly where its factor of four was lost, and the rest of this argument replaces that step.

Step 2: the shadow is budgeted. Fix an ordered pair (u,v) of distinct vertices, and let $T \subseteq V$ collect v itself when $(u,v) \in A(R)$ together with every midpoint counted by $p_2(u,v)$, so $|T| = \mathbf{1}_{(u,v)\in A(R)} + p_2(u,v)$. Every $t \in T$ is a head of at least one hyperedge whose tail is u. Form the bipartite graph joining each $t \in T$ to each such hyperedge containing it, and let M be a maximum matching in it.

Then $|M| \ge |T|/(r-1)$. To see it, take any $t \in T$ left unmatched by M. Every hyperedge containing t must already be matched, since otherwise M could be extended along that pair. So every element of T, matched or not, lies in some hyperedge used by M, and each hyperedge holds at most $r-1$ elements of T, giving $|T| \le |M|\,(r-1)$.

Now read M as a family of routes. For a matched pair (v,e) take the one-step route $u \to v$ through e, and for a matched pair (x,e) with x a midpoint take $u \to x \to v$, entering through e and leaving through any hyperedge whose tail is x and which has v as a head, one of which exists because $(x,v) \in A(R)$. These routes are pairwise hyperedge-disjoint: the entering hyperedges are distinct because M is a matching, the leaving hyperedges have pairwise distinct tails and so are distinct from each other, and no leaving hyperedge is an entering one, since the entering ones have tail u and the leaving ones have tail a midpoint, which is not u. They are also internally vertex-disjoint, since the one-step route has empty interior and each two-step route has as its interior the single vertex it is matched to, and M matches distinct targets. Hence

$$\frac{\mathbf{1}_{(u,v)\in A(R)} + p_2(u,v)}{r-1} \;\le\; |M| \;\le\; \kappa_{\mathcal{H}}(u,v) \;\le\; m-1,$$

so R is $(r-1)(m-1)$-budgeted in the sense of Definition A.32.

Step 3: assembling. Lemma A.33 with $C = (r-1)(m-1)$ gives $|A(R)| \le \lfloor n^2/4 \rfloor + 4(r-1)(m-1)(n-1)$, and substituting that into Step 1,

$$|E(\mathcal{H})| \;\le\; \frac{m-1}{r-1}\left[\left\lfloor \frac{n^2}{4} \right\rfloor + 4(r-1)(m-1)(n-1)\right] \;=\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)^2(n-1).$$

The lower bound is Proposition A.60, whose bipartite family has

$$\max_{1\le\alpha\le n-r+1} \alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor \;=\; \frac{(m-1)n^2}{4(r-1)} - O_{m,r}(n)$$

hyperedges, where the balanced split $\alpha = \lfloor n/2 \rfloor$ already attains the right-hand side on its own, and which is feasible for both separations at once, since it is λ-feasible and $\kappa \le \lambda$. (The integer maximum is occasionally taken a little off balance, since the floor can favour an unbalanced split by a unit or two, but never by enough to move the n^2 term.) The upper and lower bounds agree in their n^2 term, which is the asymptotic claim. □

Three remarks on the statement. First, the two upper bounds are not comparable and neither supersedes the other: the bound of Proposition A.60 is the smaller one until n is around $16(m-1)(r-1)/3$, since the linear term here carries a constant of $4(m-1)^2$, and the theorem is an asymptotic improvement rather than a uniform one. Second, at $r = 2$ the shadow is the digraph itself and cod is the multiplicity, so the theorem recovers the quadratic branch of Theorem 1.13 up to its linear error term, which is the consistency check one wants, though it does not reprove that theorem, which is exact.

Third, the general orientation model can look out of reach, but the obstacle and the way past it are both short to state. The proof above uses the forward model twice, once for the entering hyperedges and once for the leaving ones, in both cases to know that a hyperedge has a single tail. That is what makes Step 2's matching legitimate: the leaving hyperedges have pairwise distinct tails and so are automatically distinct from each other, and none of them can be an entering hyperedge, since the entering ones have tail u and the leaving ones do not. Allow several tails and both statements fail at once. Two midpoints may leave through one shared hyperedge, and from $r \ge 4$ a single hyperedge with two tails and two heads may serve as one target's entrance and another's exit. So the conflicts run both ways and no matching on one side controls them.

The repair is to stop building a large family and instead take a maximum one and ask what stopped it. A maximum family cannot be extended, so every target it leaves unserved must be obstructed by a hyperedge the family has already spent, and a hyperedge has only r vertices in all to obstruct with. That counts the leftovers without ever needing the two sides to be independent, which is exactly what the general model denies.

Theorem A.64 (The general orientation model shares the constant, [AI]). *Let $r \ge 2$ and $m \ge 2$, and let a directed r-uniform hyperedge be any split (T, H) into a non-empty tail set and a non-empty head set, as in Proposition A.62. Every such multihypergraph $\mathcal{H}$ on $n \ge 2$ vertices with $\kappa^{\max}_{\mathcal{H}} \le m-1$ satisfies*

$$|E(\mathcal{H})| \le \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor + \frac{4(2r+1)(m-1)^2}{r-1}(n-1).$$

The same bound therefore holds under $\lambda^{\max}_{\mathcal{H}} \le m-1$. Hence for fixed r and m the general model has the same leading term $\big(1+o(1)\big)(m-1)n^2/\big(4(r-1)\big)$ as the forward model, under both separations, and the bipartite construction of Proposition A.60 is asymptotically optimal for it too.

In words: how a hyperedge is oriented does not move the leading term. Whether one vertex points at the other $r-1$ of them, or several do, the maximum carries the same $\frac{m-1}{r-1}\lfloor n^2/4\rfloor$ in front, and only the linear term notices the difference, through the factor $2r+1$. With Lemma A.61 this closes the orientation axis to first order.

Proof. Let R be the one-step shadow read in the general model, so $(u,v)\in A(R)$ exactly when some hyperedge has u among its tails and v among its heads, and $\operatorname{cod}(u,v)$ counts the hyperedges realising it. A hyperedge's tail set and head set are disjoint, so R has no loops.

Step 1: hyperedges against shadow arcs. The $\operatorname{cod}(u,v)$ hyperedges realising an arc give that many one-step routes, one hyperedge apiece and different ones, with no interior vertex, so they are pairwise hyperedge-disjoint and internally vertex-disjoint and $\operatorname{cod}(u,v)\le\kappa_{\mathcal{H}}(u,v)\le m-1$. A hyperedge (T,H) is counted by exactly the $|T|\,|H|$ ordered pairs it fills, and $|T|+|H|=r$ with both parts non-empty gives $|T|\,|H|\ge r-1$. Hence

$$(r-1)\,|E(\mathcal{H})| \;\le\; \sum_{(T,H)\in E(\mathcal{H})}|T|\,|H| \;=\; \sum_{(u,v)\in A(R)}\operatorname{cod}(u,v) \;\le\; (m-1)\,|A(R)|.$$

This is the count of Proposition A.62 stopped one step early, exactly as Step 1 above stopped Proposition A.60 one step early.

Step 2: the shadow is budgeted. Fix an ordered pair (u,v) of distinct vertices and let $\mathcal{T}$ collect v itself when $(u,v)\in A(R)$ together with every midpoint counted by $p_2(u,v)$, so $|\mathcal{T}|=\mathbf{1}_{(u,v)\in A(R)}+p_2(u,v)$. Call a hyperedge e an *entrance* for $t\in\mathcal{T}$ if u is a tail of e and t a head, and an *exit* for a midpoint t if t is a tail of e and v a head. Every $t\in\mathcal{T}$ has at least one entrance, and every midpoint at least one exit, by the definition of R. No hyperedge is both an entrance and an exit for the *same* t, since that would put t in its tail set and its head set at once.

Consider families of routes indexed by distinct targets: for the target v the one-step route through an entrance, and for a midpoint t the two-step route $u\to t\to v$ through an entrance and an exit. Such a family is admissible when its hyperedges are pairwise distinct. Let $\mathcal{F}$ be one of maximum size M, serving the target set S, and let U be the hyperedges it uses, so $|U|\le 2M$. The routes of $\mathcal{F}$ are pairwise hyperedge-disjoint by construction and internally vertex-disjoint because their interiors are the empty set and distinct singletons, so

$$M \;\le\; \kappa_{\mathcal{H}}(u,v) \;\le\; m-1.$$

Now take $t\in\mathcal{T}\setminus S$. If some entrance $e\notin U$ existed and, for a midpoint, some exit $f\notin U$ as well, then $e\ne f$ by the paragraph above and the route through them could be added to $\mathcal{F}$, serving a target not yet served and using no hyperedge already spent. That contradicts maximality. So every entrance of t lies in U, or t is a midpoint all of whose exits lie in U. Either way t lies in the tail set or the head set of some member of U, and therefore

$$\mathcal{T}\setminus S \;\subseteq\; \bigcup_{e\in U}(T_e\cup H_e), \qquad \text{so} \qquad |\mathcal{T}| \;\le\; M+\sum_{e\in U}\bigl(|T_e|+|H_e|\bigr) \;=\; M+r\,|U|.$$

The last equality is where the two sides are counted together rather than separately, and it is what keeps the constant at r rather than $2(r-1)$. With $|U| \le 2M$ this gives $|\mathcal{T}| \le (2r+1)M \le (2r+1)(m-1)$, so R is $(2r+1)(m-1)$-budgeted in the sense of Definition A.32.

Step 3: assembling. Lemma A.33 with $C = (2r+1)(m-1)$ gives $|A(R)| \le \lfloor n^2/4 \rfloor + 4(2r+1)(m-1)(n-1)$, and substituting into Step 1 gives the stated bound. The hypothesis $\lambda^{\max} \le m-1$ implies $\kappa^{\max} \le m-1$ by Whitney, so the bound holds there too. For the lower bound, the bipartite family of Proposition A.60 consists of general hyperedges, so it witnesses $\approx (m-1)n^2/(4(r-1))$ here as well, and the upper and lower bounds agree in their n^2 term. □

Two remarks on that proof. The constant $2r+1$ is not optimised and is not claimed to be sharp: it is what the crude estimate $|U| \le 2M$ gives, and a search over deliberately adversarial instances, in which every target shares both its entrance and its exit with another, never drives $|\mathcal{T}|/\kappa(u,v)$ above $r-1$, the forward model's own constant. Nothing rests on the difference, since Lemma A.33 places this constant in the linear term, where it never touches the $n^2/4$ that carries the leading constant. The second point is what the theorem does to the orientation axis of Section 3.3.2. Forward and backward were already known equal, exactly, by Lemma A.61. The general model was known only to sit between the same two bounds a factor of four apart, which left room for it to be denser to first order. It is not. All three orientation models have the same leading term, and the computed differences between them at $n \approx r$ are a small-size effect that the asymptotics cannot see.

Part IV. The audit

The bookkeeping that keeps the search's connectivity checks cheap, and the record of what was computed, with what, and to what standard.

A.10 The search's cheap connectivity bookkeeping

Chapter 2 states that the search avoids most calls to the checker by tracking a bound on $\lambda^{\max}$ rather than recomputing it. This section records how the shortcut works and how it is checked against the value the naive implementation would have returned.

Proposition A.4's two parts settle almost every step without a flow. The energy depends on $\lambda^{\max}$ only through the excess $\max(0,\ \lambda^{\max} - (m-1))$, which is zero exactly when the graph is feasible. So a removal applied to a feasible graph leaves it feasible by part (1), with excess still zero, and needs no measurement at all. An addition made while the graph sits strictly below the bound, at $\lambda^{\max} \le m-2$, also stays feasible. Part (2) caps the rise from a single added unit at one, so the graph can climb at most to $\lambda^{\max} = m-1$, which is still inside the feasible region. The excess is zero again, and again no measurement is needed. Only an addition made right at the boundary $\lambda^{\max} = m-1$ can tip the graph over, and only there does the search run a capped pairwise sweep, stopping at the first pair for which it finds one route too many. The exact connectivity value is recomputed only on the rare step whose proposal is genuinely infeasible, where the penalty term needs it.

To know which case applies, the search carries a running upper bound on $\lambda^{\max}$, raised by one whenever it accepts an addition (part (2)) and left untouched when it accepts a removal (part (1)), and resynchronised to the exact value at a fixed step cadence. The bound can only ever sit at or above the truth, so any step it clears is certainly feasible.

The energy returned on every step is, by Proposition A.4, exactly the value the naive implementation would have computed. Every accept-or-reject decision, every random draw, and the graph the walk finally returns are therefore unchanged, and the speed-up is invisible to every number the thesis reports. A regression test runs both implementations, the fast one and a reference that measures exactly at every step, across all four graph variants and both separations, and checks that the two agree step for step in energy, in every accept-or-reject decision, and in the final graph, and that the returned graph is feasible under the exact checker. The witness the search reports is always certified exactly, either by the monotonicity deductions above or by the capped predicate, and the full binding-connectivity value is recomputed only when the energy or a recorded trace needs it.

Where the method is needed differs by variant. For undirected graphs one could resynchronise the exact value cheaply from the Gomory–Hu shortcut of Theorem A.2, which encodes every pairwise edge-connectivity in $n-1$ flows. The directed and vertex variants admit no such tree, so there the monotone bound and the capped predicate are the only inexpensive handle on feasibility, and they are what let the same matrix-search machinery serve every graph variant at speed.

A.11 Computational audit

The authoritative code and audit instructions are maintained in the versioned repository at `https://github.com/louisvandenbruwaene/thesis`. A submitted build records its precise snapshot in `revision.tex`. The status in this build is: not recorded in this build. When that status contains a commit, the recorded commit is the immutable reference for every number reported here, and anything later in the repository is a continuation rather than part of the submission. A build whose status instead says that no revision was recorded is a working build, not a submission snapshot. From `program/`, `python erdos915_unified.py` runs the principal checks and `python -m unittest discover -s tests -v` runs the regression suite. `python make_figures.py` regenerates the computational figures. The several hundred `solve` calls behind the four variant grids of Figures 3.4 to 3.7 are read from `figures/machine_values.json`, a readable record of what each of those runs returned, so the grids redraw in seconds and the plotted points are the ones this document was built from rather than whatever a fresh timed search happens to reach. Deleting that file, or passing `--refresh`, recomputes every entry, which is how the record is checked. With `--refresh`, fresh outcomes are reconciled with the stored record: a rerun can confirm or improve an entry but cannot replace it with a weaker result merely because a timed search or exhaustion achieved a weaker result on that machine. Deleting the file is the stronger operation that discards the record before recomputing from scratch. The block sweep does not run from this file. It lives in `research_notes/scripts/simple_vertex_blocks.py`, is self-contained, and was written separately, so that the values it reports are an independent check on the main program rather than a second reading of it. It requires nauty's `geng` on the path. The quoted range is reproduced from the repository root by `BMAX=9 .venv/bin/python3 research_notes/scripts/simple_vertex_blocks.py`. The recorded environment used nauty 2.9.3. The dependency ranges the program needs are in `program/requirements.txt`, and the exact versions these runs were last verified against are pinned in `program/requirements-lock.txt`: Python 3.14.6 with NumPy 2.4.6, SciPy 1.18.0 and PuLP 3.3.2, plus Matplotlib 3.11.0 and NetworkX 3.6.1 for the two optional roles. Installing from the lock file reproduces that environment exactly.

The base cases of the directed induction (Theorem 1.6) run to $n = 7$, and are settled by a pruned exhaustive search over all simple digraphs with $\lambda^{\max} \leq 1$. That search is exact and complete at these sizes, so each completed row is an upper-bound proof at that n. The vertex problem needs its own base cases, since Remark A.22 shows they cannot be inherited from the arc ones, and the same driver supplies them with only the inner feasibility test swapped. Table A.1 records both runs. The two separations agree at every size, which is what closes Theorem 1.7, and that agreement is a computed coincidence of the two searches rather than a formal consequence of Whitney's inequality.

n	arc-disjoint	vertex-disjoint	$M(n)$	nodes	status
2	2	2	2	5	complete
3	4	4	4	22	complete
4	6	6	6	537	complete
5	8	8	8	17 823	complete
6	10	10	10	788 101	complete
7	12	12	12	45 122 129	complete

Table A.1. The directed $m = 2$ base cases under both separations: exhaustive branch and bound over all simple digraphs on n vertices with $\lambda^{\max} \leq 1$, and the same run with $\kappa^{\max} \leq 1$. Same driver, same prunes, only the inner feasibility test changes. Both columns agree with $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ at every size, and every run completed. The node counts coincide as a computed fact, not by construction: the feasibility prune switches between the arc and the vertex flow test with the separation, and at these sizes the two happen to cut the tree in the same places. The wall times do not coincide, with the vertex run at $n = 7$ taking 3125 s against the arc run's 718 s. Full logs: `figures/basecase_search_log.txt` and `figures/basecase_search_vertex_log.txt`.

The historical cut-counting optimisation closes the directed multigraph checks through $n = 6$ but emits no independently replayable matching bound, so no theorem rests on it. Its raw log is `figures/certificate_log.txt`. Exact flow measures a supplied object, completed enumeration proves a finite optimum, and heuristic search supplies only a checked witness and lower bound. The file `program/README.md` records the dependencies, commands, entry points, and archival checklist.

Bibliography

[1] Kristóf Bérczi, Karthekeyan Chandrasekaran, Tamás Király, and Shubhang Kulkarni. Splitting-off in hypergraphs. *Journal of Combinatorial Theory, Series B*, 176:319–383, 2026.

[2] Louis Vandenbruwaene. Maximizing Edge Density Subject to Connectivity Constraints: Erdős Problem 915, from Proof to Search. https://github.com/louisvandenbruwaene/thesis, 2026. GitHub repository, accessed 28 August 2026.

[3] Béla Bollobás and Pál Erdős. Extremal problems in graph theory. *Matematikai Lapok*, 13:143–152, 1962. Original statement of Problem 915.

[4] Ralph E. Gomory and Tien Chung Hu. Multi-terminal network flows. *Journal of the Society for Industrial and Applied Mathematics*, 9(4):551–570, 1961. Introduces the Gomory-Hu tree structure.

[5] Pál Bártfai. Schweitzer verseny megoldás (Erdős gráffeladatára). *Matematikai Lapok*, 11:175–176, 1960. In Hungarian; solves Erdős's graph problem for the $m = 3$ vertex-disjoint case.

[6] Béla Bollobás. On graphs with at most three independent paths connecting any two vertices. *Studia Scientiarum Mathematicarum Hungarica*, 1:137–140, 1966.

[7] John L. Leonard. On graphs with at most four line-disjoint paths connecting any two vertices. *Journal of Combinatorial Theory, Series B*, 13:242–250, 1972. "Line-disjoint" is edge-disjoint and "at most four" is $\lambda^{\max} \leq 4$, so the title names the edge problem at $m = 5$. The paper also proves $\ell_m(n) = k_m(n)$ for $2 \leq m \leq 4$, which is the equality stated in Chapter 1.

[8] B. A. Sørensen and Carsten Thomassen. On k-rails in graphs. *Journal of Combinatorial Theory, Series B*, 17(2):143–159, 1974. Theorem 4 determines $f_5(n) = \lfloor 8n/3 \rfloor - 3$ for $n \geq 6$, $n \neq 7$, $n \neq 12$. Their $f_k(n)$ is the least edge count that forces a k-rail, one more than this thesis's $k_m(n)$, so the value here is $\lfloor 8n/3 \rfloor - 4$. The same paper gives the general lower bound on c_m and proves that the Bollobás–Erdős bound does hold for 3-connected graphs.

[9] Karl Menger. Zur allgemeinen Kurventheorie. *Fundamenta Mathematicae*, 10:96–115, 1927. The original statement of Menger's theorem.

[10] Thomas F. Bloom. Erdős problem #915. https://www.erdosproblems.com/915. Accessed 28 August 2026.

[11] Wolfgang Mader. Ein Extremalproblem des Zusammenhangs von Graphen. *Mathematische Zeitschrift*, 131:223–231, 1973. Carries both halves used here. It establishes $\ell_m(n)$ for every $m \geq 2$, which solves Erdős Problem 915 for edge-disjoint paths, and it disproves the

Bollobás–Erdős conjecture for the vertex problem, giving for every $m \geq 6$ and every C an n with $k_m(n) > mn/2 + C$.

[12] Hassler Whitney. Congruent graphs and the connectivity of graphs. *American Journal of Mathematics*, 54(1):150–168, 1932.

[13] L. R. Ford and D. R. Fulkerson. Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404, 1956.

[14] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60:94–112, 2014. Describes the nauty/Traces canonical-labelling tools underlying geng.

[15] Fred Glover. Tabu search—part I. *ORSA Journal on Computing*, 1(3):190–206, 1989. Introduces tabu search, the tenure, and the tabu list.

[16] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi. Optimization by simulated annealing. *Science*, 220(4598):671–680, 1983. Introduces simulated annealing as a combinatorial method.

[17] Nicholas Metropolis, Arianna W. Rosenbluth, Marshall N. Rosenbluth, Augusta H. Teller, and Edward Teller. Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21(6):1087–1092, 1953. Introduces the Metropolis acceptance rule.

[18] John L. Leonard. Graphs with 6-ways. *Canadian Journal of Mathematics*, 25:687–692, 1973. Determines $\ell_6(n)$, the edge value one step past Leonard's own $m = 5$.

[19] John L. Leonard. On a conjecture of Bollobás and Erdős. *Periodica Mathematica Hungarica*, 3(3–4):281–284, 1973. Gives the explicit $m = 5$ counterexample, on 57 vertices and 141 edges, and a constant $c > 0$ with $k_5(n) > (5/2 + c)n - O(1)$. The $m \leq 4$ equality is from Leonard's 1972 paper, not this one.

[20] Wolfgang Mader. Existenz n-fach zusammenhängender teilgraphen in Graphen genügend großer Kantendichte. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, 37:86–97, 1972. Average degree at least $4k$ forces a $(k+1)$-connected subgraph.

[21] W. Mantel. Problem 28. *Wiskundige Opgaven*, 10:60–61, 1907. The triangle-free extremal bound, the $r = 2$ case of Turán's theorem.

[22] Pál Turán. On an extremal problem in graph theory. *Matematikai és Fizikai Lapok*, 48:436–452, 1941.

[23] Chandra Chekuri and Chao Xu. Computing minimum cuts in hypergraphs. In *Proceedings of the 28th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1085–1100, 2017. Develops cut algorithms for hypergraphs.

[24] Megan Dewar, David Pike, and John Proos. Connectivity in hypergraphs. *Canadian Mathematical Bulletin*, 61(2):252–271, 2018. Studies hypergraph connectivity concepts.

[25] Saeid Hanifehnezhad and Ardeshir Dolati. Symmetric submodular system: Contractions and Gomory–Hu tree. *Information and Computation*, 271:104479, 2020. Gomory-Hu trees for symmetric submodular cut functions.

[26] Zsolt Baranyai. On the factorization of the complete uniform hypergraph. In *Infinite and Finite Sets (Colloq., Keszthely, 1973), Vol. I*, volume 10 of *Colloq. Math. Soc. Janos Bolyai*, pages 91–108. North-Holland, 1975.

[27] Reinhard Diestel. *Graph Theory*, volume 173 of *Graduate Texts in Mathematics*. Springer, 5th edition, 2017. Standard reference for graph theory. Chapter 3 carries Menger's theorem in both its local and its global form.

[28] W. T. Tutte. *Connectivity in Graphs*. University of Toronto Press, Toronto, 1966.

[29] John E. Hopcroft and Robert E. Tarjan. Dividing a graph into triconnected components. *SIAM Journal on Computing*, 2(3):135–158, 1973.

DEPARTMENT OF MATHEMATICS
Celestijnenlaan 200B
3001 Leuven, Belgium
wis.kuleuven.be